

STAR RX72N - rx_spi_comm Library
1.0.0

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Chapter 1

Directory Hierarchy

1.1 Directories

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Chapter 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

libs/rx_spi_comm/inc/ rx_spi_comm.h	
High-Level SPI Frame Protocol Communication Layer for RX72N Peripheral Mode	7
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Chapter 3

Directory Documentation

3.1 libs/rx_spi_comm/inc Directory Reference

Directory dependency graph for inc:

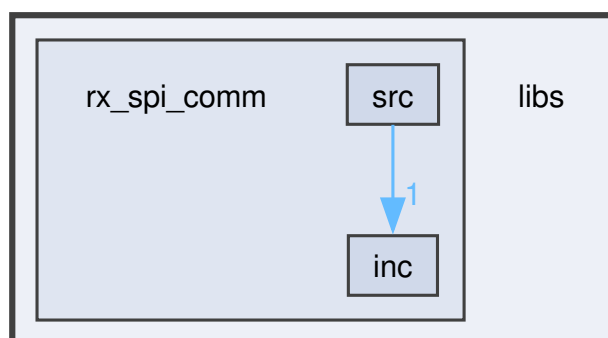


Files

- file [rx_spi_comm.h](#)
High-Level SPI Frame Protocol Communication Layer for RX72N Peripheral Mode.

3.2 libs Directory Reference

Directory dependency graph for libs:

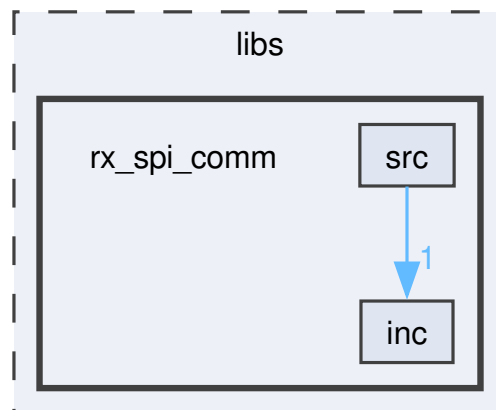


Directories

- directory [rx_spi_comm](#)

3.3 libs/rx_spi_comm Directory Reference

Directory dependency graph for rx_spi_comm:

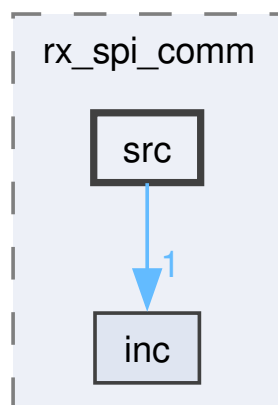


Directories

- directory [inc](#)
- directory [src](#)

3.4 libs/rx_spi_comm/src Directory Reference

Directory dependency graph for src:



Files

- file [rx_spi_comm.c](#)
High-Level SPI Frame Protocol Communication Layer Implementation.

Chapter 4

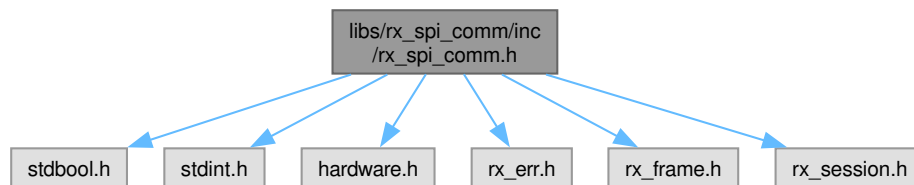
File Documentation

4.1 libs/rx_spi_comm/inc/rx_spi_comm.h File Reference

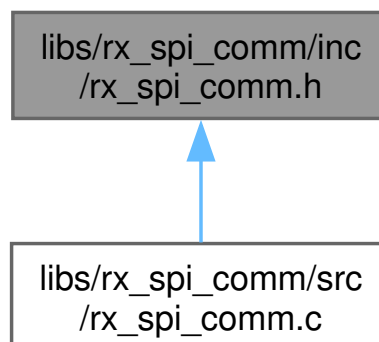
High-Level SPI Frame Protocol Communication Layer for RX72N Peripheral Mode.

```
#include <stdbool.h>
#include <stdint.h>
#include "hardware.h"
#include "rx_err.h"
#include "rx_frame.h"
#include "rx_session.h"
```

Include dependency graph for rx_spi_comm.h:



This graph shows which files directly or indirectly include this file:



Data Structures

- struct [rx_spi_comm_retransmit_config_t](#)
Configuration for automatic retransmission behavior.
- struct [rx_spi_comm_handle_t](#)
SPI communication handle containing all state for frame protocol.
- struct [rx_spi_comm_config_t](#)
Configuration structure for SPI communication initialization.

Enumerations

- enum [rx_spi_comm_constants_t](#): `uint16_t { k_spi_comm_default_mode = 0, k_spi_comm_rx_buffer_size = 2048, k_spi_comm_tx_buffer_size = 2048, k_spi_comm_default_timeout = 1000 }`
SPI communication configuration constants and buffer sizes.
- enum [rx_retransmit_defaults_retries_t](#): `uint8_t { k_retransmit_default_max_retries = 3 }`
Default retry count for automatic retransmission.
- enum [rx_retransmit_defaults_timing_t](#): `uint16_t { k_retransmit_default_ack_timeout_ms = 50, k_retransmit_default_max_backoff_ms = 400 }`
Default timing values for automatic retransmission.

Functions

- `rx_err_t rx_spi_comm_init (rx_spi_comm_handle_t *handle, const rx_spi_comm_config_t *config)`
Initialize SPI communication handle with frame protocol support.
- `rx_err_t rx_spi_comm_deinit (rx_spi_comm_handle_t *handle)`
Deinitialize SPI communication handle and release resources.
- `rx_err_t rx_spi_comm_send (rx_spi_comm_handle_t *handle, rx_frame_type_t type, uint8_t flags, const uint8_t *payload, uint32_t payload_len)`
Send frame on SPI with automatic sequencing and CRC.
- `rx_err_t rx_spi_comm_send_ack (rx_spi_comm_handle_t *handle, uint16_t sequence)`
Send ACK frame for received frame (convenience wrapper).
- `rx_err_t rx_spi_comm_send_nack (rx_spi_comm_handle_t *handle, uint16_t sequence, uint8_t flags)`
Send NACK frame for received frame with error indication.
- `rx_err_t rx_spi_comm_receive (rx_spi_comm_handle_t *handle, rx_frame_t *frame, uint32_t timeout_ms)`
Receive and decode frame from RPi5 with CRC validation.
- `rx_err_t rx_spi_comm_data_available (const rx_spi_comm_handle_t *handle, bool *available)`
Check if data is available for reading (non-blocking query).
- `rx_err_t rx_spi_comm_set_control_callbacks (rx_spi_comm_handle_t *handle, void(*on_ping_cb)(const rx_frame_t *frame, void *ctx), void(*on_reset_cb)(const rx_frame_t *frame, void *ctx), void *cb_ctx)`
Register callbacks for PING and RESET control frames.
- `rx_err_t rx_spi_comm_send_pong (rx_spi_comm_handle_t *handle, const uint8_t *payload, uint32_t payload_len)`
Send PONG frame with payload (response to PING).
- `rx_err_t rx_spi_comm_send_reset_ack (rx_spi_comm_handle_t *handle)`
Send RESET_ACK frame (response to RESET).
- `rx_err_t rx_spi_comm_process_retransmits (rx_spi_comm_handle_t *handle, uint32_t current_time_ms)`

Check and process automatic retransmissions.

- `rx_err_t rx_spi_comm_set_auto_retransmit(rx_spi_comm_handle_t *handle, bool enabled, const rx_spi_comm_retransmit_config_t *config)`

Enable or disable automatic retransmission at runtime.

- `rx_err_t rx_spi_comm_set_retransmit_callbacks(rx_spi_comm_handle_t *handle, void(*on_ack_cb)(uint16_t sequence, void *ctx), void(*on_nack_cb)(uint16_t sequence, void *ctx), void *ctx)`

Register ACK/NACK notification callbacks for retransmission.

4.1.1 Detailed Description

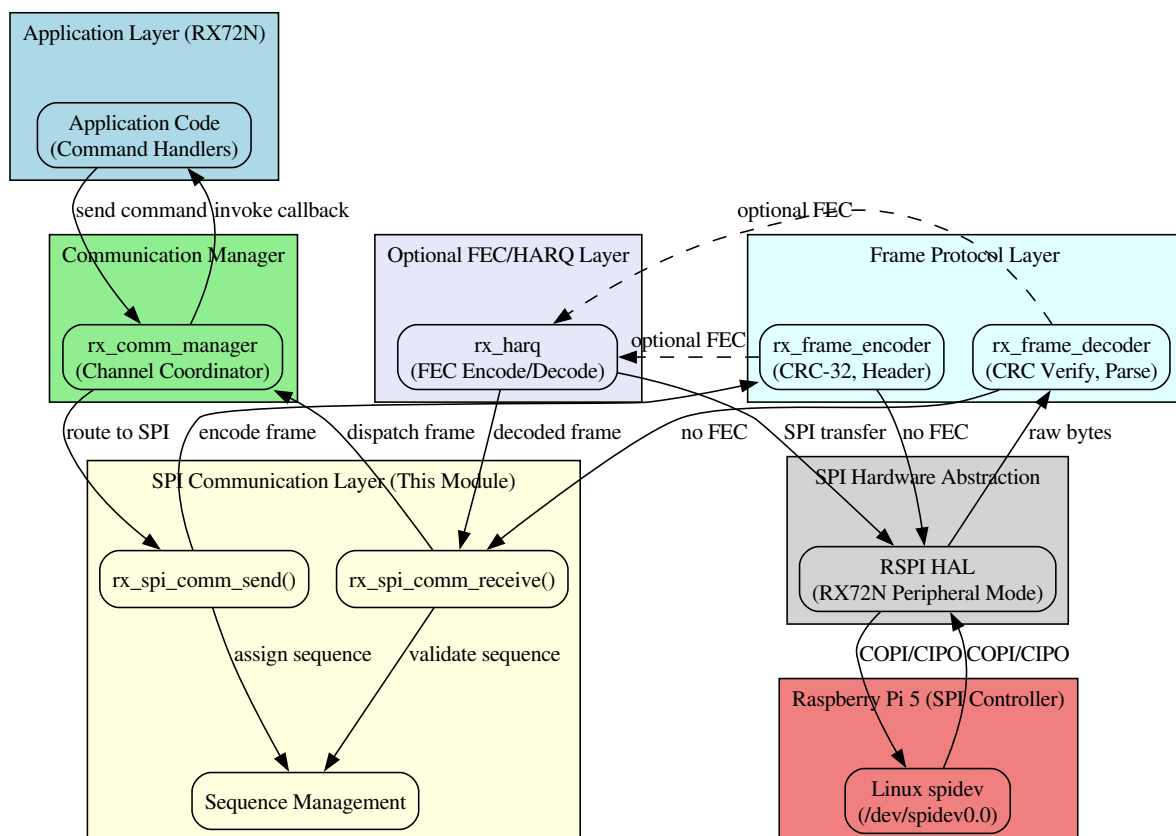
High-Level SPI Frame Protocol Communication Layer for RX72N Peripheral Mode.

4.1.1.1 Overview

Production-quality SPI communication layer providing reliable frame-based protocol between RX72N MCU (SPI peripheral) and Raspberry Pi 5 (SPI controller). Integrates frame encoding/decoding with CRC-32 validation, automatic sequence number management, optional Forward Error Correction (FEC), and HARQ (Hybrid Automatic Repeat reQuest) retransmission support.

Purpose: Enable reliable, high-throughput bidirectional communication over SPI hardware interface with automatic error detection, correction, and recovery mechanisms.

4.1.1.2 System Architecture



4.1.1.3 SPI Communication Protocol

Hardware Configuration:

- RX72N Role: SPI Peripheral
- RPi5 Role: SPI Controller
- RSPI Channel: RSPI0 (configurable)
- SPI Mode: Mode 0 (CPOL=0, CPHA=0) default
- Clock Speed: Up to 10 MHz (limited by RPi5 spidev)
- Data Width: 8-bit transfers
- Endianness: Little-endian (all multi-byte fields)

Frame Format (rx_frame protocol):

Sync	Sequence	Length	Type	Flags	Payload (0-255)	CRC-32
2B	2B	2B	1B	1B	variable	4B
0xAA55	uint16	uint16	uint8	uint8	uint8[]	uint32

Sequence Number Management (shared session):

- TX/RX sequences managed by shared rx_session_state_t
- Cross-transport continuity: USB and SPI share the same sequence counters
- Gap tolerance: Accepts gaps up to 10 frames (matching Go gateway)
- Sequence reset: Via rx_session_reset() on the shared session

Error Detection and Correction:

- CRC-32: Mandatory for all frames (polynomial 0x04C11DB7)
- FEC (optional): Reed-Solomon or convolutional coding
- HARQ (optional): Soft-combining retransmissions

4.1.1.4 Performance Characteristics

Operation	Avg Time	Worst Case	Notes
<code>rx_spi_comm_init()</code>	~10 us	~20 us	Frame encoder/decoder setup
<code>rx_spi_comm_send()</code>	~150-500 us	~1 ms	Depends on payload size, FEC
<code>rx_spi_comm_receive()</code>	~100-400 us	~800 us	Depends on payload size, FEC
<code>rx_spi_comm_send_ack()</code>	~50 us	~100 us	Minimal frame (no payload)
SPI transfer (HW)	payload_len x 0.8 us	-	@ 10 MHz SPI clock
CRC-32 calculation	payload_len x 0.05 us	-	Software CRC (hardware CRC faster)

Operation	Avg Time	Worst Case	Notes
FEC encoding	payload_len x 2 us	-	Optional, adds ~50% overhead
FEC decoding	payload_len x 3 us	-	Optional, with error correction

Throughput Analysis:

- Theoretical max: 10 Mbps (SPI clock) x 8 bits = 80 Mbps
- Practical (no FEC): ~6 Mbps (frame overhead + processing)
- With FEC: ~4 Mbps (encoding/decoding overhead)
- Latency (256B frame): ~300 us (send) + ~250 us (receive) = ~550 us round-trip

4.1.1.5 Memory Usage

Memory Type	Size	Usage
<code>rx_spi_comm_handle_t</code>	~4116 bytes	Main state structure
RX buffer	2048 bytes	Receive staging buffer
TX buffer	2048 bytes	Transmit staging buffer
Frame encoder	~8 bytes	Frame encoding state
Frame decoder	~8 bytes	Frame decoding state
Stack (send)	~64 bytes	Function locals + encoder
Stack (receive)	~80 bytes	Function locals + decoder + rx_frame_t

Memory Layout (`rx_spi_comm_handle_t`):

Offset	Size	Field	Purpose
0x00	8	encoder	Frame encoder state
0x08	8	decoder	Frame decoder state
0x10	2048	rx_buffer	RX staging buffer
0x810	2048	tx_buffer	TX staging buffer
0x1010	4/8	session	Pointer to shared session state
0x1014	1	channel	RSPI channel (0-2)
0x1015	1	fec_enabled	FEC enable flag
0x1016	1	initialized	Initialization flag
0x1017	1	(padding)	Alignment
Total: 4120 bytes (0x1018)			

4.1.1.6 Thread Safety Analysis

Conditional Thread Safety - Safe with external synchronization:

Scenario	Thread Safety	Mitigation Required
Single-threaded	[PASS] Safe	None
Multi-threaded (same handle)	[FAIL] Unsafe	External mutex required
Multi-threaded (different handles)	[PASS] Safe	None (independent state)
Send from multiple threads	[FAIL] Unsafe	Mutex around send operations

Scenario	Thread Safety	Mitigation Required
Receive from multiple threads	[FAIL] Unsafe	Dedicate one thread to polling
Send from ISR	[WARN] Conditional	OK if SPI HAL is ISR-safe
Receive from ISR	[WARN] Conditional	OK if SPI HAL is ISR-safe

Recommended Architecture:

- Dedicate one thread to SPI communication (poll + send)
- Use communication manager (rx_comm_manager) for multi-channel coordination
- External mutex if multiple threads must send on same handle

4.1.1.7 Hardware Requirements

Resource	Requirement	Usage
CPU	RX72N @ 240 MHz	SPI transfer + CRC calculation
RAM	~4.2 KB per handle	State + buffers (see Memory Usage)
RSPI	RSPI0, RSPI1, or RSPI2	Configurable channel (default: RSPI0)
SPI Pins	4 pins (COPI/CIPO/CLK/CS)	See pinout in docs/sections/03_hardware_↩ pinout.tex
DMA (optional)	DMAC channel	For high-throughput SPI transfers
Interrupts	RSPI RX/TX interrupts	For non-blocking transfers

Pin Configuration (RSPI0 example):

- COPI (Controller Out Peripheral In): PD3 (RX72N input from RPi5)
- CIPO (Controller In Peripheral Out): PD2 (RX72N output to RPi5)
- SCLK (Serial Clock): PD1 (Clock from RPi5 controller)
- CS (Chip Select): PD0 (Active low, from RPi5)

4.1.1.8 Usage Examples

Complete Setup and Integration:

```
#include "rx_spi_comm.h"
#include "hardware.h"

// Global SPI communication handle
static rx_spi_comm_handle_t g_spi_handle;

// Initialize SPI communication
rx_err_t init_spi_communication(void)
{
    // Step 1: Initialize SPI hardware (RSPI peripheral mode)
    rspi_config_t rspi_cfg = {
        .channel = 0,           // RSPI0
        .mode = 0,             // SPI mode 0 (CPOL=0, CPHA=0)
        .bit_rate = 10000000,   // 10 MHz
        .peripheral_mode = true, // RX72N is SPI peripheral
    };
```

```

rx_err_t err = rspi_init_peripheral(&rspi_cfg);
if (err != k_rx_ok) {
    return err;
}

// Step 2: Initialize shared session state
static rx_session_state_t g_session;
err = rx_session_init(&g_session);
if (err != k_rx_ok) {
    return err;
}

// Step 3: Initialize SPI communication layer
rx_spi_comm_config_t comm_cfg = {
    .session = &g_session,
    .channel = 0,           // RSPI0
    .spi_mode = 0,         // SPI mode 0
    .fec_enabled = false,  // Disable FEC for now
};
err = rx_spi_comm_init(&g_spi_handle, &comm_cfg);
if (err != k_rx_ok) {
    return err;
}

return k_rx_ok;
}

// Send telemetry response
rx_err_t send_telemetry(void)
{
    uint8_t telemetry[16];

    // Fill telemetry data
    telemetry[0] = get_temperature_celsius();
    telemetry[1] = get_motor_current();
    // ... fill remaining fields ...

    // Send telemetry frame
    rx_err_t err = rx_spi_comm_send(&g_spi_handle,
                                    k_frame_type_response,
                                    k_frame_flag_none,
                                    telemetry,
                                    sizeof(telemetry));

    if (err != k_rx_ok) {
        // Handle send error
        return err;
    }

    return k_rx_ok;
}

// Receive command from RPi5
void poll_spi_commands(void)
{
    rx_frame_t frame;

    // Poll with 0ms timeout (non-blocking)
    rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);

    if (err == k_rx_ok) {
        // Frame received successfully
        handle_command(&frame);

        // Send ACK
        rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
    } else if (err == k_rx_err_timeout) {
        // No data available - this is normal
    } else if (err == k_rx_err_crc_mismatch) {
        // CRC error - send NACK for retransmission
        rx_spi_comm_send_nack(&g_spi_handle, frame.header.sequence, k_frame_flag_none);
    } else {
        // Other error - log and continue
        log_error("SPI receive error: %d", err);
    }
}

```

Example - Sequence Number Management:

```

// Reset sequence counters (e.g., after connection reset)
rx_spi_comm_reset_sequence(&g_spi_handle);

// Query current TX sequence
uint16_t tx_seq;
rx_err_t err = rx_spi_comm_get_tx_sequence(&g_spi_handle, &tx_seq);

```

```
printf("Current TX sequence: %u\n", tx_seq);

// Query expected RX sequence
uint16_t rx_seq;
err = rx_spi_comm_get_rx_sequence(&g_spi_handle, &rx_seq);
printf("Expected RX sequence: %u\n", rx_seq);
```

4.1.1.9 NASA Power of 10 Compliance

Rule	Compliance	Implementation Notes
Rule 1: Control Flow	[PASS]	No goto, setjmp, longjmp, or recursion
Rule 2: Loop Bounds	[PASS]	All loops have statically provable bounds
Rule 3: No Dynamic Allocation	[PASS]	Zero malloc/free - all static buffers
Rule 4: Function Size	[PASS]	All functions < 60 lines
Rule 5: Assertions	[PASS]	Minimum 2 checks per function (nullptr, initialized)
Rule 6: Data Scope	[PASS]	Variables declared at smallest scope
Rule 7: Check Returns	[PASS]	All return values validated
Rule 8: Preprocessor	[PASS]	Typed enums for constants, minimal macros
Rule 9: Pointers	[PASS]	Single-level dereferencing only
Rule 10: Compiler Warnings	[PASS]	-Wall -Wextra -Werror enabled

4.1.1.10 SOLID Principles

Principle	Implementation
Single Responsibility	Module responsible ONLY for SPI frame protocol; hardware abstraction delegated to RSPI HAL
Open/Closed	Extensible via configuration (FEC enable/disable); new frame types handled by frame layer
Liskov Substitution	Interchangeable with rx_usb_comm (same rx_frame_t interface)
Interface Segregation	Small focused API (9 functions); send/receive/utility separated
Dependency Inversion	Depends on abstract rx_frame_t and RSPI HAL, not concrete implementations

4.1.1.11 Module Dependencies

Includes:

- rx_err.h - Error code definitions
- rx_frame.h - Frame protocol definitions and encoder/decoder
- stdbool.h - Boolean type
- stdint.h - Fixed-width integer types

Links Against:

- rx_frame - Frame encoding/decoding with CRC-32
- rx_harq (optional) - FEC encoding/decoding with HARQ
- rspi HAL - SPI hardware abstraction (peripheral mode)
- rx_crc32 - CRC-32 calculation (via rx_frame)

Referenced Documentation:

- docs/sections/01_nanopb_protocol.tex - Complete protocol specification
- docs/sections/03_hardware_pinout.tex - SPI pin assignments
- RX72N_Manual_Chapters/Ch25_SPI.txt - RSPI peripheral specification

Date

2026-01-27

Copyright

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Since

Version 1.0.0

See also

rx_frame.h Frame protocol layer
rx_harq.h HARQ layer (optional)
hardware.h SPI HAL interface
rx_comm_manager.h Multi-channel communication coordinator
docs/sections/01_nanopb_protocol.tex Protocol specification

Definition in file [rx_spi_comm.h](#).

4.1.2 Data Structure Documentation

4.1.2.1 struct rx`spi`comm`retransmit`config`

Configuration for automatic retransmission behavior.

Controls retransmission parameters when auto_retransmit is enabled. Zero values are replaced with defaults during initialization.

Field Constraints:

Field	Min	Max	Default	Notes
max_retries	1	10	3	0 = use default
ack_timeout_ms	10	1000	50	0 = use default
max_backoff_ms	50	5000	400	0 = use default

Since

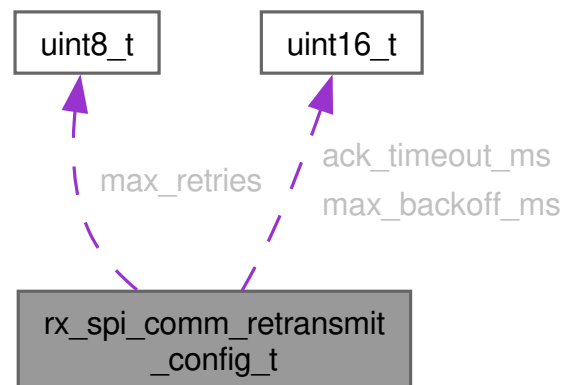
Version 1.0.0

See also

[rx_spi_comm_set_auto_retransmit\(\)](#) Runtime configuration

Definition at line 517 of file [rx_spi_comm.h](#).

Collaboration diagram for rx_spi_comm_retransmit_config_t:



Data Fields

uint16_t	ack_timeout_ms	Initial ACK timeout in ms (default 50, 0 = use default)
uint16_t	max_backoff_ms	Max backoff cap in ms (default 400, 0 = use default)
uint8_t	max_retries	Max retransmit attempts (default 3, 0 = use default)

4.1.2.2 struct rx_spi_comm_handle_t

SPI communication handle containing all state for frame protocol.

Complete state for SPI frame-based communication including frame encoder/decoder, sequence number tracking, and staging buffers. This structure must be initialized via [rx_spi_comm_init\(\)](#) before use.

Lifecycle:

1. Allocate handle (typically static or global)

2. Initialize via [rx_spi_comm_init\(\)](#)
3. Use for send/receive operations
4. Deinitialize via [rx_spi_comm_deinit\(\)](#) when done

Memory Footprint: 4120 bytes (see file-level Memory Usage section)

Thread Safety: Not thread-safe - use external mutex if accessed from multiple threads.

Memory Layout:

See file-level Memory Usage section for detailed offset and size breakdown.

Usage Example:

```
static rx_spi_comm_handle_t g_spi_handle;
static rx_session_state_t g_session;

void init(void) {
    rx_session_init(&g_session);
    rx_spi_comm_config_t cfg = {
        .session = &g_session,
        .channel = 0,
        .spi_mode = 0,
        .fec_enabled = false
    };
    rx_err_t err = rx_spi_comm_init(&g_spi_handle, &cfg);
    // ... check error ...
}
```

Since

Version 1.0.0

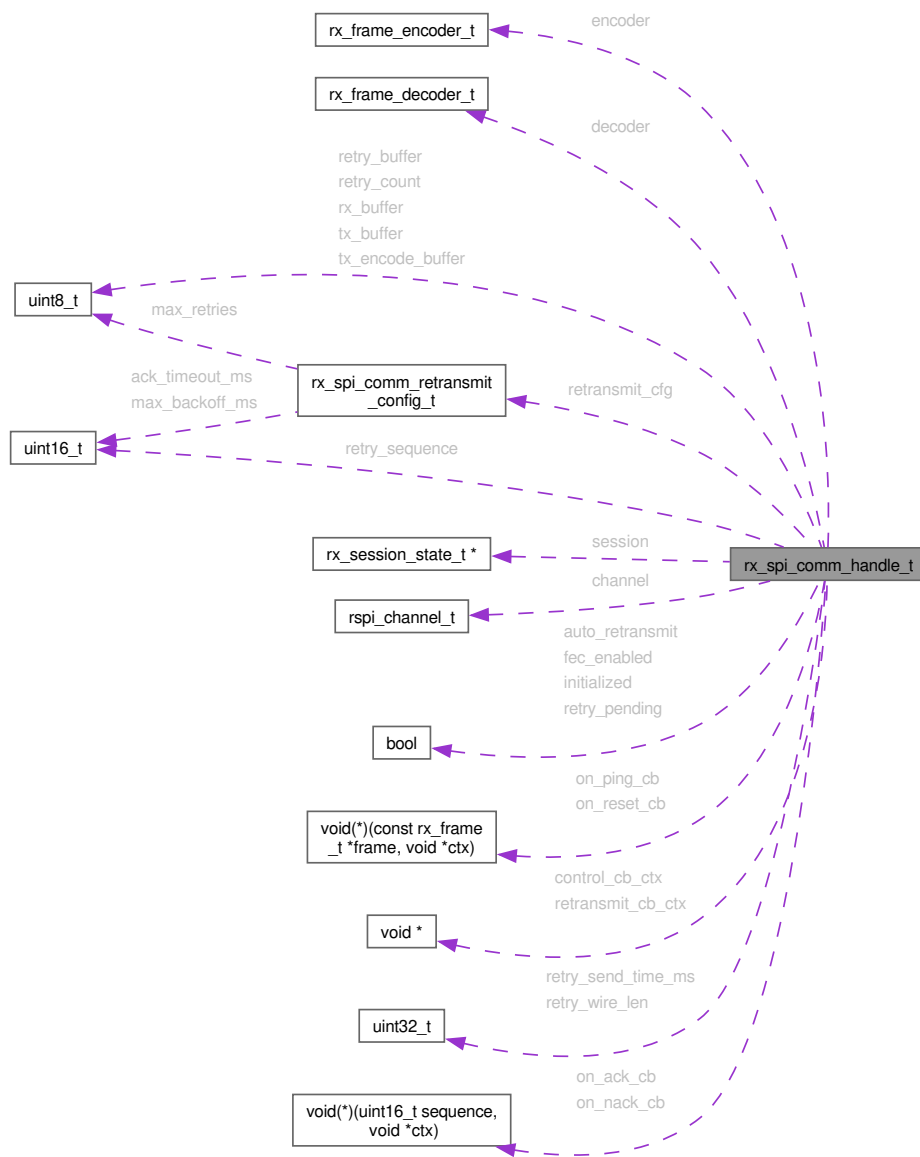
See also

[rx_spi_comm_init\(\)](#) Initialization function

[rx_spi_comm_deinit\(\)](#) Cleanup function

Definition at line 573 of file [rx_spi_comm.h](#).

Collaboration diagram for rx_spi_comm_handle_t:



Data Fields

	bool	auto_retransmit	Enable auto-matic retransmission of sent frames. When true, sent frames are buffered and retransmitted on NACK or time-out.
--	------	-----------------	---

rspi_channel_t	channel	RSPI channel (0-2). Hardware RSPI peripheral channel. Valid values: <code>k__↵rspi__↵channel__↵_0</code>, <code>k__↵rspi__↵channel__↵_1</code>, <code>k__↵rspi__↵channel__↵_2</code>. Default: <code>k__↵rspi__↵channel__↵_0</code> (RSPI0 on PD0-PD3). See also <code>rspi__↵channel__↵_t</code> for channel enumeration
void *	control_cb_ctx	User context pointer passed to control frame callbacks (PING, RESET).

rx_frame_decoder_t	decoder	Frame decoder for incoming frames. Decodes wire format into payload + header, validates CRC-32. State: Maintains decoding context (minimal ~8 bytes). See also rx↵ _↵ frame.↵ h rx↵ _↵ frame↵ _↵ decoder↵ _↵ t def- i- ni- tion
--------------------	---------	--

<code>rx_frame_encoder_t</code>	encoder	Frame encoder for outgoing frames. Encodes payload + header into wire format with CRC-32. State: Maintains encoding context (minimal ~8 bytes). See also <code>rx_frame_encoder_definitio</code>
---------------------------------	---------	--

	bool	fec_enabled	FEC (Forward Error Correction) enabled flag. If true, FEC encoding/decoding applied to frames. Impact: Adds ~50% overhead, improves reliability in noisy environments. Default: false (FEC disabled for speed) Warning FEC requires both sides (RX72N + RPi5) to agree on encoding
--	------	-------------	---

	bool	initialized	Initialization flag. Set to true by <code>rx_↔spi_↔comm_↔_init()</code> , cleared by <code>rx_↔spi_↔comm_↔_↔deinit()</code> . Purpose: Prevents operations on uninitialized handle. Validation: Checked by all public functions.
<code>void(*) (uint16_t sequence, void *ctx)</code>		<code>on_ack_cb</code>	Optional callback invoked when ACK received for a pending frame.
<code>void(*) (uint16_t sequence, void *ctx)</code>		<code>on_nack_cb</code>	Optional callback invoked when NACK received for a pending frame.

void(*) (const rx_frame_t *frame, void *ctx)	on_ping_cb	Callback invoked when PING control frame received. Called with decoded frame and user context during rx_↔ spi_↔ comm_↔ _↔ receive.
void(*) (const rx_frame_t *frame, void *ctx)	on_reset_cb	Callback invoked when RESET control frame received. Called with decoded frame and user context during rx_↔ spi_↔ comm_↔ _↔ receive.
void *	retransmit_cb_ctx	User context pointer passed to re-transmit call-backs (ACK, NACK).

rx_spi_comm_retransmit_config_t	retransmit_cfg	Active retransmit configuration (copied from config or set at run-time).
uint8_t	retry_buffer[k_frame_max_size]	Retry buffer holding last sent encoded wire frame. Sized to k_frame_max_size (1036 bytes) for maximum frame.
uint8_t	retry_count	Current retry attempt count (0 = first send, 1+ = retransmissions).
bool	retry_pending	True when a frame is buffered and awaiting ACK.
uint32_t	retry_send_time_ms	Timestamp (ms) of last send or retransmit of the buffered frame.

uint16_t	retry_sequence	Sequence number of the frame in retry_ _↔ buffer.
uint32_t	retry_wire_len	Actual encoded wire length of buffered frame in retry_ _↔ buffer.
uint8_t	rx_buffer_ _↔ [k_spi_comm_rx_buffer_size]	RX staging buffer for incoming SPI data. Temporary buffer for receiving frames from RPi5 before decoding. Size: 2048 bytes (k_ _↔ spi_ _↔ comm_ _↔ rx_ _↔ buffer_ _↔ size) Capacity: ~7 maximum-size frames Access: Internal only (not exposed to application)

<code>rx_session_state_t *</code>	<code>session</code>	Pointer to shared session state (cross-transport sequence continuity). Points to a single <code>rx_session_state_t</code> owned by <code>rx_comm_manager_t</code>, shared with the USB CDC transport. When the active transport switches (e.g., USB at seq=105 to SPI), SPI continues at seq=106 because both reference the same session state. Lifetime: Must remain valid for the lifetime of this handle. Thread Safety: Session functions are thread-
-----------------------------------	----------------------	--

uint8_t	tx_buffer↵ [k_spi_comm_tx_buffer_size]	TX staging buffer for out- going SPI data. Temporary buffer for en- coding frames before SPI trans- mission to RPi5. Size: 2048 bytes (k_↵ spi_↵ comm_↵ _tx_↵ buffer_↵ _size) Capac- ity: ~7 maximum- size frames Ac- cess: Internal only (not exposed to appli- cation)
---------	---	--

uint8_t	tx_encode_buffer[k_frame_max_size]	TX encode scratch buffer (handle-owned, not on caller stack). Sized to k_↔frame↔_max↔_size (1036 bytes). Replaces a 1036-byte stack allocation in rx_↔_spi_↔comm_↔_↔send/↔_send↔_pong that previously consumed ~50% of a 2 KB ThreadX task stack. Single-producer per handle, so the caller-side mutex serializing SPI transfers also serializes use of this scratch buffer.
---------	------------------------------------	---

4.1.2.3 struct rx_spi_comm_config_t

Configuration structure for SPI communication initialization.

Specifies SPI hardware parameters and optional FEC enable. Used during [rx_spi_comm_init\(\)](#) to configure the communication layer.

IMPORTANT: config is REQUIRED (must not be NULL). The session pointer in the config is mandatory for shared sequence state.

Field Constraints:

Field	Min	Max	Default	Notes
session	non-NULL	-	-	Required: shared session state
channel	0	2	0	RX72N has 3 RSPI channels
spi_mode	0	3	0	Mode 0 recommended for RPi5
fec_enabled	false	true	false	Enable for noisy environments

Usage Example:

```
rx_spi_comm_handle_t handle;
rx_session_state_t session;
rx_session_init(&session);

rx_spi_comm_config_t cfg = {
    .session = &session,
    .channel = k_rspi_channel_0,
    .spi_mode = k_spi_comm_default_mode,
    .fec_enabled = false
};
rx_err_t err = rx_spi_comm_init(&handle, &cfg);
```

Since

Version 1.0.0

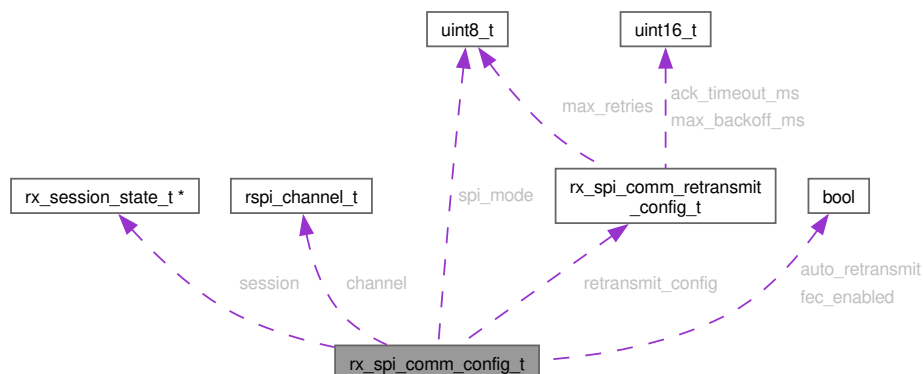
See also

[rx_spi_comm_init\(\)](#) Initialization function

[rx_spi_comm_constants_t](#) Default values

Definition at line 780 of file [rx_spi_comm.h](#).

Collaboration diagram for rx_spi_comm_config_t:



Data Fields

bool	auto_retransmit	Enable automatic retransmission (default: false). When true, sent frames with <code>k_frame_flag_requires_ack</code> are buffered and automatically retransmitted on NACK or ACK timeout. Default: false (fire-and-forget behavior)
rspi_channel_t	channel	RSPI channel (0-2). Hardware peripheral channel. RX72N has 3 RSPI channels. Default: <code>k_rspi_channel_0</code> (RSPI0) See also <code>rspi_channel_t</code> for channel enumeration
bool	fec_enabled	Enable Forward Error Correction. If true, applies FEC encoding to outgoing frames and FEC decoding to incoming frames. Improves reliability at cost of throughput. Default: false Overhead: ~50% (e.g., 100 bytes -> 150 bytes with FEC)
rx_spi_comm_retransmit_config_t	retransmit_config	Retransmit configuration (only used if <code>auto_retransmit</code> true). Zero fields use defaults. Ignored when <code>auto_retransmit</code> is false.
rx_session_state_t *	session	Shared session state (REQUIRED, must not be NULL). Pointer to shared session state for cross-transport sequence continuity. Owned by <code>rx_comm_manager_t</code> , shared with USB CDC transport. Lifetime: Must remain valid for the lifetime of the SPI comm handle. See also <code>rx_session.h</code> Shared session API
uint8_t	spi_mode	SPI mode (0-3). Clock polarity and phase configuration. Mode 0: CPOL=0, CPHA=0 (clock idles low, sample on rising edge) Default: 0 Valid range: [0, 3]

4.1.3 Enumeration Type Documentation

4.1.3.1 `rx_retransmit_defaults_retries_t`

```
enum rx\_retransmit\_defaults\_retries\_t : uint8_t
```

Default retry count for automatic retransmission.

Maximum number of retransmission attempts before giving up. Applied when `auto_retransmit` is enabled but `max_retries` is zero.

Since

Version 1.0.0

Enumerator

<code>k_retransmit_default_max_retries</code>	Default max retransmit attempts
---	---------------------------------

Definition at line 480 of file [rx_spi_comm.h](#).

```
00480         : uint8_t {
00481   k_retransmit_default_max_retries = 3, /**< Default max retransmit attempts */
00482 } rx_retransmit_defaults_retries_t;
```

4.1.3.2 rx`retransmit`defaults`timing`

```
enum rx_retransmit_defaults_timing_t : uint16_t
```

Default timing values for automatic retransmission.

Timing parameters for ACK timeout and exponential backoff cap. Applied when `auto_retransmit` is enabled but timing values are zero.

Since

Version 1.0.0

Enumerator

<code>k_retransmit_default_ack_timeout_ms</code>	Initial ACK wait in ms
<code>k_retransmit_default_max_backoff_ms</code>	Max exponential backoff cap in ms

Definition at line 494 of file [rx_spi_comm.h](#).

```
00494         : uint16_t {
00495   k_retransmit_default_ack_timeout_ms = 50, /**< Initial ACK wait in ms */
00496   k_retransmit_default_max_backoff_ms = 400, /**< Max exponential backoff cap in ms */
00497 } rx_retransmit_defaults_timing_t;
```

4.1.3.3 rx`spi`comm`constants`

```
enum rx_spi_comm_constants_t : uint16_t
```

SPI communication configuration constants and buffer sizes.

Defines compile-time constants for SPI channel selection, buffer sizes, and default timeout values. All sizes chosen to accommodate maximum frame size (`k_frame_max_total_size` = 263 bytes) with safety margin.

Design Rationale:

- 2048-byte buffers: Allows ~7 maximum-size frames to be buffered, providing tolerance for burst traffic and preventing frame loss.
- 10 MHz SPI clock: Maximum reliable speed for RPi5 spidev with 1m cable.
- 1 second timeout: Conservative default for reliable operation.

Since

Version 1.0.0

Enumerator

k_spi_comm_default_mode	Default SPI mode (CPOL=0, CPHA=0). SPI Mode 0: Clock idles low, data sampled on rising edge. RPi5 spidev mode: mode=0 (matches this setting) Value: 0 Modes: 0 (0,0), 1 (0,1), 2 (1,0), 3 (1,1)
k_spi_comm_rx_buffer_size	RX staging buffer size in bytes. Buffer for incoming frames before decoding. Size allows buffering multiple frames to handle burst traffic from RPi5. Capacity: ~7 maximum-size frames (263 bytes each) Memory: Static allocation in rx_spi_comm_handle_t Value: 2048 bytes Rationale: Power-of-2 for alignment, sufficient for burst tolerance
k_spi_comm_tx_buffer_size	TX staging buffer size in bytes. Buffer for outgoing frames before SPI transmission. Same size as RX buffer for symmetry and burst tolerance. Capacity: ~7 maximum-size frames (263 bytes each) Memory: Static allocation in rx_spi_comm_handle_t Value: 2048 bytes Rationale: Symmetric with RX buffer, burst tolerance
k_spi_comm_default_timeout	Default timeout in milliseconds for blocking operations. Used for rx_spi_comm_receive() when non-zero timeout specified. Conservative value: Allows reliable operation even with CPU load. Note: Most operations use 0ms timeout (non-blocking polling). Value: 1000 milliseconds (1 second) Rationale: Conservative for reliable operation

Definition at line 418 of file [rx_spi_comm.h](#).

```
00418         : uint16_t {
```

```

00419 /**
00420  * @brief Default SPI mode (CPOL=0, CPHA=0)
00421  * @details
00422  * SPI Mode 0: Clock idles low, data sampled on rising edge.
00423  * **RPi5 spidev mode**: mode=0 (matches this setting)
00424  * @par Value: 0
00425  * @par Modes: 0 (0,0), 1 (0,1), 2 (1,0), 3 (1,1)
00426  */
00427 k_spi_comm_default_mode = 0,
00428
00429 /**
00430  * @brief RX staging buffer size in bytes
00431  * @details
00432  * Buffer for incoming frames before decoding. Size allows buffering
00433  * multiple frames to handle burst traffic from RPi5.
00434  * **Capacity**: ~7 maximum-size frames (263 bytes each)
00435  * **Memory**: Static allocation in rx_spi_comm_handle_t
00436  * @par Value: 2048 bytes
00437  * @par Rationale: Power-of-2 for alignment, sufficient for burst tolerance
00438  */
00439 k_spi_comm_rx_buffer_size = 2048,
00440
00441 /**
00442  * @brief TX staging buffer size in bytes
00443  * @details
00444  * Buffer for outgoing frames before SPI transmission. Same size as RX
00445  * buffer for symmetry and burst tolerance.
00446  * **Capacity**: ~7 maximum-size frames (263 bytes each)
00447  * **Memory**: Static allocation in rx_spi_comm_handle_t
00448  * @par Value: 2048 bytes
00449  * @par Rationale: Symmetric with RX buffer, burst tolerance
00450  */
00451 k_spi_comm_tx_buffer_size = 2048,
00452
00453 /**
00454  * @brief Default timeout in milliseconds for blocking operations
00455  * @details
00456  * Used for rx_spi_comm_receive() when non-zero timeout specified.
00457  * **Conservative value**: Allows reliable operation even with CPU load.
00458  * **Note**: Most operations use 0ms timeout (non-blocking polling).
00459  * @par Value: 1000 milliseconds (1 second)
00460  * @par Rationale: Conservative for reliable operation
00461  */
00462 k_spi_comm_default_timeout = 1000,
00463 } rx_spi_comm_constants_t;

```

4.1.4 Function Documentation

4.1.4.1 rx_spi_comm_data_available()

```

rx_err_t rx_spi_comm_data_available (
    const rx_spi_comm_handle_t * handle,
    bool * available) [nodiscard]

```

Check if data is available for reading (non-blocking query).

Queries SPI hardware to determine if data is available in receive buffer without actually reading the data. Useful for polling loops to avoid unnecessary receive attempts.

Use Case: Optimize polling by checking availability before calling [rx_spi_comm_receive\(\)](#), avoiding timeout overhead.

Algorithm:

1. Validate handle and available pointer
2. Query RSPI hardware for RX buffer status
3. Set *available = true if data waiting, false otherwise

Performance: ~2 us (fast hardware register read)

Parameters

in	handle	Initialized SPI communication handle - Valid range: Non-nullptr to initialized handle - Constraints: Must be initialized via rx_spi_comm_init()
out	available	Pointer to receive availability status - Valid range: Non-nullptr to bool - Output values: true (data available), false (no data) - On error: Set to false

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Query successful, available flag updated
<code>k_rx_err_invalid_arg</code>	handle or available is nullptr
<code>k_rx_err_invalid_state</code>	handle not initialized

Precondition

handle must be non-NULL and initialized

available must be non-NULL

Postcondition

*available set to true or false based on hardware state

On error, *available set to false (safe default)

Note

This function does NOT read or modify receive buffer

Availability can change between check and actual receive

Fast operation - suitable for high-frequency polling

Example - Optimized Polling Loop:

```
void poll_spi(void)
{
    bool available = false;
    rx_err_t err = rx_spi_comm_data_available(&g_spi_handle, &available);

    if (err == k_rx_ok && available) {
        // Data is available - proceed with receive
        rx_frame_t frame;
        err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);
        if (err == k_rx_ok) {
            handle_frame(&frame);
        }
    }
}
```

See also

[rx_spi_comm_receive\(\)](#) Receive frame

Since

Version 1.0.0

Check if data is available for reading (non-blocking query).

Definition at line 1998 of file [rx_spi_comm.c](#).

```
01999 {
02000     if (handle == nullptr || available == nullptr) {
02001         return k_rx_err_invalid_arg;
02002     }
02003
02004     if (!handle->initialized) {
02005         return k_rx_err_invalid_state;
02006     }
02007
02008     return rspi_peripheral_read_available(handle->channel, available);
02009 }
```

References [rx_spi_comm_handle_t::channel](#), and [rx_spi_comm_handle_t::initialized](#).

4.1.4.2 rx_spi_comm_deinit()

```
rx_err_t rx_spi_comm_deinit (
    rx\_spi\_comm\_handle\_t * handle) [nodiscard]
```

Deinitialize SPI communication handle and release resources.

Marks SPI communication handle as uninitialized, preventing further operations. This function does NOT deinitialize RSPI hardware - caller must separately call [rspi_deinit\(\)](#) if needed.

Algorithm:

1. Validate handle pointer (NULL check)
2. Validate initialized flag (must be initialized to deinitialize)
3. Clear initialized flag (marks handle as invalid)

Design Note: Minimal deinitialization - no dynamic resources to free (all memory is statically allocated in handle structure). Function primarily serves as safety guard to prevent use-after-deinit.

Parameters

in,out	handle	Pointer to SPI communication handle - Valid range: Non-nullptr to initialized handle - Constraints: Must have been initialized via rx_spi_comm_init() - Side effects: Clears handle->initialized flag
--------	--------	--

Returns

`rx_err_t` Error code

Return values

<code>k_rx_ok</code>	Success - handle deinitialized
<code>k_rx_err_invalid_arg</code>	handle is nullptr
<code>k_rx_err_invalid_state</code>	handle was not initialized

Precondition

handle must be non-NULL

handle must be initialized (handle->initialized == true)

Postcondition

handle->initialized = false

Subsequent calls to `rx_spi_comm_send/receive` will return `k_rx_err_invalid_state`

Note

Does NOT free memory (handle is statically allocated)

Does NOT deinitialize RSPI hardware

Thread-safe if no concurrent operations on same handle

Example:

```
// Deinitialize SPI communication
rx_err_t err = rx_spi_comm_deinit(&g_spi_handle);
if (err == k_rx_ok) {
    // Handle no longer usable

    // Separately deinitialize RSPI hardware if needed
    rspi_deinit(g_spi_handle.channel);
}
```

See also

[rx_spi_comm_init\(\)](#) Initialization function
[rspi_deinit\(\)](#) Deinitialize RSPI hardware

Since

Version 1.0.0

Deinitialize SPI communication handle and release resources.

Tears down the encoder and decoder, clearing internal state. Saves the first error encountered from sub-deinit calls and returns it after completing all teardown steps.

Precondition

handle != nullptr
 handle was previously initialized via [rx_spi_comm_init\(\)](#)

Postcondition

handle->initialized == false
 Encoder and decoder resources released

Note

Not thread-safe; caller must ensure no concurrent SPI operations.

See also

[rx_spi_comm_init\(\)](#) Corresponding initialization function

Since

Version 1.0.0

Definition at line 1056 of file [rx_spi_comm.c](#).

```
01057 {
01058     if (handle == nullptr) {
01059         return k_rx_err_invalid_arg;
01060     }
01061     rx_err_t err = k_rx_ok;
01062     if (handle->initialized) {
01063         /* Deinit encoder/decoder: can only fail with nullptr pointer, which is
01064          * impossible here since they are fields of a valid handle. */
01065         (void)rx_frame_encoder_deinit(&handle->encoder);
01066         (void)rx_frame_decoder_deinit(&handle->decoder);
01067         handle->initialized = false;
01068     }
01069     rx_log_debug(s_tag, "SPI comm deinitialized");
01070     return err;
01071 }
```

References [rx_spi_comm_handle_t::decoder](#), [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), and [s_tag](#).

4.1.4.3 rx_spi_comm_init()

```
rx_err_t rx_spi_comm_init (
    rx_spi_comm_handle_t * handle,
    const rx_spi_comm_config_t * config) [nodiscard]
```

Initialize SPI communication handle with frame protocol support.

Performs complete initialization of SPI communication handle including frame encoder/decoder setup, sequence number initialization, and optional FEC configuration. This function **MUST** be called before any send/receive operations.

IMPORTANT: This function does NOT initialize SPI hardware. Caller must initialize RSPI peripheral via `rspi_init_peripheral()` **BEFORE** calling this function.

Algorithm:

1. Validate parameters: Check handle pointer for nullptr (NASA Rule 5)
2. Clear handle: Zero-fill entire structure (4120 bytes)
3. Apply configuration: Copy config fields if non-NULL, else use defaults
4. Initialize frame encoder: Setup CRC-32 and header encoding
5. Initialize frame decoder: Setup CRC-32 validation and header parsing
6. Initialize sequences: Set `tx_sequence = 0`, `rx_sequence = 0`
7. Mark initialized: Set `handle->initialized = true`

Configuration Options (config is **REQUIRED**, not optional):

- session: **REQUIRED**, pointer to initialized `rx_session_state_t`
- channel: 0-2 (RSPI0/1/2), default 0
- spi_mode: 0-3 (SPI modes), default 0
- fec_enabled: true/false, default false

Parameters

out	handle	Pointer to SPI communication handle to initialize • Valid range: Non-nullptr to allocated <code>rx_spi_comm_handle_t</code> (4120 bytes) • Constraints: Must be allocated by caller (typically static or global) • Side effects: Entire structure zero-filled, then populated from config • Lifetime: Must remain valid until <code>rx_spi_comm_deinit()</code> called
-----	--------	---

in	config	Required configuration structure (must not be NULL) - Valid range: Non-nullptr to valid rx_spi_comm_config_t - Constraints: session must be non-NULL, channel must be 0-2, spi_mode must be 0-3 - Lifetime: Not stored - values copied into handle (except session ptr)
----	--------	---

Returns

rx_err_t Error code indicating initialization success or failure

Return values

k_rx_ok	Success - handle fully initialized and ready for operation
k_rx_err_invalid_arg	handle is nullptr; config is nullptr; config->session is nullptr; config->channel > 2 (invalid RSPI channel); config->spi_mode > 3 (invalid SPI mode)

Precondition

handle must point to allocated memory (uninitialized content OK)
 RSPI hardware must already be initialized via `rspi_init_peripheral()`

Postcondition

handle->initialized = true on success
 handle->session points to config->session
 All buffers zero-filled (rx_buffer, tx_buffer)

Note

This function does NOT initialize SPI hardware - caller responsible
 On success, handle is ready for rx_spi_comm_send/receive operations
 Thread-safe ONLY if each thread uses a different handle

Warning

Must call `rspi_init_peripheral()` BEFORE this function

Performance:

- Execution time: ~10 us @ 240 MHz
- Dominated by memset (4120 bytes)

Example:

```
static rx_spi_comm_handle_t g_spi_handle;
static rx_session_state_t g_session;

// Step 1: Initialize shared session
rx_err_t err = rx_session_init(&g_session);
if (err != k_rx_ok) return err;

// Step 2: Initialize SPI communication layer
rx_spi_comm_config_t comm_cfg = {
    .session = &g_session,
    .channel = 0,
    .spi_mode = 0,
    .fec_enabled = false
};
err = rx_spi_comm_init(&g_spi_handle, &comm_cfg);
if (err != k_rx_ok) {
    log_error("SPI comm init failed: %d", err);
    return err;
}
```

See also

[rx_spi_comm_deinit\(\)](#) Cleanup and resource release
[rx_spi_comm_send\(\)](#) Send frame
[rx_spi_comm_receive\(\)](#) Receive frame
[rspi_init_peripheral\(\)](#) Initialize SPI hardware (must call first)

Since

Version 1.0.0

Initialize SPI communication handle with frame protocol support.

Initializes SPI communication handle for frame-based communication between RX72N (peripheral) and RPi5 (controller). Sets up frame encoder/decoder, clears staging buffers, and configures channel and FEC settings.

Initialization Steps:

1. Validate handle: Check handle pointer non-nullptr
2. Zero handle: memset entire structure to clear old state
3. Apply configuration: Use config values or defaults (RSPI0, no FEC)
4. Initialize encoder: Call rx_frame_encoder_init() for TX encoding
5. Initialize decoder: Call rx_frame_decoder_init() for RX decoding
6. Reset sequence counters: Set TX/RX sequence to 0
7. Mark initialized: Set handle->initialized = true

Configuration (config must be non-nullptr):

- Channel: Must specify RSPI channel (e.g., k_rspi_channel_2 for host)
- FEC: fec_enabled flag (true/false)
- Passing config == nullptr returns k_rx_err_invalid_arg

Memory Allocation: This function does NOT allocate memory (NASA Rule 3 compliance). Caller must provide pre-allocated handle (typically static or on task stack).

Precondition

handle must point to valid `rx_spi_comm_handle_t` storage (4120 bytes)
 config != nullptr
 config->session != nullptr
 config->channel must be a valid `rspi_channel_t` value (0-2)

Postcondition

On success, handle->initialized = true
 On success, handle->tx_sequence = 0, handle->rx_sequence = 0
 On success, handle->encoder and handle->decoder initialized
 On failure (encoder init), handle state is zeroed but not usable
 On failure (decoder init), encoder is cleaned up, handle unusable

Note

Call this ONCE per handle before any send/receive operations
 This function is NOT thread-safe (single-threaded init only)

Warning

Do not call multiple times without `rx_spi_comm_deinit()` first
 RSPI peripheral hardware (rspi HAL) must be initialized separately

Performance:

Execution time: ~50 us @ 240 MHz (memset + encoder/decoder init)

Example - Basic Configuration:

```
// Allocate handle (static or on task stack, NOT malloc)
static rx_spi_comm_handle_t g_spi_handle;

rx_spi_comm_config_t config = {
    .session = &session,
    .channel = k_rspi_channel_0,
    .fec_enabled = false,
};
rx_err_t err = rx_spi_comm_init(&g_spi_handle, &config);
if (err != k_rx_ok) {
    rx_log_error("init", "SPI comm init failed");
    return;
}

// Now ready to send/receive frames
rx_spi_comm_send(&g_spi_handle, k_frame_type_data, 0, payload, len);
```

Example - FEC Enabled:

```
static rx_spi_comm_handle_t g_spi_handle;

// Enable FEC for noisy environments
rx_spi_comm_config_t config = {
    .channel = 0,           // RSPi0
    .fec_enabled = true,    // Enable Forward Error Correction
};

rx_err_t err = rx_spi_comm_init(&g_spi_handle, &config);
if (err == k_rx_ok) {
    rx_log_info("init", "SPI comm initialized with FEC");
}
```

Example - Multiple RSPI Channels:

```
// Use RSPI1 instead of default RSPI0
static rx_spi_comm_handle_t g_spi1_handle;

rx_spi_comm_config_t config = {
    .channel = 1,           // RSPI1 (pins PE0-PE3)
    .fec_enabled = false,
};

rx_spi_comm_init(&g_spi1_handle, &config);
```

See also

[rx_spi_comm_deinit\(\)](#) Clean up resources when done
[rx_spi_comm_config_t](#) Configuration structure definition
[rx_frame_encoder_init\(\)](#) Frame encoder initialization
[rx_frame_decoder_init\(\)](#) Frame decoder initialization

Since

Version 1.0.0

Definition at line 990 of file [rx_spi_comm.c](#).

```
00991 {
00992     if (handle == nullptr) {
00993         return k_rx_err_invalid_arg;
00994     }
00995
00996     if (config == nullptr || config->session == nullptr) {
00997         return k_rx_err_invalid_arg;
00998     }
00999
01000     *handle = s_zero_handle;
01001
01002     /* Apply configuration */
01003     handle->session = config->session;
01004     handle->channel = config->channel;
01005     handle->fec_enabled = config->fec_enabled;
01006     handle->auto_retransmit = config->auto_retransmit;
01007     handle->retransmit_cfg = config->retransmit_config;
01008
01009     /* Apply retransmit defaults for zero fields */
01010     if (handle->auto_retransmit) {
01011         if (handle->retransmit_cfg.max_retries == 0) {
01012             handle->retransmit_cfg.max_retries = k_retransmit_default_max_retries;
01013         }
01014         if (handle->retransmit_cfg.ack_timeout_ms == 0) {
01015             handle->retransmit_cfg.ack_timeout_ms = k_retransmit_default_ack_timeout_ms;
01016         }
01017         if (handle->retransmit_cfg.max_backoff_ms == 0) {
01018             handle->retransmit_cfg.max_backoff_ms = k_retransmit_default_max_backoff_ms;
01019         }
01020     }
01021
01022     /* Initialize frame encoder: can only fail if pointer is nullptr, which is
01023      * impossible here since &handle->encoder is always non-null. */
01024     (void)rx_frame_encoder_init(&handle->encoder);
01025
01026     /* Initialize frame decoder: same invariant as encoder above. */
01027     (void)rx_frame_decoder_init(&handle->decoder);
01028
01029     /* Sequence counters managed by shared session (config->session) */
01030
01031     handle->initialized = true;
01032
01033     rx_log_debug(s_tag, "SPI comm initialized");
01034     return k_rx_ok;
01035 }
```

References [rx_spi_comm_retransmit_config_t::ack_timeout_ms](#), [rx_spi_comm_config_t::auto_retransmit](#), [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_config_t::channel](#), [rx_spi_comm_handle_t::channel](#), [rx_spi_comm_handle_t::decoder](#), [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_config_t::fec_enabled](#), [rx_spi_comm_handle_t::fec_enabled](#), [rx_spi_comm_handle_t::initialized](#), [k_retransmit_default_ack_timeout_ms](#), [k_retransmit_default_max_backoff_ms](#), [k_retransmit_default_max_retries](#), [rx_spi_comm_retransmit_config_t::max_retries](#), [rx_spi_comm_handle_t::retransmit_cfg](#), [rx_spi_comm_config_t::s_tag](#), [s_zero_handle](#), [rx_spi_comm_config_t::session](#), and [rx_spi_comm_handle_t::session](#).

4.1.4.4 rx'spi'comm'process'retransmits()

```
rx_err_t rx_spi_comm_process_retransmits (
    rx_spi_comm_handle_t * handle,
    const uint32_t current_time_ms) [nodiscard]
```

Check and process automatic retransmissions.

Checks whether an ACK timeout has elapsed for the most recent sent frame. If `auto_retransmit` is enabled and a retransmit is pending, resends the buffered frame (up to `max_retries` with exponential backoff).

Parameters

in,out	handle	Initialized SPI communication handle
in	current_time_ms	Current system time in milliseconds

Returns

rx_err_t Error code

Return values

k_rx_ok	No action needed or retransmit sent
k_rx_err_invalid_arg	handle is nullptr
k_rx_err_invalid_state	handle not initialized

Precondition

handle must be non-NULL and initialized

Postcondition

retry_count incremented on retransmit, pending cleared on limit

Note

Safe to call when `auto_retransmit` is disabled (no-op)

See also

[rx_spi_comm_set_auto_retransmit\(\)](#) Enable/disable retransmission

Since

Version 1.0.0

Check and process automatic retransmissions.

Precondition

handle must be non-NULL and initialized

Postcondition

retry_count incremented on retransmit, pending cleared on limit

Note

Safe to call when auto_retransmit is disabled (no-op)

Since

Version 1.0.0

Definition at line 2239 of file [rx_spi_comm.c](#).

```

02241 {
02242     if (handle == nullptr) {
02243         return k_rx_err_invalid_arg;
02244     }
02245     if (!handle->initialized) {
02246         return k_rx_err_invalid_state;
02247     }
02248 }
02249
02250 /* No-op when retransmit disabled or nothing pending */
02251 if (!handle->auto_retransmit || !handle->retry_pending) {
02252     return k_rx_ok;
02253 }
02254
02255 /* Compute exponential backoff: ack_timeout_ms * 2^retry_count, capped.
02256  * Clamp shift to 31 to avoid undefined behavior when retry_count >= 32. */
02257 uint8_t shift = handle->retry_count;
02258 if (handle->retry_count >= k_max_backoff_shift) {
02259     shift = k_max_backoff_shift;
02260 }
02261 uint32_t backoff_ms = (uint32_t)handle->retransmit_cfg.ack_timeout_ms « shift;
02262
02263 if (backoff_ms > handle->retransmit_cfg.max_backoff_ms) {
02264     backoff_ms = handle->retransmit_cfg.max_backoff_ms;
02265 }
02266
02267 /* Check if timeout has elapsed */
02268 const uint32_t elapsed_ms = current_time_ms - handle->retry_send_time_ms;
02269 if (elapsed_ms < backoff_ms) {
02270     return k_rx_ok; /* Not yet time to retry */
02271 }
02272
02273 const rx_err_t err = internal_retransmit_frame(handle);
02274
02275 /* Override send time with caller-provided time (more accurate than internal timestamp) */
02276 handle->retry_send_time_ms = current_time_ms;
02277
02278 return err;
02279 }

```

References [rx_spi_comm_retransmit_config_t::ack_timeout_ms](#), [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_handle_t::initialized](#), [internal_retransmit_frame\(\)](#), [k_max_backoff_shift](#), [rx_spi_comm_retransmit_cfg](#), [rx_spi_comm_handle_t::retransmit_cfg](#), [rx_spi_comm_handle_t::retry_count](#), [rx_spi_comm_handle_t::retry_pending](#) and [rx_spi_comm_handle_t::retry_send_time_ms](#).

Here is the call graph for this function:



4.1.4.5 rx_spi_comm_receive()

```
rx_err_t rx_spi_comm_receive (
    rx_spi_comm_handle_t * handle,
    rx_frame_t * frame,
    uint32_t timeout_ms) [nodiscard]
```

Receive and decode frame from RPi5 with CRC validation.

Reads raw bytes from SPI hardware, validates CRC-32, decodes frame header and payload, and optionally applies FEC decoding with HARQ soft-combining. Returns complete decoded frame to caller.

Algorithm:

1. Validate parameters: Check handle, frame pointer (NASA Rule 5)
2. Poll SPI hardware: Read available bytes (non-blocking if timeout=0)
3. Optional FEC decode: If enabled, apply FEC decoding with HARQ
4. Decode frame: Parse header, extract payload
5. Validate CRC: Compute CRC-32 and compare with frame CRC
6. Validate sequence: Check if sequence matches expected rx_sequence
7. Update state: Increment rx_sequence on success
8. Return frame: Copy decoded frame to output parameter

Blocking Behavior:

- timeout_ms = 0: Non-blocking poll (returns immediately)
- timeout_ms > 0: Blocks up to timeout_ms waiting for data
- timeout_ms = k_spi_comm_default_timeout: Use default (1 second)

Performance:

- Small frame (16 bytes): ~100 us
- Medium frame (128 bytes): ~250 us
- Large frame (255 bytes): ~400 us
- FEC overhead: +100 us

Parameters

in,out	handle	Initialized SPI communication handle - Valid range: Non-nullptr to initialized handle - Constraints: Must be initialized via <code>rx_spi_comm_init()</code> - Side effects: Updates shared session RX sequence via <code>rx_session_validate_rx()</code>
out	frame	Pointer to frame structure to receive decoded frame - Valid range: Non-nullptr to allocated <code>rx_frame_t</code> - Constraints: Must have sufficient space (263 bytes) - Lifetime: Filled by this function, owned by caller afterward
in	timeout_ms	Timeout in milliseconds - Valid range: [0, 60000] (0 = non-blocking, up to 1 minute) - Units: Milliseconds - Common values: 0 (poll), <code>k_spi_comm_default_timeout</code> (1 second)

Returns

`rx_err_t` Error code indicating receive result

Return values

<code>k_rx_ok</code>	Frame received and decoded successfully, <code>rx_sequence</code> incremented
<code>k_rx_err_invalid_arg</code>	handle or frame is nullptr
<code>k_rx_err_invalid_state</code>	handle not initialized
<code>k_rx_err_timeout</code>	No data received within timeout (not an error if timeout=0)
<code>k_rx_err_crc_mismatch</code>	CRC validation failed (frame corrupted)
<code>k_rx_err_protocol_error</code>	Frame format invalid (bad sync, length, etc.)
<code>k_rx_err_sequence_error</code>	Unexpected sequence number (out-of-order)
<code>k_rx_err_hw</code>	SPI hardware error

Precondition

handle must be non-NULL and initialized
 frame must be non-NULL and allocated
 RSPI hardware must be initialized and ready

Postcondition

On `k_rx_ok`: frame filled with decoded data, `rx_sequence` incremented
 On timeout: frame unchanged, `rx_sequence` unchanged
 On CRC error: frame may be partially filled, `rx_sequence` unchanged

Note

timeout=0 is recommended for polling loops (non-blocking)
 CRC errors should trigger NACK and retransmission request
 Sequence errors indicate lost frames or out-of-order delivery

Warning

Must call [rx_spi_comm_send_ack\(\)](#) or [rx_spi_comm_send_nack\(\)](#) after receive

Example - Non-Blocking Poll:

```
rx_frame_t frame;

// Poll with 0ms timeout (non-blocking)
rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);

if (err == k_rx_ok) {
    // Frame received successfully
    handle_frame(&frame);
    rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
} else if (err == k_rx_err_timeout) {
    // No data available - this is normal
} else if (err == k_rx_err_crc_mismatch) {
    // CRC error - request retransmission
    rx_spi_comm_send_nack(&g_spi_handle, frame.header.sequence, k_frame_flag_none);
} else {
    // Other error - log and continue
    log_error("Receive error: %d", err);
}
```

Example - Blocking Receive with Timeout:

```
rx_frame_t frame;

// Wait up to 1 second for frame
rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 1000);

if (err == k_rx_ok) {
    // Frame received within timeout
    process_frame(&frame);
    rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
} else if (err == k_rx_err_timeout) {
    // No frame received within 1 second - RPi5 may be idle
    log_warn("No frame received within timeout");
}
```

See also

[rx_spi_comm_send\(\)](#) Send frames to RPi5
[rx_spi_comm_send_ack\(\)](#) Send ACK for received frame
[rx_spi_comm_send_nack\(\)](#) Send NACK on error
[rx_spi_comm_data_available\(\)](#) Check if data waiting
[rx_frame.h](#) Frame structure definition

Since

Version 1.0.0

Definition at line 1958 of file `rx_spi_comm.c`.

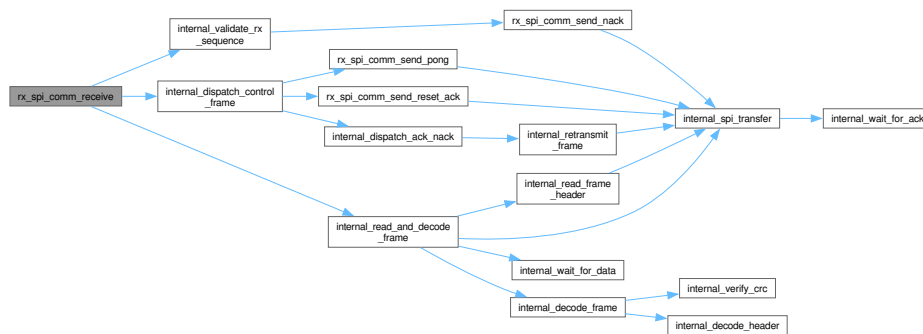
```

01959 {
01960     if (handle == nullptr || frame == nullptr) {
01961         return k_rx_err_invalid_arg;
01962     }
01963
01964     if (!handle->initialized) {
01965         return k_rx_err_invalid_state;
01966     }
01967
01968     /* Loop to handle control frames internally before returning data frames.
01969      * Bounded to k_max_control_frames_per_receive iterations (NASA Rule 2). */
01970     for (uint8_t ctrl_count = 0; ctrl_count < k_max_control_frames_per_receive; ctrl_count++) {
01971         rx_err_t err = internal_read_and_decode_frame(handle, frame, timeout_ms);
01972         if (err != k_rx_ok) {
01973             return err;
01974         }
01975
01976         /* Dispatch control frames (PING, RESET, ACK/NACK).
01977          * internal_dispatch_control_frame always returns k_rx_ok; errors from
01978          * sub-calls (PONG/RESET/ACK send failures) are logged and swallowed
01979          * internally. This is an invariant. */
01980         ctrl_dispatch_result_t ctrl_result = k_ctrl_not_handled;
01981         (void)internal_dispatch_control_frame(handle, frame, &ctrl_result);
01982         if (ctrl_result == k_ctrl_consumed) {
01983             continue; /* Control frame handled; loop for next frame */
01984         }
01985
01986         /* Data frame: validate sequence (NACK on rejection per Audit F-05) */
01987         return internal_validate_rx_sequence(handle, frame);
01988     }
01989
01990     /* Exceeded max control frames without receiving a data frame */
01991     rx_log_error(s_tag, "Too many control frames, no data frame received");
01992     return k_rx_err_timeout;
01993 }

```

References `rx_spi_comm_handle_t::initialized`, `internal_dispatch_control_frame()`, `internal_read_and_decode_frame`, `internal_validate_rx_sequence()`, `k_ctrl_consumed`, `k_ctrl_not_handled`, `k_max_control_frames_per_receive`, and `s_tag`.

Here is the call graph for this function:



4.1.4.6 rx_spi_comm_send()

```

rx_err_t rx_spi_comm_send (
    rx_spi_comm_handle_t * handle,
    rx_frame_type_t type,
    uint8_t flags,
    const uint8_t * payload,
    uint32_t payload_len) [nodiscard]

```

Send frame on SPI with automatic sequencing and CRC.

Encodes payload into frame format with automatic sequence number assignment, CRC-32 calculation, and optional FEC encoding. Transmits frame to RPi5 via SPI hardware. This is the primary send function for all frame types.

Algorithm:

1. Validate parameters: Check handle, payload, payload_len (NASA Rule 5)
2. Assign sequence: Get next TX sequence from shared rx_session_state_t
3. Build frame: Create frame header (type, flags, length, sequence)
4. Encode frame: Call frame encoder to add CRC-32
5. Optional FEC: If enabled, apply FEC encoding
6. SPI transfer: Transmit encoded frame via RSPI peripheral
7. Update state: TX sequence already incremented by rx_session_next_tx()

Performance:

- Small payload (16 bytes): ~150 us
- Medium payload (128 bytes): ~300 us
- Large payload (255 bytes): ~500 us
- FEC overhead: +50% time

Parameters

in,out	handle	Initialized SPI communication handle • Valid range: Non-nullptr to initialized handle • Constraints: Must be initialized via rx_spi_comm_init() • Side effects: Increments shared session TX sequence via rx_session_next_tx()
in	type	Frame type (command, request, response, etc.) • Valid range: Any rx_frame_type_t value • Common values: k_frame_type_response (most common for RX72N) • Units: Enumeration value
in	flags	Frame flags (ACK, NACK, etc.) • Valid range: Bitmask of rx_frame_flag_t values • Common values: k_frame_flag_none (0x00) • Units: Bitmask

in	payload	Payload data to send - Valid range: Non-NULL if payload_len > 0, may be NULL if payload_len == 0 - Constraints: Must contain valid data for payload_len bytes - Lifetime: Copied immediately, does not need to persist - Max size: k_frame_max_payload (255 bytes)
in	payload_len	Payload length in bytes - Valid range: [0, k_frame_max_payload] (0-255 bytes) - Units: Bytes - Constraints: Must be <= k_frame_max_payload

Returns

rx_err_t Error code indicating send success or failure

Return values

k_rx_ok	Frame sent successfully, tx_sequence incremented
k_rx_err_invalid_arg	handle or payload is nullptr; payload_len > k_frame_max_payload (255 bytes); payload is nullptr but payload_len > 0
k_rx_err_invalid_state	handle not initialized
k_rx_err_hw	SPI hardware error (RSPI transfer failed)
k_rx_err_timeout	SPI transfer timeout

Precondition

handle must be non-NULL and initialized

If payload_len > 0, payload must point to valid data

RSPI hardware must be initialized and ready

Postcondition

Shared session TX sequence incremented (wraps at 65535)

Frame transmitted to RPi5 if successful

Note

Sequence number managed automatically - caller does not specify

CRC-32 calculated automatically by frame encoder

FEC applied automatically if handle->fec_enabled = true

Warning

Ensure RPi5 is ready to receive (chip select asserted)

Example - Send Telemetry Response:

```
uint8_t telemetry[16];
// ... fill telemetry data ...

rx_err_t err = rx_spi_comm_send(&g_spi_handle,
                                k_frame_type_response,
                                k_frame_flag_none,
                                telemetry,
                                sizeof(telemetry));

if (err == k_rx_ok) {
    // Telemetry sent successfully
} else {
    log_error("Send failed: %d", err);
}
```

Example - Send Empty ACK (No Payload):

```
// Send empty frame with ACK flag
rx_err_t err = rx_spi_comm_send(&g_spi_handle,
                                k_frame_type_response,
                                k_frame_flag_ack,
                                NULL, // No payload
                                0);   // Zero length
```

See also

[rx_spi_comm_send_ack\(\)](#) Convenience wrapper for ACK frames
[rx_spi_comm_send_nack\(\)](#) Convenience wrapper for NACK frames
[rx_spi_comm_receive\(\)](#) Receive frames from RPi5
[rx_frame.h](#) Frame protocol definitions

Since

Version 1.0.0

Definition at line 1370 of file [rx_spi_comm.c](#).

```
01375 {
01376     if (handle == nullptr) {
01377         return k_rx_err_invalid_arg;
01378     }
01379
01380     if (!handle->initialized) {
01381         rx_log_error(s_tag, "Handle not initialized");
01382         return k_rx_err_invalid_state;
01383     }
01384
01385     /* Validate payload before consuming a sequence number (Issue #9). */
01386     if (payload == nullptr && payload_len > 0) {
01387         return k_rx_err_invalid_arg;
01388     }
01389     if (payload_len > k_frame_max_payload) {
01390         rx_log_error(s_tag, "Payload too large");
01391         return k_rx_err_invalid_size;
01392     }
01393
01394     /* Get next TX sequence from shared session. Invariant: cannot fail. */
01395     uint16_t sequence = 0;
01396     (void)(rx_session_next_tx(handle->session, &sequence));
01397
01398     /* Build frame. Cannot fail: handle and &frame are non-null, payload validated. */
01399     rx_frame_t frame;
01400     (void)(internal_build_frame(handle, sequence, type, flags, payload, payload_len, &frame));
01401 }
```

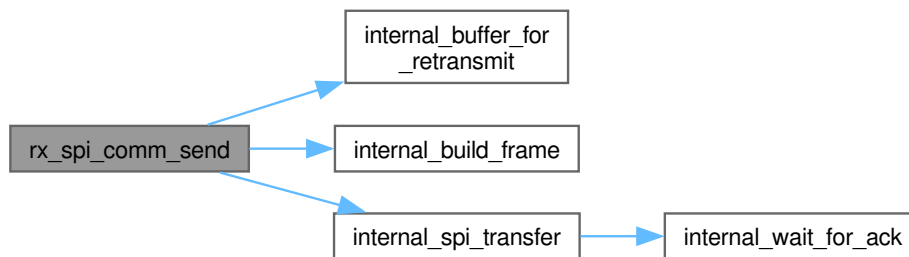
```

01402 /* Encode frame into handle-owned scratch buffer (not stack -- see comment
01403 * on tx_encode_buffer in rx_spi_comm.h). Cannot fail: encoder initialized,
01404 * pointers valid. */
01405 uint32_t wire_len = 0;
01406 (void)(rx_frame_encode(&handle->encoder, &frame, handle->tx_encode_buffer, &wire_len));
01407
01408 /* Transfer via SPI (waits for host ACK internally) */
01409 rx_err_t xfer_err = internal_spi_transfer(handle, handle->tx_encode_buffer, wire_len, nullptr, 0);
01410 if (xfer_err != k_rx_ok) {
01411     rx_log_error(s_tag, "SPI transfer failed");
01412     return xfer_err;
01413 }
01414
01415 /* Buffer frame for retransmission if enabled and requires-ACK flag set */
01416 if ((int)handle->auto_retransmit && ((flags & k_frame_flag_requires_ack) != 0)) {
01417     internal_buffer_for_retransmit(handle, handle->tx_encode_buffer, wire_len, sequence);
01418 }
01419
01420 return k_rx_ok;
01421 }

```

References [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::init](#), [internal_buffer_for_retransmit\(\)](#), [internal_build_frame\(\)](#), [internal_spi_transfer\(\)](#), [s_tag](#), [rx_spi_comm_handle_t::sequence](#) and [rx_spi_comm_handle_t::tx_encode_buffer](#).

Here is the call graph for this function:



4.1.4.7 rx_spi_comm_send_ack()

```

rx_err_t rx_spi_comm_send_ack (
    rx_spi_comm_handle_t * handle,
    const uint16_t sequence) [nodiscard]

```

Send ACK frame for received frame (convenience wrapper).

Convenience function for sending acknowledgment frames. Automatically sets frame type to `k_frame_type_response` and flags to `k_frame_flag_ack`. Includes the sequence number being acknowledged in the frame.

Use Case: Acknowledge successful receipt of a frame from RPi5.

Algorithm:

1. Build ACK frame with type=response, flags=ack
2. Include sequence number in payload (2 bytes, little-endian)
3. Call [rx_spi_comm_send\(\)](#) with constructed frame

Performance: ~50 us (minimal frame with 2-byte payload)

Parameters

in,out	handle	Initialized SPI communication handle - Valid range: Non-nullptr to initialized handle - Constraints: Must be initialized via rx_spi_comm_init()
in	sequence	Sequence number to acknowledge - Valid range: [0, 65535] - Units: Sequence number (uint16_t) - Purpose: Tells RPi5 which frame is being acknowledged

Returns

rx_err_t Error code (same as rx_spi_comm_send)

Return values

k_rx_ok	ACK sent successfully
k_rx_err_invalid_arg	handle is nullptr
k_rx_err_invalid_state	handle not initialized
(other)	Errors propagated from rx_spi_comm_send()

Precondition

Same preconditions as [rx_spi_comm_send\(\)](#)

Postcondition

ACK frame transmitted with specified sequence number

Note

Equivalent to calling [rx_spi_comm_send\(\)](#) with type=response, flags=ack

This is a thin wrapper - no additional validation performed

Example:

```
// Receive frame and send ACK
rx_frame_t frame;
rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);
if (err == k_rx_ok) {
    // Process frame
    handle_frame(&frame);

    // Send ACK for this frame
    err = rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
}
```

See also

[rx_spi_comm_send\(\)](#) Full send function
[rx_spi_comm_send_nack\(\)](#) Send NACK for errors

Since

Version 1.0.0

Send ACK frame for received frame (convenience wrapper).

Definition at line 1426 of file [rx_spi_comm.c](#).

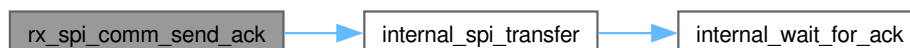
```

01427 {
01428     if (handle == nullptr) {
01429         return k_rx_err_invalid_arg;
01430     }
01431
01432     if (!handle->initialized) {
01433         return k_rx_err_invalid_state;
01434     }
01435
01436     /* Create ACK frame: cannot fail since &ack_frame is non-null. */
01437     rx_frame_t ack_frame;
01438     (void)(rx_frame_create_ack(&ack_frame, sequence));
01439
01440     /* Encode and send: encoder initialized at rx_spi_comm_init; all pointers valid. */
01441     uint8_t wire_buffer[k_frame_min_size];
01442     uint32_t wire_len = 0;
01443
01444     (void)(rx_frame_encode(&handle->encoder, &ack_frame, wire_buffer, &wire_len));
01445
01446     /* Post-condition: ACK frame encoded to minimum size */
01447
01448     /* Transfer via SPI (waits for host ACK internally) */
01449     return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
01450 }

```

References [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), and [internal_spi_transfer\(\)](#).

Here is the call graph for this function:



4.1.4.8 rx_spi_comm_send_nack()

```

rx_err_t rx_spi_comm_send_nack (
    rx_spi_comm_handle_t * handle,
    const uint16_t sequence,
    uint8_t flags) [nodiscard]

```

Send NACK frame for received frame with error indication.

Convenience function for sending negative acknowledgment frames. Used to request retransmission when frame reception fails (CRC error, corruption, etc.).

Use Cases:

- CRC validation failure (hard NACK)
- FEC decoding failure (soft NACK if HARQ enabled)

- Out-of-order sequence detection
- Protocol violation

Algorithm:

1. Build NACK frame with type=response, flags=nack | additional_flags
2. Include sequence number in payload (2 bytes, little-endian)
3. Call `rx_spi_comm_send()` with constructed frame

Performance: ~50 us (minimal frame with 2-byte payload)

Parameters

in,out	handle	Initialized SPI communication handle • Valid range: Non-nullptr to initialized handle • Constraints: Must be initialized via <code>rx_spi_comm_init()</code>
in	sequence	Sequence number being NACKed • Valid range: [0, 65535] • Units: Sequence number (uint16_t) • Purpose: Tells RPi5 which frame failed and needs retransmission
in	flags	Additional flags (e.g., k_frame_flag_soft_nack) • Valid range: Bitmask of rx_frame_flag_t values • Common values: k_frame_flag_none (hard NACK), k_frame_flag_soft↵_nack (HARQ) • Units: Bitmask

Returns

rx_err_t Error code (same as rx_spi_comm_send)

Return values

k_rx_ok	NACK sent successfully
k_rx_err_invalid_arg	handle is nullptr
k_rx_err_invalid_state	handle not initialized
(other)	Errors propagated from <code>rx_spi_comm_send()</code>

Precondition

Same preconditions as `rx_spi_comm_send()`

Postcondition

NACK frame transmitted with specified sequence number and flags

Note

RPi5 should retransmit frame with specified sequence number

Use `k_frame_flag_soft_nack` for FEC soft-combining (HARQ)

Example - Hard NACK on CRC Error:

```
rx_frame_t frame;
rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);
if (err == k_rx_err_crc_mismatch) {
    // CRC validation failed - send hard NACK
    rx_spi_comm_send_nack(&g_spi_handle,
        frame.header.sequence,
        k_frame_flag_none); // Hard NACK
}
```

Example - Soft NACK for HARQ:

```
// FEC decoding failed but HARQ soft-combining available
if (fec_decode_failed && harq_enabled) {
    rx_spi_comm_send_nack(&g_spi_handle,
        frame.header.sequence,
        k_frame_flag_soft_nack); // Soft NACK for HARQ
}
```

See also

[rx_spi_comm_send\(\)](#) Full send function

[rx_spi_comm_send_ack\(\)](#) Send ACK for success

[rx_frame.h](#) Frame flag definitions

Since

Version 1.0.0

Send NACK frame for received frame with error indication.

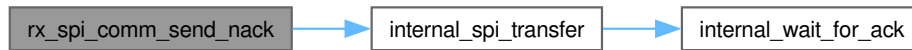
Definition at line 1455 of file [rx_spi_comm.c](#).

```
01456 {
01457     if (handle == nullptr) {
01458         return k_rx_err_invalid_arg;
01459     }
01460
01461     if (!handle->initialized) {
01462         return k_rx_err_invalid_state;
01463     }
01464
01465     /* Create NACK frame: cannot fail since &nack_frame is non-null. */
01466     rx_frame_t nack_frame;
01467     (void)(rx_frame_create_nack(&nack_frame, sequence, flags));
01468
01469     /* Encode and send: encoder initialized at rx_spi_comm_init; all pointers valid. */
01470     uint8_t wire_buffer[k_frame_min_size];
01471     uint32_t wire_len = 0;
01472
01473     (void)(rx_frame_encode(&handle->encoder, &nack_frame, wire_buffer, &wire_len));
01474
01475     /* Post-condition: NACK frame encoded to minimum size */
01476
01477     /* Transfer via SPI (waits for host ACK internally) */
01478     return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
01479 }
```

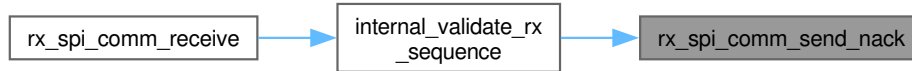
References [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), and [internal_spi_transfer\(\)](#).

Referenced by [internal_validate_rx_sequence\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.1.4.9 rx_spi_comm_send_pong()

```

rx_err_t rx_spi_comm_send_pong (
    rx_spi_comm_handle_t * handle,
    const uint8_t * payload,
    uint32_t payload_len) [nodiscard]
  
```

Send PONG frame with payload (response to PING).

Constructs and transmits a PONG frame echoing the provided payload. Used internally by [rx_spi_comm_receive\(\)](#) for auto-PONG, but also available for manual PONG sending.

Parameters

in,out	handle	Initialized SPI communication handle
in	payload	Payload data to echo (may be NULL if payload_len is 0)
in	payload_len	Length of payload in bytes

Returns

rx_err_t Error code

Return values

k_rx_ok	PONG frame sent successfully
k_rx_err_invalid_arg	handle is nullptr
k_rx_err_invalid_state	handle not initialized
(other)	Errors propagated from frame creation or SPI transfer

Precondition

handle must be non-NULL and initialized

Postcondition

PONG frame transmitted with echoed payload

Note

Gets next TX sequence from shared session state for the PONG frame

See also

[rx_frame_create_pong\(\)](#) Frame creation for PONG

[rx_spi_comm_receive\(\)](#) Auto-sends PONG on PING reception

Since

Version 1.0.0

Send PONG frame with payload (response to PING).

Constructs a PONG frame echoing the given payload, acquires the next TX sequence number from the shared session, encodes the frame, and transfers it over SPI. Typically called automatically by the receive path in response to a PING.

Precondition

handle != nullptr && handle->initialized

payload != NULL when payload_len > 0

Postcondition

TX sequence incremented on success

PONG frame sent over SPI

Note

Not thread-safe; called from the SPI receive context.

See also

[rx_frame_create_pong\(\)](#) Frame construction helper

Since

Version 1.0.0

Definition at line 2089 of file [rx_spi_comm.c](#).

```

02090 {
02091     if (handle == nullptr) {
02092         return k_rx_err_invalid_arg;
02093     }
02094     if (!handle->initialized) {
02095         return k_rx_err_invalid_state;
02096     }
02097 }
02098
02099 /* Get next TX sequence from shared session.
02100  * rx_session_next_tx only fails for null/uninitialized session; handle->session
02101  * is valid (set at init time) and the handle is initialized above. Invariant. */
02102 uint16_t sequence = 0;
02103 (void)(rx_session_next_tx(handle->session, &sequence));

```

```

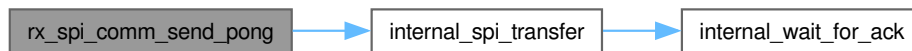
02104
02105 /* Create PONG frame: cannot fail since &pong_frame is non-null and payload
02106  * is validated above (nullptr allowed when payload_len == 0). */
02107 rx_frame_t pong_frame;
02108 (void)(rx_frame_create_pong(&pong_frame, sequence, payload, payload_len));
02109
02110 /* Encode into handle-owned scratch buffer (not stack -- see comment on
02111  * tx_encode_buffer in rx_spi_comm.h). encoder initialized at
02112  * rx_spi_comm_init; all pointers valid. */
02113 uint32_t wire_len = 0;
02114 (void)(rx_frame_encode(&handle->encoder, &pong_frame, handle->tx_encode_buffer, &wire_len));
02115
02116 /* Transfer via SPI */
02117 return internal_spi_transfer(handle, handle->tx_encode_buffer, wire_len, nullptr, 0);
02118 }

```

References [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), [internal_spi_transfer\(\)](#), [rx_spi_comm_handle_t::session](#), and [rx_spi_comm_handle_t::tx_encode_buffer](#).

Referenced by [internal_dispatch_control_frame\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.1.4.10 rx_spi_comm_send_reset_ack()

```

rx_err_t rx_spi_comm_send_reset_ack (
    rx_spi_comm_handle_t * handle) [nodiscard]

```

Send RESET_ACK frame (response to RESET).

Constructs and transmits a RESET_ACK frame. Used internally by [rx_spi_comm_receive\(\)](#) for auto-RESET_ACK, but also available for manual sending.

Parameters

in,out	handle	Initialized SPI communication handle
--------	--------	--------------------------------------

Returns

rx_err_t Error code

Return values

k_rx_ok	RESET_ACK frame sent successfully
k_rx_err_invalid_arg	handle is nullptr

<code>k_rx_err_invalid_state</code>	handle not initialized
(other)	Errors propagated from frame creation or SPI transfer

Precondition

handle must be non-NULL and initialized

Postcondition

RESET_ACK frame transmitted

Note

Does NOT reset sequence numbers (caller must do that if desired)

See also

`rx_frame_create_reset_ack()` Frame creation for RESET_ACK

[rx_spi_comm_receive\(\)](#) Auto-sends RESET_ACK on RESET reception

Since

Version 1.0.0

Send RESET_ACK frame (response to RESET).

Constructs a RESET_ACK frame (empty payload), acquires the next TX sequence number, encodes, and transfers over SPI. Called automatically by the receive path after processing a RESET frame. The RESET_ACK itself does NOT reset sequence numbers; that is handled by the session layer in response to the RESET.

Precondition

handle != nullptr && handle->initialized

A RESET frame was received and session state cleared

Postcondition

TX sequence incremented on success

RESET_ACK frame sent over SPI

Note

Not thread-safe; called from the SPI receive context.

See also

`rx_frame_create_reset_ack()` Frame construction helper

[rx_spi_comm_set_control_callbacks\(\)](#) Register RESET notification

Since

Version 1.0.0

Definition at line 2142 of file [rx_spi_comm.c](#).

```

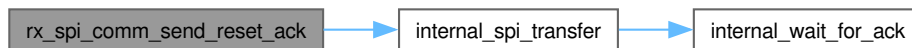
02143 {
02144     if (handle == nullptr) {
02145         return k_rx_err_invalid_arg;
02146     }
02147     if (!handle->initialized) {
02148         return k_rx_err_invalid_state;
02149     }
02150 }
02151 /* Get next TX sequence from shared session.
02152  * rx_session_next_tx only fails for null/uninitialized session; handle->session
02153  * is valid and the handle is initialized above. Invariant. */
02154 uint16_t sequence = 0;
02155 (void)rx_session_next_tx(handle->session, &sequence);
02156 /* Create RESET_ACK frame: cannot fail since &reset_ack_frame is non-null. */
02157 rx_frame_t reset_ack_frame;
02158 (void)rx_frame_create_reset_ack(&reset_ack_frame, sequence);
02159 /* Encode: encoder initialized at rx_spi_comm_init; all pointers valid. */
02160 uint8_t wire_buffer[k_frame_min_size];
02161 uint32_t wire_len = 0;
02162 (void)rx_frame_encode(&handle->encoder, &reset_ack_frame, wire_buffer, &wire_len);
02163 /* Post-condition: RESET_ACK frame encoded to minimum size */
02164 /* Transfer via SPI */
02165 return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
02166 }

```

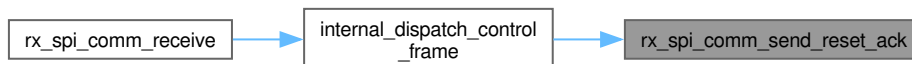
References [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), [internal_spi_transfer\(\)](#), and [rx_spi_comm_handle_t::session](#).

Referenced by [internal_dispatch_control_frame\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.1.4.11 rx_spi_comm_set_auto_retransmit()

```

rx_err_t rx_spi_comm_set_auto_retransmit (
    rx_spi_comm_handle_t * handle,
    const bool enabled,
    const rx_spi_comm_retransmit_config_t * config) [nodiscard]

```

Enable or disable automatic retransmission at runtime.

Allows the RPi5 to configure retransmission behavior via protobuf command. When disabling, any pending retry state is cleared.

Parameters

in,out	handle	Initialized SPI communication handle
in	enabled	true to enable, false to disable
in	config	Retransmit configuration (may be NULL to keep current or use defaults)

Returns

rx_err_t Error code

Return values

k_rx_ok	Configuration applied successfully
k_rx_err_invalid_arg	handle is nullptr
k_rx_err_invalid_state	handle not initialized

Precondition

handle must be non-NULL and initialized

Postcondition

auto_retransmit flag and config updated
If disabling, pending retry state cleared

Since

Version 1.0.0

Precondition

handle must be non-NULL and initialized

Postcondition

auto_retransmit flag and config updated

Since

Version 1.0.0

Definition at line 2291 of file [rx_spi_comm.c](#).

```
02294 {  
02295     if (handle == nullptr) {  
02296         return k_rx_err_invalid_arg;  
02297     }  
02298  
02299     if (!handle->initialized) {  
02300         return k_rx_err_invalid_state;  
02301     }  
02302 }
```

```

02303 handle->auto_retransmit = enabled;
02304
02305 /* Apply new config if provided */
02306 if (config != nullptr) {
02307     handle->retransmit_cfg = *config;
02308 }
02309
02310 /* Apply defaults for zero fields */
02311 if (handle->retransmit_cfg.max_retries == 0) {
02312     handle->retransmit_cfg.max_retries = k_retransmit_default_max_retries;
02313 }
02314 if (handle->retransmit_cfg.ack_timeout_ms == 0) {
02315     handle->retransmit_cfg.ack_timeout_ms = k_retransmit_default_ack_timeout_ms;
02316 }
02317 if (handle->retransmit_cfg.max_backoff_ms == 0) {
02318     handle->retransmit_cfg.max_backoff_ms = k_retransmit_default_max_backoff_ms;
02319 }
02320
02321 /* Clear pending retry state when disabling */
02322 if (!enabled) {
02323     handle->retry_pending = false;
02324     handle->retry_count = 0;
02325 }
02326
02327 return k_rx_ok;
02328 }

```

References [rx_spi_comm_retransmit_config_t::ack_timeout_ms](#), [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_handle_t::initialized](#), [k_retransmit_default_ack_timeout_ms](#), [k_retransmit_default_max_backoff_ms](#), [k_retransmit_default_max_retries](#), [rx_spi_comm_retransmit_config_t::max_backoff_ms](#), [rx_spi_comm_retransmit_config_t::max_retries](#), [rx_spi_comm_retransmit_config_t::ack_timeout_ms](#), [rx_spi_comm_handle_t::retransmit_cfg](#), [rx_spi_comm_handle_t::retry_count](#), and [rx_spi_comm_handle_t::retry_pending](#).

4.1.4.12 rx_spi_comm_set_control_callbacks()

```

rx_err_t rx_spi_comm_set_control_callbacks (
    rx_spi_comm_handle_t * handle,
    void(* on_ping_cb)(const rx_frame_t *frame, void *ctx),
    void(* on_reset_cb)(const rx_frame_t *frame, void *ctx),
    void * cb_ctx) [nodiscard]

```

Register callbacks for PING and RESET control frames.

Sets callback functions that are invoked when [rx_spi_comm_receive\(\)](#) encounters PING or RESET control frames. Control frames are handled internally (auto PONG / auto RESET_ACK) and callbacks provide notification to the application layer.

Parameters

in,out	handle	Initialized SPI communication handle
in	on_ping_cb	Callback for PING frames (may be NULL to disable)
in	on_reset_cb	Callback for RESET frames (may be NULL to disable)
in	cb_ctx	User context pointer passed to callbacks

Returns

rx_err_t Error code

Return values

k_rx_ok	Callbacks registered successfully
---------	-----------------------------------

k_rx_err_invalid_arg	handle is nullptr
----------------------	-------------------

Precondition

handle must be non-NULL

Postcondition

on_ping_cb, on_reset_cb, control_cb_ctx stored in handle

Note

Callbacks are invoked from within [rx_spi_comm_receive\(\)](#) context

Not thread-safe, set callbacks before starting receive loop

See also

[rx_spi_comm_receive\(\)](#) Invokes callbacks on control frames

Since

Version 1.0.0

Stores function pointers that the receive path invokes when it decodes a PING or RESET frame. Passing NULL for a callback disables notification for that frame type while still performing the automatic PONG / RESET_ACK response.

Precondition

handle != nullptr

handle->initialized == true

Postcondition

Callbacks stored; invoked on next matching control frame

Previous callbacks overwritten

Note

Not thread-safe; register before starting the receive loop.

See also

[rx_spi_comm_send_pong\(\)](#) Auto-response sent on PING

[rx_spi_comm_send_reset_ack\(\)](#) Auto-response sent on RESET

Since

Version 1.0.0

Definition at line 2048 of file [rx_spi_comm.c](#).

```

02052 {
02053     if (handle == nullptr) {
02054         return k_rx_err_invalid_arg;
02055     }
02056     if (!handle->initialized) {
02057         return k_rx_err_invalid_state;
02058     }
02059 }
02060
02061 handle->on_ping_cb    = on_ping_cb;
02062 handle->on_reset_cb   = on_reset_cb;
02063 handle->control_cb_ctx = cb_ctx;
02064
02065 return k_rx_ok;
02066 }
```

References [rx_spi_comm_handle_t::control_cb_ctx](#), [rx_spi_comm_handle_t::initialized](#), [rx_spi_comm_handle_t::on_ping_cb](#) and [rx_spi_comm_handle_t::on_reset_cb](#).

4.1.4.13 rx_spi_comm_set_retransmit_callbacks()

```

rx_err_t rx_spi_comm_set_retransmit_callbacks (
    rx_spi_comm_handle_t * handle,
    void(* on_ack_cb )(uint16_t sequence, void *ctx),
    void(* on_nack_cb )(uint16_t sequence, void *ctx),
    void * ctx) [nodiscard]
```

Register ACK/NACK notification callbacks for retransmission.

Sets optional callbacks invoked when ACK or NACK control frames are consumed during [rx_spi_comm_receive\(\)](#). Useful for telemetry and diagnostics.

Parameters

in,out	handle	Initialized SPI communication handle
in	on_ack_cb	Callback for ACK frames (may be NULL to disable)
in	on_nack_cb	Callback for NACK frames (may be NULL to disable)
in	ctx	User context pointer passed to callbacks

Returns

rx_err_t Error code

Return values

k_rx_ok	Callbacks registered
k_rx_err_invalid_arg	handle is nullptr

Precondition

handle must be non-NULL

Postcondition

Callback pointers stored in handle

Note

Callbacks invoked from within `rx_spi_comm_receive()` context

Since

Version 1.0.0

Register ACK/NACK notification callbacks for retransmission.

Since

Version 1.0.0

Definition at line 2337 of file `rx_spi_comm.c`.

```

02341 {
02342     if (handle == nullptr) {
02343         return k_rx_err_invalid_arg;
02344     }
02345
02346     if (!handle->initialized) {
02347         return k_rx_err_invalid_state;
02348     }
02349
02350     handle->on_ack_cb      = on_ack_cb;
02351     handle->on_nack_cb     = on_nack_cb;
02352     handle->retransmit_cb_ctx = ctx;
02353
02354     return k_rx_ok;
02355 }
```

References `rx_spi_comm_handle_t::initialized`, `rx_spi_comm_handle_t::on_ack_cb`, `rx_spi_comm_handle_t::on_nack_cb`, and `rx_spi_comm_handle_t::retransmit_cb_ctx`.

4.2 rx_spi_comm.h

[Go to the documentation of this file.](#)

```

00001 /**
00002  * @file rx_spi_comm.h
00003  * @brief High-Level SPI Frame Protocol Communication Layer for RX72N Peripheral Mode
00004  *
00005  * @details
00006  * ## Overview
00007  * Production-quality SPI communication layer providing reliable frame-based protocol
00008  * between RX72N MCU (SPI peripheral) and Raspberry Pi 5 (SPI controller). Integrates
00009  * frame encoding/decoding with CRC-32 validation, automatic sequence number management,
00010  * optional Forward Error Correction (FEC), and HARQ (Hybrid Automatic Repeat reQuest)
00011  * retransmission support.
00012  *
00013  * **Purpose**: Enable reliable, high-throughput bidirectional communication over SPI
00014  * hardware interface with automatic error detection, correction, and recovery mechanisms.
00015  *
00016  * ## System Architecture
00017  *
00018  * @dot
00019  * digraph spi_comm_architecture {
00020  *     rankdir=TB;
00021  *     node [shape=box, style=rounded];
00022  *
00023  *     subgraph cluster_app {
00024  *         label="Application Layer (RX72N)";
00025  *         style=filled;
```

```

00026 *    fillcolor=lightblue;
00027 *    app [label="Application Code\n(Command Handlers)"];
00028 * }
00029 *
00030 * subgraph cluster_comm {
00031 *     label="Communication Manager";
00032 *     style=filled;
00033 *     fillcolor=lightgreen;
00034 *     comm_mgr [label="rx_comm_manager\n(Channel Coordinator)"];
00035 * }
00036 *
00037 * subgraph cluster_spi {
00038 *     label="SPI Communication Layer (This Module)";
00039 *     style=filled;
00040 *     fillcolor=lightyellow;
00041 *     spi_send [label="rx_spi_comm_send()"];
00042 *     spi_rcv [label="rx_spi_comm_receive()"];
00043 *     seq_mgmt [label="Sequence Management"];
00044 * }
00045 *
00046 * subgraph cluster_frame {
00047 *     label="Frame Protocol Layer";
00048 *     style=filled;
00049 *     fillcolor=lightcyan;
00050 *     encoder [label="rx_frame_encoder\n(CRC-32, Header)"];
00051 *     decoder [label="rx_frame_decoder\n(CRC Verify, Parse)"];
00052 * }
00053 *
00054 * subgraph cluster_harq {
00055 *     label="Optional FEC/HARQ Layer";
00056 *     style=filled;
00057 *     fillcolor=lavender;
00058 *     fec [label="rx_harq\n(FEC Encode/Decode)"];
00059 * }
00060 *
00061 * subgraph cluster_hal {
00062 *     label="SPI Hardware Abstraction";
00063 *     style=filled;
00064 *     fillcolor=lightgray;
00065 *     rspi [label="RSPI HAL\n(RX72N Peripheral Mode)"];
00066 * }
00067 *
00068 * subgraph cluster_rpi5 {
00069 *     label="Raspberry Pi 5 (SPI Controller)";
00070 *     style=filled;
00071 *     fillcolor=lightcoral;
00072 *     rpi_spi [label="Linux spidev\n(/dev/spidev0.0)"];
00073 * }
00074 *
00075 * // Data flow: Send path
00076 * app -> comm_mgr [label="send command"];
00077 * comm_mgr -> spi_send [label="route to SPI"];
00078 * spi_send -> seq_mgmt [label="assign sequence"];
00079 * spi_send -> encoder [label="encode frame"];
00080 * encoder -> fec [label="optional FEC", style=dashed];
00081 * fec -> rspi [label="SPI transfer"];
00082 * encoder -> rspi [label="no FEC"];
00083 * rspi -> rpi_spi [label="COPI/CIPO"];
00084 *
00085 * // Data flow: Receive path
00086 * rpi_spi -> rspi [label="COPI/CIPO"];
00087 * rspi -> decoder [label="raw bytes"];
00088 * decoder -> fec [label="optional FEC", style=dashed];
00089 * fec -> spi_rcv [label="decoded frame"];
00090 * decoder -> spi_rcv [label="no FEC"];
00091 * spi_rcv -> seq_mgmt [label="validate sequence"];
00092 * spi_rcv -> comm_mgr [label="dispatch frame"];
00093 * comm_mgr -> app [label="invoke callback"];
00094 * }
00095 * @enddot
00096 *
00097 * ## SPI Communication Protocol
00098 *
00099 * **Hardware Configuration:**
00100 * - **RX72N Role**: SPI Peripheral
00101 * - **RPi5 Role**: SPI Controller
00102 * - **RSPI Channel**: RSPI0 (configurable)
00103 * - **SPI Mode**: Mode 0 (CPOL=0, CPHA=0) default
00104 * - **Clock Speed**: Up to 10 MHz (limited by RPi5 spidev)
00105 * - **Data Width**: 8-bit transfers
00106 * - **Endianness**: Little-endian (all multi-byte fields)
00107 *
00108 * **Frame Format** (rx_frame protocol):
00109 * ```
00110 * +-----+-----+-----+-----+-----+
00111 * | Sync  | Sequence | Length | Type  | Flags | Payload (0-255) | CRC-32 |
00112 * | 2B    | 2B      | 2B    | 1B   | 1B   | variable        | 4B    |

```

```

00113 * +-----+-----+-----+-----+-----+-----+
00114 * | 0xAA55 | uint16 | uint16 | uint8 | uint8 | uint8[] | uint32 |
00115 * +-----+-----+-----+-----+-----+-----+
00116 * ```
00117 *
00118 * **Sequence Number Management (shared session):**
00119 * - TX/RX sequences managed by shared rx_session_state_t
00120 * - Cross-transport continuity: USB and SPI share the same sequence counters
00121 * - Gap tolerance: Accepts gaps up to 10 frames (matching Go gateway)
00122 * - Sequence reset: Via rx_session_reset() on the shared session
00123 *
00124 * **Error Detection and Correction:**
00125 * - **CRC-32**: Mandatory for all frames (polynomial 0x04C11DB7)
00126 * - **FEC (optional)**: Reed-Solomon or convolutional coding
00127 * - **HARQ (optional)**: Soft-combining retransmissions
00128 *
00129 * ## Performance Characteristics
00130 *
00131 * | Operation | Avg Time | Worst Case | Notes |
00132 * |-----|-----|-----|-----|
00133 * | **rx_spi_comm_init()** | ~10 us | ~20 us | Frame encoder/decoder setup |
00134 * | **rx_spi_comm_send()** | ~150-500 us | ~1 ms | Depends on payload size, FEC |
00135 * | **rx_spi_comm_receive()** | ~100-400 us | ~800 us | Depends on payload size, FEC |
00136 * | **rx_spi_comm_send_ack()** | ~50 us | ~100 us | Minimal frame (no payload) |
00137 * | **SPI transfer (HW)** | payload_len x 0.8 us | - | @ 10 MHz SPI clock |
00138 * | **CRC-32 calculation** | payload_len x 0.05 us | - | Software CRC (hardware CRC faster) |
00139 * | **FEC encoding** | payload_len x 2 us | - | Optional, adds ~50% overhead |
00140 * | **FEC decoding** | payload_len x 3 us | - | Optional, with error correction |
00141 *
00142 * **Throughput Analysis**:
00143 * - **Theoretical max**: 10 Mbps (SPI clock) x 8 bits = 80 Mbps
00144 * - **Practical (no FEC)**: ~6 Mbps (frame overhead + processing)
00145 * - **With FEC**: ~4 Mbps (encoding/decoding overhead)
00146 * - **Latency (256B frame)**: ~300 us (send) + ~250 us (receive) = ~550 us round-trip
00147 *
00148 * ## Memory Usage
00149 *
00150 * | Memory Type | Size | Usage |
00151 * |-----|-----|-----|
00152 * | **rx_spi_comm_handle_t** | ~4116 bytes | Main state structure |
00153 * | **RX buffer** | 2048 bytes | Receive staging buffer |
00154 * | **TX buffer** | 2048 bytes | Transmit staging buffer |
00155 * | **Frame encoder** | ~8 bytes | Frame encoding state |
00156 * | **Frame decoder** | ~8 bytes | Frame decoding state |
00157 * | **Stack (send)** | ~64 bytes | Function locals + encoder |
00158 * | **Stack (receive)** | ~80 bytes | Function locals + decoder + rx_frame_t |
00159 *
00160 * **Memory Layout** (rx_spi_comm_handle_t):
00161 * ```
00162 * | Offset | Size | Field | Purpose |
00163 * |-----|-----|-----|-----|
00164 * | 0x00 | 8 | encoder | Frame encoder state |
00165 * | 0x08 | 8 | decoder | Frame decoder state |
00166 * | 0x10 | 2048 | rx_buffer | RX staging buffer |
00167 * | 0x810 | 2048 | tx_buffer | TX staging buffer |
00168 * | 0x1010 | 4/8 | session | Pointer to shared session state |
00169 * | 0x1014 | 1 | channel | RSPI channel (0-2) |
00170 * | 0x1015 | 1 | fec_enabled | FEC enable flag |
00171 * | 0x1016 | 1 | initialized | Initialization flag |
00172 * | 0x1017 | 1 | (padding) | Alignment |
00173 * |-----|-----|-----|-----|
00174 * | Total: 4120 bytes (0x1018) |
00175 * ```
00176 *
00177 * ## Thread Safety Analysis
00178 *
00179 * **Conditional Thread Safety** - Safe with external synchronization:
00180 *
00181 * | Scenario | Thread Safety | Mitigation Required |
00182 * |-----|-----|-----|
00183 * | **Single-threaded** | [PASS] Safe | None |
00184 * | **Multi-threaded (same handle)** | [FAIL] Unsafe | External mutex required |
00185 * | **Multi-threaded (different handles)** | [PASS] Safe | None (independent state) |
00186 * | **Send from multiple threads** | [FAIL] Unsafe | Mutex around send operations |
00187 * | **Receive from multiple threads** | [FAIL] Unsafe | Dedicate one thread to polling |
00188 * | **Send from ISR** | [WARN] Conditional | OK if SPI HAL is ISR-safe |
00189 * | **Receive from ISR** | [WARN] Conditional | OK if SPI HAL is ISR-safe |
00190 *
00191 * **Recommended Architecture**:
00192 * - Dedicate one thread to SPI communication (poll + send)
00193 * - Use communication manager (rx_comm_manager) for multi-channel coordination
00194 * - External mutex if multiple threads must send on same handle
00195 *
00196 * ## Hardware Requirements
00197 *
00198 * | Resource | Requirement | Usage |
00199 * |-----|-----|-----|

```

```

00200 * | **CPU** | RX72N @ 240 MHz | SPI transfer + CRC calculation |
00201 * | **RAM** | ~4.2 KB per handle | State + buffers (see Memory Usage) |
00202 * | **RSPi** | RSPi0, RSPi1, or RSPi2 | Configurable channel (default: RSPi0) |
00203 * | **SPI Pins** | 4 pins (COPI/CIPO/CLK/CS) | See pinout in docs/sections/03_hardware_pinout.tex |
00204 * | **DMA (optional)** | DMAC channel | For high-throughput SPI transfers |
00205 * | **Interrupts** | RSPi RX/TX interrupts | For non-blocking transfers |
00206 *
00207 * **Pin Configuration** (RSPi0 example):
00208 * - **COPI (Controller Out Peripheral In)**: PD3 (RX72N input from RPi5)
00209 * - **CIPO (Controller In Peripheral Out)**: PD2 (RX72N output to RPi5)
00210 * - **SCLK (Serial Clock)**: PD1 (Clock from RPi5 controller)
00211 * - **CS (Chip Select)**: PD0 (Active low, from RPi5)
00212 *
00213 * ## Usage Examples
00214 *
00215 * @par Complete Setup and Integration:
00216 * @code{.c}
00217 * #include "rx_spi_comm.h"
00218 * #include "hardware.h"
00219 *
00220 * // Global SPI communication handle
00221 * static rx_spi_comm_handle_t g_spi_handle;
00222 *
00223 * // Initialize SPI communication
00224 * rx_err_t init_spi_communication(void)
00225 * {
00226 *     // Step 1: Initialize SPI hardware (RSPi peripheral mode)
00227 *     rspi_config_t rspi_cfg = {
00228 *         .channel = 0,           // RSPi0
00229 *         .mode = 0,             // SPI mode 0 (CPOL=0, CPHA=0)
00230 *         .bit_rate = 10000000,  // 10 MHz
00231 *         .peripheral_mode = true, // RX72N is SPI peripheral
00232 *     };
00233 *     rx_err_t err = rspi_init_peripheral(&rspi_cfg);
00234 *     if (err != k_rx_ok) {
00235 *         return err;
00236 *     }
00237 *
00238 *     // Step 2: Initialize shared session state
00239 *     static rx_session_state_t g_session;
00240 *     err = rx_session_init(&g_session);
00241 *     if (err != k_rx_ok) {
00242 *         return err;
00243 *     }
00244 *
00245 *     // Step 3: Initialize SPI communication layer
00246 *     rx_spi_comm_config_t comm_cfg = {
00247 *         .session = &g_session,
00248 *         .channel = 0,           // RSPi0
00249 *         .spi_mode = 0,          // SPI mode 0
00250 *         .fec_enabled = false,   // Disable FEC for now
00251 *     };
00252 *     err = rx_spi_comm_init(&g_spi_handle, &comm_cfg);
00253 *     if (err != k_rx_ok) {
00254 *         return err;
00255 *     }
00256 *
00257 *     return k_rx_ok;
00258 * }
00259 *
00260 * // Send telemetry response
00261 * rx_err_t send_telemetry(void)
00262 * {
00263 *     uint8_t telemetry[16];
00264 *
00265 *     // Fill telemetry data
00266 *     telemetry[0] = get_temperature_celsius();
00267 *     telemetry[1] = get_motor_current();
00268 *     // ... fill remaining fields ...
00269 *
00270 *     // Send telemetry frame
00271 *     rx_err_t err = rx_spi_comm_send(&g_spi_handle,
00272 *                                     k_frame_type_response,
00273 *                                     k_frame_flag_none,
00274 *                                     telemetry,
00275 *                                     sizeof(telemetry));
00276 *     if (err != k_rx_ok) {
00277 *         // Handle send error
00278 *         return err;
00279 *     }
00280 *
00281 *     return k_rx_ok;
00282 * }
00283 *
00284 * // Receive command from RPi5
00285 * void poll_spi_commands(void)
00286 * {

```

```

00287 * rx_frame_t frame;
00288 *
00289 * // Poll with 0ms timeout (non-blocking)
00290 * rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);
00291 *
00292 * if (err == k_rx_ok) {
00293 *     // Frame received successfully
00294 *     handle_command(&frame);
00295 *
00296 *     // Send ACK
00297 *     rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
00298 * } else if (err == k_rx_err_timeout) {
00299 *     // No data available - this is normal
00300 * } else if (err == k_rx_err_crc_mismatch) {
00301 *     // CRC error - send NACK for retransmission
00302 *     rx_spi_comm_send_nack(&g_spi_handle, frame.header.sequence, k_frame_flag_none);
00303 * } else {
00304 *     // Other error - log and continue
00305 *     log_error("SPI receive error: %d", err);
00306 * }
00307 * }
00308 * @endcode
00309 *
00310 * @par Example - Sequence Number Management:
00311 * @code{.c}
00312 * // Reset sequence counters (e.g., after connection reset)
00313 * rx_spi_comm_reset_sequence(&g_spi_handle);
00314 *
00315 * // Query current TX sequence
00316 * uint16_t tx_seq;
00317 * rx_err_t err = rx_spi_comm_get_tx_sequence(&g_spi_handle, &tx_seq);
00318 * printf("Current TX sequence: %u\n", tx_seq);
00319 *
00320 * // Query expected RX sequence
00321 * uint16_t rx_seq;
00322 * err = rx_spi_comm_get_rx_sequence(&g_spi_handle, &rx_seq);
00323 * printf("Expected RX sequence: %u\n", rx_seq);
00324 * @endcode
00325 *
00326 * ## NASA Power of 10 Compliance
00327 *
00328 * | Rule | Compliance | Implementation Notes |
00329 * |-----|-----|-----|
00330 * | **Rule 1: Control Flow** | [PASS] | No goto, setjmp, longjmp, or recursion |
00331 * | **Rule 2: Loop Bounds** | [PASS] | All loops have statically provable bounds |
00332 * | **Rule 3: No Dynamic Allocation** | [PASS] | Zero malloc/free - all static buffers |
00333 * | **Rule 4: Function Size** | [PASS] | All functions < 60 lines |
00334 * | **Rule 5: Assertions** | [PASS] | Minimum 2 checks per function (nullptr, initialized) |
00335 * | **Rule 6: Data Scope** | [PASS] | Variables declared at smallest scope |
00336 * | **Rule 7: Check Returns** | [PASS] | All return values validated |
00337 * | **Rule 8: Preprocessor** | [PASS] | Typed enums for constants, minimal macros |
00338 * | **Rule 9: Pointers** | [PASS] | Single-level dereferencing only |
00339 * | **Rule 10: Compiler Warnings** | [PASS] | -Wall -Wextra -Werror enabled |
00340 *
00341 * ## SOLID Principles
00342 *
00343 * | Principle | Implementation |
00344 * |-----|-----|
00345 * | **Single Responsibility** | Module responsible ONLY for SPI frame protocol; hardware abstraction delegated to RSPI HAL |
00346 * | **Open/Closed** | Extensible via configuration (FEC enable/disable); new frame types handled by frame layer |
00347 * | **Liskov Substitution** | Interchangeable with rx_usb_comm (same rx_frame_t interface) |
00348 * | **Interface Segregation** | Small focused API (9 functions); send/receive/utility separated |
00349 * | **Dependency Inversion** | Depends on abstract rx_frame_t and RSPI HAL, not concrete implementations |
00350 *
00351 * ## Module Dependencies
00352 *
00353 * **Includes:**
00354 * - rx_err.h - Error code definitions
00355 * - rx_frame.h - Frame protocol definitions and encoder/decoder
00356 * - stdbool.h - Boolean type
00357 * - stdint.h - Fixed-width integer types
00358 *
00359 * **Links Against:**
00360 * - rx_frame - Frame encoding/decoding with CRC-32
00361 * - rx_harq (optional) - FEC encoding/decoding with HARQ
00362 * - rspi HAL - SPI hardware abstraction (peripheral mode)
00363 * - rx_crc32 - CRC-32 calculation (via rx_frame)
00364 *
00365 * **Referenced Documentation:**
00366 * - docs/sections/01_nanopb_protocol.tex - Complete protocol specification
00367 * - docs/sections/03_hardware_pinout.tex - SPI pin assignments
00368 * - RX72N_Manual_Chapters/Ch25_SPI.txt - RSPI peripheral specification
00369 *
00370 * @date 2026-01-27
00371 * @copyright Copyright (c) 2026 Locked Inc.
00372 * SPDX-License-Identifier: MIT

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00373 * @since Version 1.0.0
00374 *
00375 * @see rx_frame.h Frame protocol layer
00376 * @see rx_harq.h HARQ layer (optional)
00377 * @see hardware.h SPI HAL interface
00378 * @see rx_comm_manager.h Multi-channel communication coordinator
00379 * @see docs/sections/01_nanopb_protocol.tex Protocol specification
00380 */
00381
00382 #pragma once
00383
00384 #include <stdbool.h>
00385 #include <stdint.h>
00386
00387 #include "hardware.h"
00388 #include "rx_err.h"
00389 #include "rx_frame.h"
00390 #include "rx_session.h"
00391
00392 #ifndef __cplusplus
00393 extern "C" {
00394 #endif
00395
00396 /*
=====
00397 * Configuration Constants
00398 *
=====
00399 */
00400
00401 /**
00402 * @enum rx_spi_comm_constants_t
00403 * @brief SPI communication configuration constants and buffer sizes
00404 *
00405 * @details
00406 * Defines compile-time constants for SPI channel selection, buffer sizes,
00407 * and default timeout values. All sizes chosen to accommodate maximum frame
00408 * size (k_frame_max_total_size = 263 bytes) with safety margin.
00409 *
00410 * **Design Rationale:**
00411 * - **2048-byte buffers**: Allows ~7 maximum-size frames to be buffered,
00412 *   providing tolerance for burst traffic and preventing frame loss.
00413 * - **10 MHz SPI clock**: Maximum reliable speed for RPi5 spidev with 1m cable.
00414 * - **1 second timeout**: Conservative default for reliable operation.
00415 *
00416 * @since Version 1.0.0
00417 */
00418 typedef enum : uint16_t {
00419 /**
00420 * @brief Default SPI mode (CPOL=0, CPHA=0)
00421 * @details
00422 * SPI Mode 0: Clock idles low, data sampled on rising edge.
00423 * **RPi5 spidev mode**: mode=0 (matches this setting)
00424 * @par Value: 0
00425 * @par Modes: 0 (0,0), 1 (0,1), 2 (1,0), 3 (1,1)
00426 */
00427 k_spi_comm_default_mode = 0,
00428
00429 /**
00430 * @brief RX staging buffer size in bytes
00431 * @details
00432 * Buffer for incoming frames before decoding. Size allows buffering
00433 * multiple frames to handle burst traffic from RPi5.
00434 * **Capacity**: ~7 maximum-size frames (263 bytes each)
00435 * **Memory**: Static allocation in rx_spi_comm_handle_t
00436 * @par Value: 2048 bytes
00437 * @par Rationale: Power-of-2 for alignment, sufficient for burst tolerance
00438 */
00439 k_spi_comm_rx_buffer_size = 2048,
00440
00441 /**
00442 * @brief TX staging buffer size in bytes
00443 * @details
00444 * Buffer for outgoing frames before SPI transmission. Same size as RX
00445 * buffer for symmetry and burst tolerance.
00446 * **Capacity**: ~7 maximum-size frames (263 bytes each)
00447 * **Memory**: Static allocation in rx_spi_comm_handle_t
00448 * @par Value: 2048 bytes
00449 * @par Rationale: Symmetric with RX buffer, burst tolerance
00450 */
00451 k_spi_comm_tx_buffer_size = 2048,
00452
00453 /**
00454 * @brief Default timeout in milliseconds for blocking operations
00455 * @details
00456 * Used for rx_spi_comm_receive() when non-zero timeout specified.
00457 * **Conservative value**: Allows reliable operation even with CPU load.

```

```

00458  * **Note**: Most operations use 0ms timeout (non-blocking polling).
00459  * @par Value: 1000 milliseconds (1 second)
00460  * @par Rationale: Conservative for reliable operation
00461  */
00462  k_spi_comm_default_timeout = 1000,
00463 } rx_spi_comm_constants_t;
00464
00465 /*
=====
00466 * Retransmit Configuration
00467 *
=====
00468 */
00469
00470 /**
00471  * @enum rx_retransmit_defaults_retries_t
00472  * @brief Default retry count for automatic retransmission
00473  *
00474  * @details
00475  * Maximum number of retransmission attempts before giving up.
00476  * Applied when auto_retransmit is enabled but max_retries is zero.
00477  *
00478  * @since Version 1.0.0
00479  */
00480 typedef enum : uint8_t {
00481  k_retransmit_default_max_retries = 3, /**< Default max retransmit attempts */
00482 } rx_retransmit_defaults_retries_t;
00483
00484 /**
00485  * @enum rx_retransmit_defaults_timing_t
00486  * @brief Default timing values for automatic retransmission
00487  *
00488  * @details
00489  * Timing parameters for ACK timeout and exponential backoff cap.
00490  * Applied when auto_retransmit is enabled but timing values are zero.
00491  *
00492  * @since Version 1.0.0
00493  */
00494 typedef enum : uint16_t {
00495  k_retransmit_default_ack_timeout_ms = 50, /**< Initial ACK wait in ms */
00496  k_retransmit_default_max_backoff_ms = 400, /**< Max exponential backoff cap in ms */
00497 } rx_retransmit_defaults_timing_t;
00498
00499 /**
00500  * @struct rx_spi_comm_retransmit_config_t
00501  * @brief Configuration for automatic retransmission behavior
00502  *
00503  * @details
00504  * Controls retransmission parameters when auto_retransmit is enabled.
00505  * Zero values are replaced with defaults during initialization.
00506  *
00507  * @par Field Constraints:
00508  * | Field | Min | Max | Default | Notes |
00509  * |-----|-----|-----|-----|-----|
00510  * | max_retries | 1 | 10 | 3 | 0 = use default |
00511  * | ack_timeout_ms | 10 | 1000 | 50 | 0 = use default |
00512  * | max_backoff_ms | 50 | 5000 | 400 | 0 = use default |
00513  *
00514  * @since Version 1.0.0
00515  * @see rx_spi_comm_set_auto_retransmit() Runtime configuration
00516  */
00517 typedef struct {
00518  uint8_t max_retries; /**< Max retransmit attempts (default 3, 0 = use default) */
00519  uint16_t ack_timeout_ms; /**< Initial ACK timeout in ms (default 50, 0 = use default) */
00520  uint16_t max_backoff_ms; /**< Max backoff cap in ms (default 400, 0 = use default) */
00521 } rx_spi_comm_retransmit_config_t;
00522
00523 /*
=====
00524 * Handle and Configuration
00525 *
=====
00526 */
00527
00528 /**
00529  * @struct rx_spi_comm_handle_t
00530  * @brief SPI communication handle containing all state for frame protocol
00531  *
00532  * @details
00533  * Complete state for SPI frame-based communication including frame encoder/decoder,
00534  * sequence number tracking, and staging buffers. This structure must be initialized
00535  * via rx_spi_comm_init() before use.
00536  *
00537  * **Lifecycle:**
00538  * 1. Allocate handle (typically static or global)
00539  * 2. Initialize via rx_spi_comm_init()
00540  * 3. Use for send/receive operations

```

```

00541 * 4. Deinitialize via rx_spi_comm_deinit() when done
00542 *
00543 * **Memory Footprint**: 4120 bytes (see file-level Memory Usage section)
00544 *
00545 * **Thread Safety**: Not thread-safe - use external mutex if accessed from
00546 * multiple threads.
00547 *
00548 * @par Memory Layout:
00549 * See file-level Memory Usage section for detailed offset and size breakdown.
00550 *
00551 * @par Usage Example:
00552 * @code{.c}
00553 * static rx_spi_comm_handle_t g_spi_handle;
00554 * static rx_session_state_t g_session;
00555 *
00556 * void init(void) {
00557 *     rx_session_init(&g_session);
00558 *     rx_spi_comm_config_t cfg = {
00559 *         .session = &g_session,
00560 *         .channel = 0,
00561 *         .spi_mode = 0,
00562 *         .fec_enabled = false
00563 *     };
00564 *     rx_err_t err = rx_spi_comm_init(&g_spi_handle, &cfg);
00565 *     // ... check error ...
00566 * }
00567 * @endcode
00568 *
00569 * @since Version 1.0.0
00570 * @see rx_spi_comm_init() Initialization function
00571 * @see rx_spi_comm_deinit() Cleanup function
00572 */
00573 typedef struct {
00574 /**
00575  * @brief Frame encoder for outgoing frames
00576  * @details
00577  * Encodes payload + header into wire format with CRC-32.
00578  * **State**: Maintains encoding context (minimal ~8 bytes).
00579  * @see rx_frame.h rx_frame_encoder_t definition
00580  */
00581 rx_frame_encoder_t encoder;
00582 /**
00583  * @brief Frame decoder for incoming frames
00584  * @details
00585  * Decodes wire format into payload + header, validates CRC-32.
00586  * **State**: Maintains decoding context (minimal ~8 bytes).
00587  * @see rx_frame.h rx_frame_decoder_t definition
00588  */
00589 rx_frame_decoder_t decoder;
00590 /**
00591  * @brief RX staging buffer for incoming SPI data
00592  * @details
00593  * Temporary buffer for receiving frames from RPi5 before decoding.
00594  * **Size**: 2048 bytes (k_spi_comm_rx_buffer_size)
00595  * **Capacity**: ~7 maximum-size frames
00596  * **Access**: Internal only (not exposed to application)
00597  */
00598 uint8_t rx_buffer[k_spi_comm_rx_buffer_size];
00599 /**
00600  * @brief TX staging buffer for outgoing SPI data
00601  * @details
00602  * Temporary buffer for encoding frames before SPI transmission to RPi5.
00603  * **Size**: 2048 bytes (k_spi_comm_tx_buffer_size)
00604  * **Capacity**: ~7 maximum-size frames
00605  * **Access**: Internal only (not exposed to application)
00606  */
00607 uint8_t tx_buffer[k_spi_comm_tx_buffer_size];
00608 /**
00609  * @brief Pointer to shared session state (cross-transport sequence continuity)
00610  * @details
00611  * Points to a single rx_session_state_t owned by rx_comm_manager_t, shared
00612  * with the USB CDC transport. When the active transport switches (e.g., USB
00613  * at seq=105 to SPI), SPI continues at seq=106 because both reference the
00614  * same session state.
00615  * **Lifetime**: Must remain valid for the lifetime of this handle.
00616  * **Thread Safety**: Session functions are thread-safe (internal mutex).
00617  * @see rx_session.h Shared session API
00618  * @see rx_comm_manager.h Owns the session instance
00619  */
00620 rx_session_state_t* session;
00621 /**
00622  * @brief RSPI channel (0-2)

```

```

00628 * @details
00629 * Hardware RSPI peripheral channel.
00630 * **Valid values**: k_rspi_channel_0, k_rspi_channel_1, k_rspi_channel_2
00631 * **Default**: k_rspi_channel_0 (RSPi0 on PD0-PD3)
00632 * @see rspi_channel_t for channel enumeration
00633 */
00634 rspi_channel_t channel;
00635
00636 /**
00637 * @brief FEC (Forward Error Correction) enabled flag
00638 * @details
00639 * If true, FEC encoding/decoding applied to frames.
00640 * **Impact**: Adds ~50% overhead, improves reliability in noisy environments.
00641 * **Default**: false (FEC disabled for speed)
00642 * @warning FEC requires both sides (RX72N + RPi5) to agree on encoding
00643 */
00644 bool fec_enabled;
00645
00646 /**
00647 * @brief Initialization flag
00648 * @details
00649 * Set to true by rx_spi_comm_init(), cleared by rx_spi_comm_deinit().
00650 * **Purpose**: Prevents operations on uninitialized handle.
00651 * **Validation**: Checked by all public functions.
00652 */
00653 bool initialized;
00654
00655 /**
00656 * @brief Callback invoked when PING control frame received
00657 * @details Called with decoded frame and user context during rx_spi_comm_receive.
00658 */
00659 void (*on_ping_cb)(const rx_frame_t* frame, void* ctx);
00660
00661 /**
00662 * @brief Callback invoked when RESET control frame received
00663 * @details Called with decoded frame and user context during rx_spi_comm_receive.
00664 */
00665 void (*on_reset_cb)(const rx_frame_t* frame, void* ctx);
00666
00667 /**
00668 * @brief User context pointer passed to control frame callbacks (PING, RESET)
00669 */
00670 void* control_cb_ctx;
00671
00672 /* ---- Retransmission state (only active when auto_retransmit enabled) ---- */
00673
00674 /**
00675 * @brief Enable automatic retransmission of sent frames
00676 * @details When true, sent frames are buffered and retransmitted on NACK or timeout.
00677 */
00678 bool auto_retransmit;
00679
00680 /**
00681 * @brief Active retransmit configuration (copied from config or set at runtime)
00682 */
00683 rx_spi_comm_retransmit_config_t retransmit_cfg;
00684
00685 /**
00686 * @brief Retry buffer holding last sent encoded wire frame
00687 * @details Sized to k_frame_max_size (1036 bytes) for maximum frame.
00688 */
00689 uint8_t retry_buffer[k_frame_max_size];
00690
00691 /**
00692 * @brief TX encode scratch buffer (handle-owned, not on caller stack)
00693 * @details Sized to k_frame_max_size (1036 bytes). Replaces a 1036-byte
00694 * stack allocation in rx_spi_comm_send/_send_pong that previously consumed
00695 * ~50% of a 2 KB ThreadX task stack. Single-producer per handle, so the
00696 * caller-side mutex serializing SPI transfers also serializes use of this
00697 * scratch buffer.
00698 */
00699 uint8_t tx_encode_buffer[k_frame_max_size];
00700
00701 /**
00702 * @brief Actual encoded wire length of buffered frame in retry_buffer
00703 */
00704 uint32_t retry_wire_len;
00705
00706 /**
00707 * @brief Sequence number of the frame in retry_buffer
00708 */
00709 uint16_t retry_sequence;
00710
00711 /**
00712 * @brief Current retry attempt count (0 = first send, 1+ = retransmissions)
00713 */
00714 uint8_t retry_count;

```

```

00715
00716 /**
00717  * @brief True when a frame is buffered and awaiting ACK
00718  */
00719 bool retry_pending;
00720
00721 /**
00722  * @brief Timestamp (ms) of last send or retransmit of the buffered frame
00723  */
00724 uint32_t retry_send_time_ms;
00725
00726 /**
00727  * @brief Optional callback invoked when ACK received for a pending frame
00728  */
00729 void (*on_ack_cb)(uint16_t sequence, void* ctx);
00730
00731 /**
00732  * @brief Optional callback invoked when NACK received for a pending frame
00733  */
00734 void (*on_nack_cb)(uint16_t sequence, void* ctx);
00735
00736 /**
00737  * @brief User context pointer passed to retransmit callbacks (ACK, NACK)
00738  */
00739 void* retransmit_cb_ctx;
00740 } rx_spi_comm_handle_t;
00741
00742 /**
00743  * @struct rx_spi_comm_config_t
00744  * @brief Configuration structure for SPI communication initialization
00745  *
00746  * @details
00747  * Specifies SPI hardware parameters and optional FEC enable. Used during
00748  * rx_spi_comm_init() to configure the communication layer.
00749  *
00750  * IMPORTANT: config is REQUIRED (must not be NULL). The session pointer
00751  * in the config is mandatory for shared sequence state.
00752  *
00753  * @par Field Constraints:
00754  * | Field | Min | Max | Default | Notes |
00755  * |-----|-----|-----|-----|-----|
00756  * | session | non-NULL | - | - | Required: shared session state |
00757  * | channel | 0 | 2 | 0 | RX72N has 3 RSPI channels |
00758  * | spi_mode | 0 | 3 | 0 | Mode 0 recommended for RPi5 |
00759  * | fec_enabled | false | true | false | Enable for noisy environments |
00760  *
00761  * @par Usage Example:
00762  * @code{.c}
00763  * rx_spi_comm_handle_t handle;
00764  * rx_session_state_t session;
00765  * rx_session_init(&session);
00766  *
00767  * rx_spi_comm_config_t cfg = {
00768  *     .session = &session,
00769  *     .channel = k_rspi_channel_0,
00770  *     .spi_mode = k_spi_comm_default_mode,
00771  *     .fec_enabled = false
00772  * };
00773  * rx_err_t err = rx_spi_comm_init(&handle, &cfg);
00774  * @endcode
00775  *
00776  * @since Version 1.0.0
00777  * @see rx_spi_comm_init() Initialization function
00778  * @see rx_spi_comm_constants_t Default values
00779  */
00780 typedef struct {
00781     /**
00782      * @brief Shared session state (REQUIRED, must not be NULL)
00783      *
00784      * @details
00785      * Pointer to shared session state for cross-transport sequence continuity.
00786      * Owned by rx_comm_manager_t, shared with USB CDC transport.
00787      * Lifetime: Must remain valid for the lifetime of the SPI comm handle.
00788      * @see rx_session.h Shared session API
00789      */
00790     rx_session_state_t* session;
00791
00792     /**
00793      * @brief RSPI channel (0-2)
00794      *
00795      * @details
00796      * Hardware peripheral channel. RX72N has 3 RSPI channels.
00797      * Default: k_rspi_channel_0 (RSPI0)
00798      * @see rspi_channel_t for channel enumeration
00799      */
00800     rspi_channel_t channel;
00801
00802     /**
00803      * @brief SPI mode (0-3)

```

```

00802 * @details
00803 * Clock polarity and phase configuration.
00804 * **Mode 0**: CPOL=0, CPHA=0 (clock idles low, sample on rising edge)
00805 * **Default**: 0
00806 * **Valid range**: [0, 3]
00807 */
00808 uint8_t spi_mode;
00809
00810 /**
00811 * @brief Enable Forward Error Correction
00812 * @details
00813 * If true, applies FEC encoding to outgoing frames and FEC decoding
00814 * to incoming frames. Improves reliability at cost of throughput.
00815 * **Default**: false
00816 * **Overhead**: ~50% (e.g., 100 bytes -> 150 bytes with FEC)
00817 */
00818 bool fec_enabled;
00819
00820 /**
00821 * @brief Enable automatic retransmission (default: false)
00822 * @details
00823 * When true, sent frames with k_frame_flag_requires_ack are buffered
00824 * and automatically retransmitted on NACK or ACK timeout.
00825 * **Default**: false (fire-and-forget behavior)
00826 */
00827 bool auto_retransmit;
00828
00829 /**
00830 * @brief Retransmit configuration (only used if auto_retransmit true)
00831 * @details Zero fields use defaults. Ignored when auto_retransmit is false.
00832 */
00833 rx_spi_comm_retransmit_config_t retransmit_config;
00834 } rx_spi_comm_config_t;
00835
00836 /*
=====
00837 * Initialization
00838 *
=====
00839 */
00840
00841 /**
00842 * @brief Initialize SPI communication handle with frame protocol support
00843 *
00844 * @details
00845 * Performs complete initialization of SPI communication handle including frame
00846 * encoder/decoder setup, sequence number initialization, and optional FEC configuration.
00847 * This function MUST be called before any send/receive operations.
00848 *
00849 * **IMPORTANT**: This function does NOT initialize SPI hardware. Caller must
00850 * initialize RSPI peripheral via rspi_init_peripheral() BEFORE calling this function.
00851 *
00852 * **Algorithm**:
00853 * 1. **Validate parameters**: Check handle pointer for nullptr (NASA Rule 5)
00854 * 2. **Clear handle**: Zero-fill entire structure (4120 bytes)
00855 * 3. **Apply configuration**: Copy config fields if non-NULL, else use defaults
00856 * 4. **Initialize frame encoder**: Setup CRC-32 and header encoding
00857 * 5. **Initialize frame decoder**: Setup CRC-32 validation and header parsing
00858 * 6. **Initialize sequences**: Set tx_sequence = 0, rx_sequence = 0
00859 * 7. **Mark initialized**: Set handle->initialized = true
00860 *
00861 * **Configuration Options** (config is REQUIRED, not optional):
00862 * - **session**: REQUIRED, pointer to initialized rx_session_state_t
00863 * - **channel**: 0-2 (RSPI0/1/2), default 0
00864 * - **spi_mode**: 0-3 (SPI modes), default 0
00865 * - **fec_enabled**: true/false, default false
00866 *
00867 * @param[out] handle Pointer to SPI communication handle to initialize
00868 * - **Valid range**: Non-nullptr to allocated rx_spi_comm_handle_t (4120 bytes)
00869 * - **Constraints**: Must be allocated by caller (typically static or global)
00870 * - **Side effects**: Entire structure zero-filled, then populated from config
00871 * - **Lifetime**: Must remain valid until rx_spi_comm_deinit() called
00872 *
00873 * @param[in] config Required configuration structure (must not be NULL)
00874 * - **Valid range**: Non-nullptr to valid rx_spi_comm_config_t
00875 * - **Constraints**: session must be non-NULL, channel must be 0-2, spi_mode must be 0-3
00876 * - **Lifetime**: Not stored - values copied into handle (except session ptr)
00877 *
00878 * @return rx_err_t Error code indicating initialization success or failure
00879 * @retval k_rx_ok Success - handle fully initialized and ready for operation
00880 * @retval k_rx_err_invalid_arg handle is nullptr; config is nullptr; config->session is nullptr; config->channel > 2
00881 * (invalid RSPI channel); config->spi_mode > 3 (invalid SPI mode)
00882 *
00883 * @pre handle must point to allocated memory (uninitialized content OK)
00884 * @pre RSPI hardware must already be initialized via rspi_init_peripheral()
00885 * @post handle->initialized = true on success
00886 * @post handle->session points to config->session

```

```

00886 * @post All buffers zero-filled (rx_buffer, tx_buffer)
00887 *
00888 * @note This function does NOT initialize SPI hardware - caller responsible
00889 * @note On success, handle is ready for rx_spi_comm_send/receive operations
00890 * @note Thread-safe ONLY if each thread uses a different handle
00891 *
00892 * @warning Must call rspi_init_peripheral() BEFORE this function
00893 *
00894 * @par Performance:
00895 * - Execution time: ~10 us @ 240 MHz
00896 * - Dominated by memset (4120 bytes)
00897 *
00898 * @par Example:
00899 * @code{.c}
00900 * static rx_spi_comm_handle_t g_spi_handle;
00901 * static rx_session_state_t g_session;
00902 *
00903 * // Step 1: Initialize shared session
00904 * rx_err_t err = rx_session_init(&g_session);
00905 * if (err != k_rx_ok) return err;
00906 *
00907 * // Step 2: Initialize SPI communication layer
00908 * rx_spi_comm_config_t comm_cfg = {
00909 *     .session = &g_session,
00910 *     .channel = 0,
00911 *     .spi_mode = 0,
00912 *     .fec_enabled = false
00913 * };
00914 * err = rx_spi_comm_init(&g_spi_handle, &comm_cfg);
00915 * if (err != k_rx_ok) {
00916 *     log_error("SPI comm init failed: %d", err);
00917 *     return err;
00918 * }
00919 * @endcode
00920 *
00921 * @see rx_spi_comm_deinit() Cleanup and resource release
00922 * @see rx_spi_comm_send() Send frame
00923 * @see rx_spi_comm_receive() Receive frame
00924 * @see rspi_init_peripheral() Initialize SPI hardware (must call first)
00925 *
00926 * @since Version 1.0.0
00927 */
00928 [[nodiscard]] rx_err_t rx_spi_comm_init(rx_spi_comm_handle_t* handle,
00929                                         const rx_spi_comm_config_t* config);
00930
00931 /**
00932 * @brief Deinitialize SPI communication handle and release resources
00933 *
00934 * @details
00935 * Marks SPI communication handle as uninitialized, preventing further operations.
00936 * This function does NOT deinitialize RSPI hardware - caller must separately call
00937 * rspi_deinit() if needed.
00938 *
00939 * **Algorithm:**
00940 * 1. Validate handle pointer (NULL check)
00941 * 2. Validate initialized flag (must be initialized to deinitialize)
00942 * 3. Clear initialized flag (marks handle as invalid)
00943 *
00944 * **Design Note:** Minimal deinitialization - no dynamic resources to free
00945 * (all memory is statically allocated in handle structure). Function primarily
00946 * serves as safety guard to prevent use-after-deinit.
00947 *
00948 * @param[in,out] handle Pointer to SPI communication handle
00949 * - **Valid range:** Non-nullptr to initialized handle
00950 * - **Constraints:** Must have been initialized via rx_spi_comm_init()
00951 * - **Side effects:** Clears handle->initialized flag
00952 *
00953 * @return rx_err_t Error code
00954 * @retval k_rx_ok Success - handle deinitialized
00955 * @retval k_rx_err_invalid_arg handle is nullptr
00956 * @retval k_rx_err_invalid_state handle was not initialized
00957 *
00958 * @pre handle must be non-NULL
00959 * @pre handle must be initialized (handle->initialized == true)
00960 * @post handle->initialized = false
00961 * @post Subsequent calls to rx_spi_comm_send/receive will return k_rx_err_invalid_state
00962 *
00963 * @note Does NOT free memory (handle is statically allocated)
00964 * @note Does NOT deinitialize RSPI hardware
00965 * @note Thread-safe if no concurrent operations on same handle
00966 *
00967 * @par Example:
00968 * @code{.c}
00969 * // Deinitialize SPI communication
00970 * rx_err_t err = rx_spi_comm_deinit(&g_spi_handle);
00971 * if (err == k_rx_ok) {
00972 *     // Handle no longer usable

```

```

00973 *
00974 * // Separately deinitialize RSPi hardware if needed
00975 * rspi_deinit(g_spi_handle.channel);
00976 * }
00977 * @endcode
00978 *
00979 * @see rx_spi_comm_init() Initialization function
00980 * @see rspi_deinit() Deinitialize RSPi hardware
00981 *
00982 * @since Version 1.0.0
00983 */
00984 [[nodiscard]] rx_err_t rx_spi_comm_deinit(rx_spi_comm_handle_t* handle);
00985
00986 /*
=====
00987 * Send API
00988 *
=====
00989 */
00990
00991 /**
00992 * @brief Send frame on SPI with automatic sequencing and CRC
00993 *
00994 * @details
00995 * Encodes payload into frame format with automatic sequence number assignment,
00996 * CRC-32 calculation, and optional FEC encoding. Transmits frame to RPi5 via
00997 * SPI hardware. This is the primary send function for all frame types.
00998 *
00999 * **Algorithm:**
01000 * 1. **Validate parameters**: Check handle, payload, payload_len (NASA Rule 5)
01001 * 2. **Assign sequence**: Get next TX sequence from shared rx_session_state_t
01002 * 3. **Build frame**: Create frame header (type, flags, length, sequence)
01003 * 4. **Encode frame**: Call frame encoder to add CRC-32
01004 * 5. **Optional FEC**: If enabled, apply FEC encoding
01005 * 6. **SPI transfer**: Transmit encoded frame via RSPi peripheral
01006 * 7. **Update state**: TX sequence already incremented by rx_session_next_tx()
01007 *
01008 * **Performance**:
01009 * - Small payload (16 bytes): ~150 us
01010 * - Medium payload (128 bytes): ~300 us
01011 * - Large payload (255 bytes): ~500 us
01012 * - FEC overhead: +50% time
01013 *
01014 * @param[in,out] handle Initialized SPI communication handle
01015 * - **Valid range**: Non-nullptr to initialized handle
01016 * - **Constraints**: Must be initialized via rx_spi_comm_init()
01017 * - **Side effects**: Increments shared session TX sequence via rx_session_next_tx()
01018 *
01019 * @param[in] type Frame type (command, request, response, etc.)
01020 * - **Valid range**: Any rx_frame_type_t value
01021 * - **Common values**: k_frame_type_response (most common for RX72N)
01022 * - **Units**: Enumeration value
01023 *
01024 * @param[in] flags Frame flags (ACK, NACK, etc.)
01025 * - **Valid range**: Bitmask of rx_frame_flag_t values
01026 * - **Common values**: k_frame_flag_none (0x00)
01027 * - **Units**: Bitmask
01028 *
01029 * @param[in] payload Payload data to send
01030 * - **Valid range**: Non-NULL if payload_len > 0, may be NULL if payload_len == 0
01031 * - **Constraints**: Must contain valid data for payload_len bytes
01032 * - **Lifetime**: Copied immediately, does not need to persist
01033 * - **Max size**: k_frame_max_payload (255 bytes)
01034 *
01035 * @param[in] payload_len Payload length in bytes
01036 * - **Valid range**: [0, k_frame_max_payload] (0-255 bytes)
01037 * - **Units**: Bytes
01038 * - **Constraints**: Must be <= k_frame_max_payload
01039 *
01040 * @return rx_err_t Error code indicating send success or failure
01041 * @retval k_rx_ok Frame sent successfully, tx_sequence incremented
01042 * @retval k_rx_err_invalid_arg handle or payload is nullptr; payload_len > k_frame_max_payload (255 bytes); payload
is nullptr but payload_len > 0
01043 * @retval k_rx_err_invalid_state handle not initialized
01044 * @retval k_rx_err_hw SPI hardware error (RSPi transfer failed)
01045 * @retval k_rx_err_timeout SPI transfer timeout
01046 *
01047 * @pre handle must be non-NULL and initialized
01048 * @pre If payload_len > 0, payload must point to valid data
01049 * @pre RSPi hardware must be initialized and ready
01050 * @post Shared session TX sequence incremented (wraps at 65535)
01051 * @post Frame transmitted to RPi5 if successful
01052 *
01053 * @note Sequence number managed automatically - caller does not specify
01054 * @note CRC-32 calculated automatically by frame encoder
01055 * @note FEC applied automatically if handle->fec_enabled = true
01056 * @warning Ensure RPi5 is ready to receive (chip select asserted)

```

```

01057 *
01058 * @par Example - Send Telemetry Response:
01059 * @code{.c}
01060 * uint8_t telemetry[16];
01061 * // ... fill telemetry data ...
01062 *
01063 * rx_err_t err = rx_spi_comm_send(&g_spi_handle,
01064 *                                k_frame_type_response,
01065 *                                k_frame_flag_none,
01066 *                                telemetry,
01067 *                                sizeof(telemetry));
01068 * if (err == k_rx_ok) {
01069 *     // Telemetry sent successfully
01070 * } else {
01071 *     log_error("Send failed: %d", err);
01072 * }
01073 * @endcode
01074 *
01075 * @par Example - Send Empty ACK (No Payload):
01076 * @code{.c}
01077 * // Send empty frame with ACK flag
01078 * rx_err_t err = rx_spi_comm_send(&g_spi_handle,
01079 *                                k_frame_type_response,
01080 *                                k_frame_flag_ack,
01081 *                                NULL, // No payload
01082 *                                0); // Zero length
01083 * @endcode
01084 *
01085 * @see rx_spi_comm_send_ack() Convenience wrapper for ACK frames
01086 * @see rx_spi_comm_send_nack() Convenience wrapper for NACK frames
01087 * @see rx_spi_comm_receive() Receive frames from RPi5
01088 * @see rx_frame.h Frame protocol definitions
01089 *
01090 * @since Version 1.0.0
01091 */
01092 [[nodiscard]] rx_err_t rx_spi_comm_send(rx_spi_comm_handle_t* handle,
01093                                         rx_frame_type_t type,
01094                                         uint8_t flags,
01095                                         const uint8_t* payload,
01096                                         uint32_t payload_len);
01097
01098 /**
01099 * @brief Send ACK frame for received frame (convenience wrapper)
01100 *
01101 * @details
01102 * Convenience function for sending acknowledgment frames. Automatically sets
01103 * frame type to k_frame_type_response and flags to k_frame_flag_ack. Includes
01104 * the sequence number being acknowledged in the frame.
01105 *
01106 * **Use Case**: Acknowledge successful receipt of a frame from RPi5.
01107 *
01108 * **Algorithm**:
01109 * 1. Build ACK frame with type=response, flags=ack
01110 * 2. Include sequence number in payload (2 bytes, little-endian)
01111 * 3. Call rx_spi_comm_send() with constructed frame
01112 *
01113 * **Performance**: ~50 us (minimal frame with 2-byte payload)
01114 *
01115 * @param[in,out] handle Initialized SPI communication handle
01116 * - **Valid range**: Non-nullptr to initialized handle
01117 * - **Constraints**: Must be initialized via rx_spi_comm_init()
01118 *
01119 * @param[in] sequence Sequence number to acknowledge
01120 * - **Valid range**: [0, 65535]
01121 * - **Units**: Sequence number (uint16_t)
01122 * - **Purpose**: Tells RPi5 which frame is being acknowledged
01123 *
01124 * @return rx_err_t Error code (same as rx_spi_comm_send)
01125 * @retval k_rx_ok ACK sent successfully
01126 * @retval k_rx_err_invalid_arg handle is nullptr
01127 * @retval k_rx_err_invalid_state handle not initialized
01128 * @retval (other) Errors propagated from rx_spi_comm_send()
01129 *
01130 * @pre Same preconditions as rx_spi_comm_send()
01131 * @post ACK frame transmitted with specified sequence number
01132 *
01133 * @note Equivalent to calling rx_spi_comm_send() with type=response, flags=ack
01134 * @note This is a thin wrapper - no additional validation performed
01135 *
01136 * @par Example:
01137 * @code{.c}
01138 * // Receive frame and send ACK
01139 * rx_frame_t frame;
01140 * rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);
01141 * if (err == k_rx_ok) {
01142 *     // Process frame
01143 *     handle_frame(&frame);

```

```

01144 *
01145 * // Send ACK for this frame
01146 * err = rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
01147 * }
01148 * @endcode
01149 *
01150 * @see rx_spi_comm_send() Full send function
01151 * @see rx_spi_comm_send_nack() Send NACK for errors
01152 *
01153 * @since Version 1.0.0
01154 */
01155 [[nodiscard]] rx_err_t rx_spi_comm_send_ack(rx_spi_comm_handle_t* handle, uint16_t sequence);
01156 /**
01157 * @brief Send NACK frame for received frame with error indication
01158 *
01159 *
01160 * @details
01161 * Convenience function for sending negative acknowledgment frames. Used to
01162 * request retransmission when frame reception fails (CRC error, corruption, etc.).
01163 *
01164 * **Use Cases**:
01165 * - CRC validation failure (hard NACK)
01166 * - FEC decoding failure (soft NACK if HARQ enabled)
01167 * - Out-of-order sequence detection
01168 * - Protocol violation
01169 *
01170 * **Algorithm**:
01171 * 1. Build NACK frame with type=response, flags=nack | additional_flags
01172 * 2. Include sequence number in payload (2 bytes, little-endian)
01173 * 3. Call rx_spi_comm_send() with constructed frame
01174 *
01175 * **Performance**: ~50 us (minimal frame with 2-byte payload)
01176 *
01177 * @param[in,out] handle Initialized SPI communication handle
01178 * - **Valid range**: Non-nullptr to initialized handle
01179 * - **Constraints**: Must be initialized via rx_spi_comm_init()
01180 *
01181 * @param[in] sequence Sequence number being NACKed
01182 * - **Valid range**: [0, 65535]
01183 * - **Units**: Sequence number (uint16_t)
01184 * - **Purpose**: Tells RPi5 which frame failed and needs retransmission
01185 *
01186 * @param[in] flags Additional flags (e.g., k_frame_flag_soft_nack)
01187 * - **Valid range**: Bitmask of rx_frame_flag_t values
01188 * - **Common values**: k_frame_flag_none (hard NACK), k_frame_flag_soft_nack (HARQ)
01189 * - **Units**: Bitmask
01190 *
01191 * @return rx_err_t Error code (same as rx_spi_comm_send)
01192 * @retval k_rx_ok NACK sent successfully
01193 * @retval k_rx_err_invalid_arg handle is nullptr
01194 * @retval k_rx_err_invalid_state handle not initialized
01195 * @retval (other) Errors propagated from rx_spi_comm_send()
01196 *
01197 * @pre Same preconditions as rx_spi_comm_send()
01198 * @post NACK frame transmitted with specified sequence number and flags
01199 *
01200 * @note RPi5 should retransmit frame with specified sequence number
01201 * @note Use k_frame_flag_soft_nack for FEC soft-combining (HARQ)
01202 *
01203 * @par Example - Hard NACK on CRC Error:
01204 * @code{.c}
01205 * rx_frame_t frame;
01206 * rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);
01207 * if (err == k_rx_err_crc_mismatch) {
01208 *     // CRC validation failed - send hard NACK
01209 *     rx_spi_comm_send_nack(&g_spi_handle,
01210 *                          frame.header.sequence,
01211 *                          k_frame_flag_none); // Hard NACK
01212 * }
01213 * @endcode
01214 *
01215 * @par Example - Soft NACK for HARQ:
01216 * @code{.c}
01217 * // FEC decoding failed but HARQ soft-combining available
01218 * if (fec_decode_failed && harq_enabled) {
01219 *     rx_spi_comm_send_nack(&g_spi_handle,
01220 *                          frame.header.sequence,
01221 *                          k_frame_flag_soft_nack); // Soft NACK for HARQ
01222 * }
01223 * @endcode
01224 *
01225 * @see rx_spi_comm_send() Full send function
01226 * @see rx_spi_comm_send_ack() Send ACK for success
01227 * @see rx_frame.h Frame flag definitions
01228 *
01229 * @since Version 1.0.0
01230 */

```

```

01231 [[nodiscard]] rx_err_t
01232 rx_spi_comm_send_nack(rx_spi_comm_handle_t* handle, uint16_t sequence, uint8_t flags);
01233
01234 /*
=====
01235  * Receive API
01236  *
=====
01237  */
01238
01239 /**
01240  * @brief Receive and decode frame from RPi5 with CRC validation
01241  *
01242  * @details
01243  * Reads raw bytes from SPI hardware, validates CRC-32, decodes frame header and
01244  * payload, and optionally applies FEC decoding with HARQ soft-combining. Returns
01245  * complete decoded frame to caller.
01246  *
01247  * **Algorithm**:
01248  * 1. **Validate parameters**: Check handle, frame pointer (NASA Rule 5)
01249  * 2. **Poll SPI hardware**: Read available bytes (non-blocking if timeout=0)
01250  * 3. **Optional FEC decode**: If enabled, apply FEC decoding with HARQ
01251  * 4. **Decode frame**: Parse header, extract payload
01252  * 5. **Validate CRC**: Compute CRC-32 and compare with frame CRC
01253  * 6. **Validate sequence**: Check if sequence matches expected rx_sequence
01254  * 7. **Update state**: Increment rx_sequence on success
01255  * 8. **Return frame**: Copy decoded frame to output parameter
01256  *
01257  * **Blocking Behavior**:
01258  * - **timeout_ms = 0**: Non-blocking poll (returns immediately)
01259  * - **timeout_ms > 0**: Blocks up to timeout_ms waiting for data
01260  * - **timeout_ms = k_spi_comm_default_timeout**: Use default (1 second)
01261  *
01262  * **Performance**:
01263  * - Small frame (16 bytes): ~100 us
01264  * - Medium frame (128 bytes): ~250 us
01265  * - Large frame (255 bytes): ~400 us
01266  * - FEC overhead: +100 us
01267  *
01268  * @param[in,out] handle Initialized SPI communication handle
01269  * - **Valid range**: Non-nullptr to initialized handle
01270  * - **Constraints**: Must be initialized via rx_spi_comm_init()
01271  * - **Side effects**: Updates shared session RX sequence via rx_session_validate_rx()
01272  *
01273  * @param[out] frame Pointer to frame structure to receive decoded frame
01274  * - **Valid range**: Non-nullptr to allocated rx_frame_t
01275  * - **Constraints**: Must have sufficient space (263 bytes)
01276  * - **Lifetime**: Filled by this function, owned by caller afterward
01277  *
01278  * @param[in] timeout_ms Timeout in milliseconds
01279  * - **Valid range**: [0, 60000] (0 = non-blocking, up to 1 minute)
01280  * - **Units**: Milliseconds
01281  * - **Common values**: 0 (poll), k_spi_comm_default_timeout (1 second)
01282  *
01283  * @return rx_err_t Error code indicating receive result
01284  * @retval k_rx_ok Frame received and decoded successfully, rx_sequence incremented
01285  * @retval k_rx_err_invalid_arg handle or frame is nullptr
01286  * @retval k_rx_err_invalid_state handle not initialized
01287  * @retval k_rx_err_timeout No data received within timeout (not an error if timeout=0)
01288  * @retval k_rx_err_crc_mismatch CRC validation failed (frame corrupted)
01289  * @retval k_rx_err_protocol_error Frame format invalid (bad sync, length, etc.)
01290  * @retval k_rx_err_sequence_error Unexpected sequence number (out-of-order)
01291  * @retval k_rx_err_hw SPI hardware error
01292  *
01293  * @pre handle must be non-NULL and initialized
01294  * @pre frame must be non-NULL and allocated
01295  * @pre RSPI hardware must be initialized and ready
01296  * @post On k_rx_ok: frame filled with decoded data, rx_sequence incremented
01297  * @post On timeout: frame unchanged, rx_sequence unchanged
01298  * @post On CRC error: frame may be partially filled, rx_sequence unchanged
01299  *
01300  * @note timeout=0 is recommended for polling loops (non-blocking)
01301  * @note CRC errors should trigger NACK and retransmission request
01302  * @note Sequence errors indicate lost frames or out-of-order delivery
01303  * @warning Must call rx_spi_comm_send_ack() or rx_spi_comm_send_nack() after receive
01304  *
01305  * @par Example - Non-Blocking Poll:
01306  * @code{.c}
01307  * rx_frame_t frame;
01308  *
01309  * // Poll with 0ms timeout (non-blocking)
01310  * rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);
01311  *
01312  * if (err == k_rx_ok) {
01313  *     // Frame received successfully
01314  *     handle_frame(&frame);
01315  *     rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);

```

```

01316 * } else if (err == k_rx_err_timeout) {
01317 * // No data available - this is normal
01318 * } else if (err == k_rx_err_crc_mismatch) {
01319 * // CRC error - request retransmission
01320 * rx_spi_comm_send_nack(&g_spi_handle, frame.header.sequence, k_frame_flag_none);
01321 * } else {
01322 * // Other error - log and continue
01323 * log_error("Receive error: %d", err);
01324 * }
01325 * @endcode
01326 *
01327 * @par Example - Blocking Receive with Timeout:
01328 * @code{.c}
01329 * rx_frame_t frame;
01330 *
01331 * // Wait up to 1 second for frame
01332 * rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 1000);
01333 *
01334 * if (err == k_rx_ok) {
01335 * // Frame received within timeout
01336 * process_frame(&frame);
01337 * rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
01338 * } else if (err == k_rx_err_timeout) {
01339 * // No frame received within 1 second - RPi5 may be idle
01340 * log_warn("No frame received within timeout");
01341 * }
01342 * @endcode
01343 *
01344 * @see rx_spi_comm_send() Send frames to RPi5
01345 * @see rx_spi_comm_send_ack() Send ACK for received frame
01346 * @see rx_spi_comm_send_nack() Send NACK on error
01347 * @see rx_spi_comm_data_available() Check if data waiting
01348 * @see rx_frame.h Frame structure definition
01349 *
01350 * @since Version 1.0.0
01351 */
01352 [[nodiscard]] rx_err_t
01353 rx_spi_comm_receive(rx_spi_comm_handle_t* handle, rx_frame_t* frame, uint32_t timeout_ms);
01354
01355 /**
01356 * @brief Check if data is available for reading (non-blocking query)
01357 *
01358 * @details
01359 * Queries SPI hardware to determine if data is available in receive buffer
01360 * without actually reading the data. Useful for polling loops to avoid
01361 * unnecessary receive attempts.
01362 *
01363 * **Use Case**: Optimize polling by checking availability before calling
01364 * rx_spi_comm_receive(), avoiding timeout overhead.
01365 *
01366 * **Algorithm**:
01367 * 1. Validate handle and available pointer
01368 * 2. Query RSPi hardware for RX buffer status
01369 * 3. Set *available = true if data waiting, false otherwise
01370 *
01371 * **Performance**: ~2 us (fast hardware register read)
01372 *
01373 * @param[in] handle Initialized SPI communication handle
01374 * - **Valid range**: Non-nullptr to initialized handle
01375 * - **Constraints**: Must be initialized via rx_spi_comm_init()
01376 *
01377 * @param[out] available Pointer to receive availability status
01378 * - **Valid range**: Non-nullptr to bool
01379 * - **Output values**: true (data available), false (no data)
01380 * - **On error**: Set to false
01381 *
01382 * @return rx_err_t Error code
01383 * @retval k_rx_ok Query successful, available flag updated
01384 * @retval k_rx_err_invalid_arg handle or available is nullptr
01385 * @retval k_rx_err_invalid_state handle not initialized
01386 *
01387 * @pre handle must be non-NULL and initialized
01388 * @pre available must be non-NULL
01389 * @post *available set to true or false based on hardware state
01390 * @post On error, *available set to false (safe default)
01391 *
01392 * @note This function does NOT read or modify receive buffer
01393 * @note Availability can change between check and actual receive
01394 * @note Fast operation - suitable for high-frequency polling
01395 *
01396 * @par Example - Optimized Polling Loop:
01397 * @code{.c}
01398 * void poll_spi(void)
01399 * {
01400 * bool available = false;
01401 * rx_err_t err = rx_spi_comm_data_available(&g_spi_handle, &available);
01402 *

```

```

01403 * if (err == k_rx_ok && available) {
01404 *     // Data is available - proceed with receive
01405 *     rx_frame_t frame;
01406 *     err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);
01407 *     if (err == k_rx_ok) {
01408 *         handle_frame(&frame);
01409 *     }
01410 * }
01411 * }
01412 * @endcode
01413 *
01414 * @see rx_spi_comm_receive() Receive frame
01415 *
01416 * @since Version 1.0.0
01417 */
01418 [[nodiscard]] rx_err_t rx_spi_comm_data_available(const rx_spi_comm_handle_t* handle,
01419                                                    bool* available);
01420
01421 /* Sequence management is now handled by the shared rx_session_state_t.
01422 * Use rx_session_next_tx(), rx_session_validate_rx(), rx_session_reset()
01423 * via the session pointer in the handle or config.
01424 * @see rx_session.h
01425 */
01426
01427 /*
=====
01428 * Control Frame API
01429 *
=====
01430 */
01431
01432 /**
01433 * @brief Register callbacks for PING and RESET control frames
01434 *
01435 * @details
01436 * Sets callback functions that are invoked when rx_spi_comm_receive()
01437 * encounters PING or RESET control frames. Control frames are handled
01438 * internally (auto PONG / auto RESET_ACK) and callbacks provide
01439 * notification to the application layer.
01440 *
01441 * @param[in,out] handle Initialized SPI communication handle
01442 * @param[in] on_ping_cb Callback for PING frames (may be NULL to disable)
01443 * @param[in] on_reset_cb Callback for RESET frames (may be NULL to disable)
01444 * @param[in] cb_ctx User context pointer passed to callbacks
01445 *
01446 * @return rx_err_t Error code
01447 * @retval k_rx_ok Callbacks registered successfully
01448 * @retval k_rx_err_invalid_arg handle is nullptr
01449 *
01450 * @pre handle must be non-NULL
01451 * @post on_ping_cb, on_reset_cb, control_cb_ctx stored in handle
01452 *
01453 * @note Callbacks are invoked from within rx_spi_comm_receive() context
01454 * @note Not thread-safe, set callbacks before starting receive loop
01455 *
01456 * @see rx_spi_comm_receive() Invokes callbacks on control frames
01457 *
01458 * @since Version 1.0.0
01459 */
01460 [[nodiscard]] rx_err_t
01461 rx_spi_comm_set_control_callbacks(rx_spi_comm_handle_t* handle,
01462                                  void (*on_ping_cb)(const rx_frame_t* frame, void* ctx),
01463                                  void (*on_reset_cb)(const rx_frame_t* frame, void* ctx),
01464                                  void* cb_ctx);
01465
01466 /**
01467 * @brief Send PONG frame with payload (response to PING)
01468 *
01469 * @details
01470 * Constructs and transmits a PONG frame echoing the provided payload.
01471 * Used internally by rx_spi_comm_receive() for auto-PONG, but also
01472 * available for manual PONG sending.
01473 *
01474 * @param[in,out] handle Initialized SPI communication handle
01475 * @param[in] payload Payload data to echo (may be NULL if payload_len is 0)
01476 * @param[in] payload_len Length of payload in bytes
01477 *
01478 * @return rx_err_t Error code
01479 * @retval k_rx_ok PONG frame sent successfully
01480 * @retval k_rx_err_invalid_arg handle is nullptr
01481 * @retval k_rx_err_invalid_state handle not initialized
01482 * @retval (other) Errors propagated from frame creation or SPI transfer
01483 *
01484 * @pre handle must be non-NULL and initialized
01485 * @post PONG frame transmitted with echoed payload
01486 *
01487 * @note Gets next TX sequence from shared session state for the PONG frame

```

```

01488 *
01489 * @see rx_frame_create_pong() Frame creation for PONG
01490 * @see rx_spi_comm_receive() Auto-sends PONG on PING reception
01491 *
01492 * @since Version 1.0.0
01493 */
01494 [[nodiscard]] rx_err_t
01495 rx_spi_comm_send_pong(rx_spi_comm_handle_t* handle, const uint8_t* payload, uint32_t payload_len);
01496
01497 /**
01498 * @brief Send RESET_ACK frame (response to RESET)
01499 *
01500 * @details
01501 * Constructs and transmits a RESET_ACK frame. Used internally by
01502 * rx_spi_comm_receive() for auto-RESET_ACK, but also available
01503 * for manual sending.
01504 *
01505 * @param[in,out] handle Initialized SPI communication handle
01506 *
01507 * @return rx_err_t Error code
01508 * @retval k_rx_ok RESET_ACK frame sent successfully
01509 * @retval k_rx_err_invalid_arg handle is nullptr
01510 * @retval k_rx_err_invalid_state handle not initialized
01511 * @retval (other) Errors propagated from frame creation or SPI transfer
01512 *
01513 * @pre handle must be non-NULL and initialized
01514 * @post RESET_ACK frame transmitted
01515 *
01516 * @note Does NOT reset sequence numbers (caller must do that if desired)
01517 *
01518 * @see rx_frame_create_reset_ack() Frame creation for RESET_ACK
01519 * @see rx_spi_comm_receive() Auto-sends RESET_ACK on RESET reception
01520 *
01521 * @since Version 1.0.0
01522 */
01523 [[nodiscard]] rx_err_t rx_spi_comm_send_reset_ack(rx_spi_comm_handle_t* handle);
01524
01525 /**
01526 * @brief Check and process automatic retransmissions
01527 *
01528 * @details
01529 * Checks whether an ACK timeout has elapsed for the most recent sent frame.
01530 * If auto_retransmit is enabled and a retransmit is pending, resends the
01531 * buffered frame (up to max_retries with exponential backoff).
01532 *
01533 * @param[in,out] handle Initialized SPI communication handle
01534 * @param[in] current_time_ms Current system time in milliseconds
01535 *
01536 * @return rx_err_t Error code
01537 * @retval k_rx_ok No action needed or retransmit sent
01538 * @retval k_rx_err_invalid_arg handle is nullptr
01539 * @retval k_rx_err_invalid_state handle not initialized
01540 *
01541 * @pre handle must be non-NULL and initialized
01542 * @post retry_count incremented on retransmit, pending cleared on limit
01543 *
01544 * @note Safe to call when auto_retransmit is disabled (no-op)
01545 *
01546 * @see rx_spi_comm_set_auto_retransmit() Enable/disable retransmission
01547 *
01548 * @since Version 1.0.0
01549 */
01550 [[nodiscard]] rx_err_t rx_spi_comm_process_retransmits(rx_spi_comm_handle_t* handle,
01551                                                         uint32_t current_time_ms);
01552
01553 /**
01554 * @brief Enable or disable automatic retransmission at runtime
01555 *
01556 * @details
01557 * Allows the RPi5 to configure retransmission behavior via protobuf command.
01558 * When disabling, any pending retry state is cleared.
01559 *
01560 * @param[in,out] handle Initialized SPI communication handle
01561 * @param[in] enabled true to enable, false to disable
01562 * @param[in] config Retransmit configuration (may be NULL to keep current or use defaults)
01563 *
01564 * @return rx_err_t Error code
01565 * @retval k_rx_ok Configuration applied successfully
01566 * @retval k_rx_err_invalid_arg handle is nullptr
01567 * @retval k_rx_err_invalid_state handle not initialized
01568 *
01569 * @pre handle must be non-NULL and initialized
01570 * @post auto_retransmit flag and config updated
01571 * @post If disabling, pending retry state cleared
01572 *
01573 * @since Version 1.0.0
01574 */

```

```

01575 [[nodiscard]] rx_err_t
01576 rx_spi_comm_set_auto_retransmit(rx_spi_comm_handle_t* handle,
01577                                bool enabled,
01578                                const rx_spi_comm_retransmit_config_t* config);
01579
01580 /**
01581  * @brief Register ACK/NACK notification callbacks for retransmission
01582  *
01583  * @details
01584  * Sets optional callbacks invoked when ACK or NACK control frames are
01585  * consumed during rx_spi_comm_receive(). Useful for telemetry and diagnostics.
01586  *
01587  * @param[in,out] handle Initialized SPI communication handle
01588  * @param[in] on_ack_cb Callback for ACK frames (may be NULL to disable)
01589  * @param[in] on_nack_cb Callback for NACK frames (may be NULL to disable)
01590  * @param[in] ctx User context pointer passed to callbacks
01591  *
01592  * @return rx_err_t Error code
01593  * @retval k_rx_ok Callbacks registered
01594  * @retval k_rx_err_invalid_arg handle is nullptr
01595  *
01596  * @pre handle must be non-NULL
01597  * @post Callback pointers stored in handle
01598  *
01599  * @note Callbacks invoked from within rx_spi_comm_receive() context
01600  *
01601  * @since Version 1.0.0
01602  */
01603 [[nodiscard]] rx_err_t
01604 rx_spi_comm_set_retransmit_callbacks(rx_spi_comm_handle_t* handle,
01605                                     void (*on_ack_cb)(uint16_t sequence, void* ctx),
01606                                     void (*on_nack_cb)(uint16_t sequence, void* ctx),
01607                                     void* ctx);
01608
01609 #ifdef __cplusplus
01610 }
01611 #endif

```

4.3 libs/rx_spi_comm/src/rx_spi_comm.c File Reference

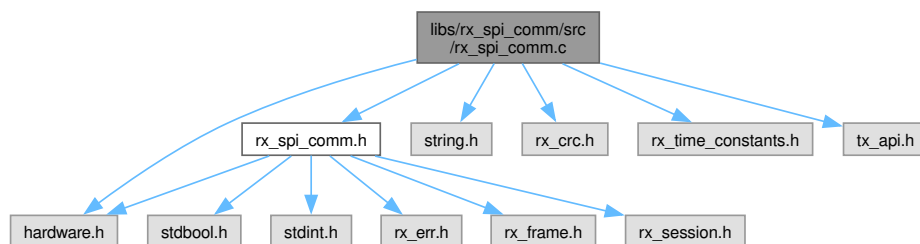
High-Level SPI Frame Protocol Communication Layer Implementation.

```

#include "rx_spi_comm.h"
#include <string.h>
#include "hardware.h"
#include "rx_crc.h"
#include "rx_time_constants.h"
#include "tx_api.h"

```

Include dependency graph for rx_spi_comm.c:



Enumerations

- enum `backoff_limit_t` : `uint8_t { k_max_backoff_shift = 31 }`
Maximum left-shift for exponential backoff (prevents UB on `uint32_t`).
- enum `frame_header_offset_t` : `uint8_t { k_frame_offset_init = 0 , k_hdr_sync_low = 1 , k_hdr_sync_high = 2 , k_hdr_seq_low = 3 , k_hdr_seq_high = 4 , k_hdr_len_low = 5 , k_hdr_len_high = 6 , k_hdr_flags = 7 }`

- Byte offsets within frame header buffer (SYNC + header fields).
- enum `spi_crc32_seed_t` : `uint32_t` { `k_spi_crc32_seed_initial` = 0U }
CRC-32 seed value (pre-initialization before `rx_crc32_ieee` writes it).
- enum `poll_timing_t` : `uint32_t` { `k_poll_sleep_ticks` = 1 }
Sleep duration for polling loop.
- enum `ack_wait_t` : `uint16_t` { `k_ack_wait_timeout_ms` = 50 }
ACK/ready wait timing constants.
- enum `receive_bounds_t` : `uint8_t` { `k_max_control_frames_per_receive` = 16 }
Receive loop bounds (NASA Power of 10 Rule 2).
- enum `polling_limits_t` : `uint16_t` { `k_max_poll_iterations` = 1000 }
Polling loop iteration limits.
- enum `ctrl_dispatch_result_t` : `uint8_t` { `k_ctrl_not_handled` = 0 , `k_ctrl_consumed` = 1 , `k_ctrl_error` = 2 }
Result of control-frame dispatch within the receive loop.

Functions

- static `rx_err_t` `internal_retransmit_frame` (`rx_spi_comm_handle_t` *handle)
Retransmit the buffered frame via SPI.
- RX_STATIC_TESTABLE `rx_err_t` `internal_decode_header` (const `uint8_t` *data, const `uint32_t` data_len, `rx_frame_t` *frame, `uint32_t` *offset_out)
Decode and validate SPI frame header from raw wire buffer.
- RX_STATIC_TESTABLE `rx_err_t` `internal_verify_crc` (const `uint8_t` *data, `uint32_t` offset, `uint32_t` *crc_out)
Verify CRC-32 for a received SPI frame.
- static `rx_err_t` `internal_wait_for_ack` (const `rx_spi_comm_handle_t` *handle, `uint32_t` timeout_ms)
Wait for RPi5 controller ready signal before transmitting data.
- RX_STATIC_TESTABLE `rx_err_t` `internal_spi_transfer` (`rx_spi_comm_handle_t` *handle, const `uint8_t` *tx_data, const `uint32_t` tx_len, `uint8_t` *rx_data, const `uint32_t` rx_len)
Perform raw SPI full-duplex transfer using handle's staging buffers.
- `rx_err_t` `rx_spi_comm_init` (`rx_spi_comm_handle_t` *handle, const `rx_spi_comm_config_t` *config)
Initialize SPI communication layer and allocate resources.
- `rx_err_t` `rx_spi_comm_deinit` (`rx_spi_comm_handle_t` *handle)
Deinitialize SPI communication layer.
- RX_STATIC_TESTABLE `rx_err_t` `internal_build_frame` (const `rx_spi_comm_handle_t` *handle, const `uint16_t` sequence, const `rx_frame_type_t` type, const `uint8_t` flags, const `uint8_t` *payload, const `uint32_t` payload_len, `rx_frame_t` *frame)
Build frame structure from application payload data.
- static void `internal_buffer_for_retransmit` (`rx_spi_comm_handle_t` *handle, const `uint8_t` *wire_data, const `uint32_t` wire_len, const `uint16_t` sequence)
Send data frame to RPi5 controller via SPI.
- `rx_err_t` `rx_spi_comm_send` (`rx_spi_comm_handle_t` *handle, const `rx_frame_type_t` type, const `uint8_t` flags, const `uint8_t` *payload, const `uint32_t` payload_len)
Send frame on SPI with automatic sequencing and CRC.
- `rx_err_t` `rx_spi_comm_send_ack` (`rx_spi_comm_handle_t` *handle, const `uint16_t` sequence)
Send ACK frame for specified sequence number.
- `rx_err_t` `rx_spi_comm_send_nack` (`rx_spi_comm_handle_t` *handle, const `uint16_t` sequence, `uint8_t` flags)
Send NACK frame for specified sequence number.

- static rx_err_t [internal_wait_for_data](#) (const rx_spi_comm_handle_t *handle, const uint32_t timeout_ms)
Wait for data to be available on SPI channel.
- static rx_err_t [internal_read_frame_header](#) (rx_spi_comm_handle_t *handle, uint8_t *header_buf, uint16_t *payload_len)
Read and validate frame header from SPI.
- static rx_err_t [internal_decode_frame](#) (const rx_spi_comm_handle_t *handle, rx_frame_t *frame, const uint32_t total_size)
Decode and verify a received SPI frame.
- static void [internal_dispatch_ack_nack](#) (rx_spi_comm_handle_t *handle, const rx_frame_t *frame)
Handle ACK or NACK frame for the retransmit subsystem.
- static rx_err_t [internal_dispatch_control_frame](#) (rx_spi_comm_handle_t *handle, rx_frame_t *frame, ctrl_dispatch_result_t *result)
Dispatch control frames (PING, RESET, ACK/NACK) during receive.
- static rx_err_t [internal_read_and_decode_frame](#) (rx_spi_comm_handle_t *handle, rx_frame_t *frame, const uint32_t timeout_ms)
Receive data frame from RPi5 controller via SPI (blocking with timeout).
- static rx_err_t [internal_validate_rx_sequence](#) (rx_spi_comm_handle_t *handle, const rx_frame_t *frame)
Validate received frame sequence and emit NACK on rejection.
- rx_err_t [rx_spi_comm_receive](#) (rx_spi_comm_handle_t *handle, rx_frame_t *frame, const uint32_t timeout_ms)
Receive and decode frame from RPi5 with CRC validation.
- rx_err_t [rx_spi_comm_data_available](#) (const rx_spi_comm_handle_t *handle, bool *available)
Check if data is available to receive on SPI.
- rx_err_t [rx_spi_comm_set_control_callbacks](#) (rx_spi_comm_handle_t *handle, void(*on_ping_cb)(const rx_frame_t *frame, void *ctx), void(*on_reset_cb)(const rx_frame_t *frame, void *ctx), void *cb_ctx)
Register callbacks for PING and RESET control frames.
- rx_err_t [rx_spi_comm_send_pong](#) (rx_spi_comm_handle_t *handle, const uint8_t *payload, uint32_t payload_len)
Send PONG frame with echoed payload.
- rx_err_t [rx_spi_comm_send_reset_ack](#) (rx_spi_comm_handle_t *handle)
Send RESET_ACK frame confirming a reset request.
- rx_err_t [rx_spi_comm_process_retransmits](#) (rx_spi_comm_handle_t *handle, const uint32_t current_time_ms)
Process pending retransmissions based on timeout.
- rx_err_t [rx_spi_comm_set_auto_retransmit](#) (rx_spi_comm_handle_t *handle, const bool enabled, const rx_spi_comm_retransmit_config_t *config)
Enable or disable automatic retransmission at runtime.
- rx_err_t [rx_spi_comm_set_retransmit_callbacks](#) (rx_spi_comm_handle_t *handle, void(*on_ack_cb)(uint16_t sequence, void *ctx), void(*on_nack_cb)(uint16_t sequence, void *ctx), void *ctx)
Register ACK/NACK notification callbacks.

Variables

- static const char [s_tag](#) [] = "rx_spi_comm"
- static const rx_spi_comm_handle_t [s_zero_handle](#) = {}
Zero-initialized SPI comm handle template for stack-safe initialization.

4.3.1 Detailed Description

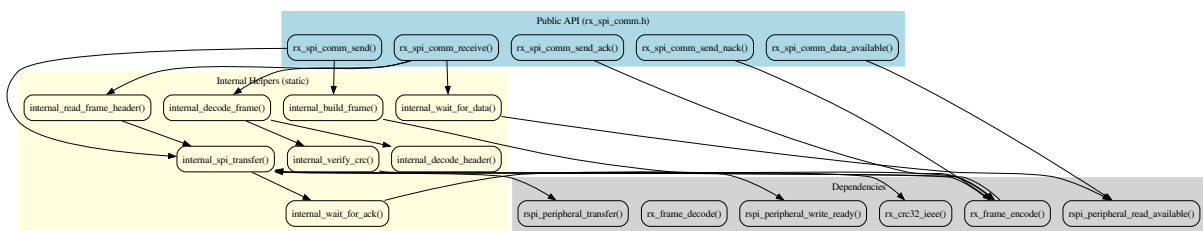
High-Level SPI Frame Protocol Communication Layer Implementation.

4.3.1.1 Overview

Production-quality implementation of SPI frame-based communication protocol for reliable data exchange between RX72N MCU (SPI peripheral) and Raspberry Pi 5 (SPI controller). Integrates frame encoding/↔ decoding (`rx_frame`), CRC-32 validation (`rx_crc`), and RSPI peripheral hardware abstraction (`rspi HAL`).

This module provides the complete implementation of the API defined in `rx_spi_comm.h`, handling all aspects of frame transmission, reception, flow control, and error recovery.

4.3.1.2 Implementation Architecture



4.3.1.3 Data Flow - Transmission

```

Application
|
| rx_spi_comm_send(payload, len)
|
v
internal_build_frame()      # Populate rx_frame_t with header + payload
|
v
rx_frame_encode()           # Encode to wire format (sync, header, CRC)
|
v
internal_spi_transfer()
|
| +---> internal_wait_for_ack() # Wait for RPi5 ready signal (50ms timeout)
|
| +---> rspi_peripheral_transfer() # DMA/IRQ-based SPI transfer
|
v
TX sequence++              # Increment for next frame
  
```

4.3.1.4 Data Flow - Reception

```

Application
|
| rx_spi_comm_receive(timeout_ms)
|
v
internal_wait_for_data()    # Poll RSPI for data (with ThreadX sleep)
|
v
internal_read_frame_header() # Read 10 bytes (SYNC + header)
|
| Validate sync word (0xAA55)
| Extract payload length
|
v
internal_spi_transfer()     # Read remaining (payload + CRC-32)
|
v
internal_decode_frame()
|
| +---> internal_decode_header() # Parse header fields
| +---> memcpy payload
| +---> internal_verify_crc()    # Validate CRC-32
|
v
RX sequence = frame.seq + 1 # Update expected next sequence
  
```

4.3.1.5 Performance Characteristics

Operation	Avg Time (us)	Worst Case (us)	Notes
rx_spi_comm_send()	150-500	1000	Depends on payload (0-1024 bytes)
rx_spi_comm_receive()	200-600	2000	Includes wait + decode + CRC
internal_build_frame()	5-20	50	Memcpy payload (0-1024 bytes)
internal_wait_for_ack()	10-100	50000	Polls every 10ms, 50ms timeout
internal_spi_transfer()	80-900	1500	SPI @ 10 MHz, 10-1038 bytes
internal_decode_header()	15	30	Byte-level parsing
internal_verify_crc()	40-500	800	CRC-32 over 10-1038 bytes
internal_wait_for_data()	10	timeout_ms*1000	Polls every 10ms until timeout

Throughput Analysis (10 MHz SPI, no FEC):

- Max payload: 1024 bytes -> 1038 bytes on wire -> 830 us SPI time
- Frame overhead: ~100 us encode + ~150 us decode + ~250 us CRC
- Total latency: ~1330 us per 1024-byte frame = ~770 KB/s effective
- Theoretical max: 10 MHz / 8 = 1.25 MB/s (raw SPI)
- Efficiency: ~62% (overhead from framing, CRC, ACK waits)

4.3.1.6 Memory Usage

Stack Usage per Function Call:

- [rx_spi_comm_send\(\)](#): ~310 bytes (rx_frame_t + locals; wire scratch is in handle)
- [rx_spi_comm_receive\(\)](#): ~350 bytes (rx_frame_t + header_buf[10])
- [internal_spi_transfer\(\)](#): ~20 bytes (local vars, no buffers)
- [internal_decode_frame\(\)](#): ~10 bytes (offset, err)

Total Handle Footprint: ~5160 bytes (handle now owns tx_encode_buffer in addition to retry_buffer; see [rx_spi_comm_handle_t](#) in .h)

4.3.1.7 Thread Safety

Function	Thread Safe?	Notes
rx_spi_comm_init()	[FAIL] No	Call once during init, not from multiple threads
rx_spi_comm_send()	[FAIL] No	Use external mutex if multiple threads send
rx_spi_comm_receive()	[FAIL] No	Use external mutex if multiple threads receive
rx_spi_comm_send_ack()	[FAIL] No	Same as send (shares TX sequence)
rx_spi_comm_send_nack()	[FAIL] No	Same as send (shares TX sequence)

Function	Thread Safe?	Notes
rx_spi_comm_data_available()	[PASS] Yes	Read-only hardware poll (safe if HAL is safe)

Typical pattern: Dedicate one ThreadX task to SPI communication, no locking needed.

4.3.1.8 Hardware Requirements

RX72N RSPI Configuration (handled by rspi HAL):

- Mode: Peripheral
- Clock: External (provided by RPi5 controller)
- Data bits: 8
- Bit order: MSB-first
- Clock polarity: CPOL=0 (idle low)
- Clock phase: CPHA=0 (sample on first edge)
- Max frequency: 10 MHz (reliable with 1m cable)

Pin Connections (RSPI0 example):

- CS: PD0 (chip select, active low)
- CLK: PD1 (clock from RPi5)
- CIPO: PD2 (Controller In Peripheral Out, RX72N -> RPi5)
- COPI: PD3 (Controller Out Peripheral In, RPi5 -> RX72N)

4.3.1.9 Error Handling Strategy

Recoverable Errors (return error code, caller retries):

- `k_rx_err_timeout`: No data available, ACK timeout, normal in polling
- `k_rx_err_crc_mismatch`: Frame corrupted, send NACK, expect retransmit
- `k_rx_err_protocol_error`: Invalid sync word, resync by discarding

Fatal Errors (return error code, caller should reinitialize):

- `k_rx_err_invalid_arg`: nullptr pointer, programming error
- `k_rx_err_invalid_state`: Not initialized, must call [rx_spi_comm_init\(\)](#)
- `k_rx_err_invalid_size`: Payload > 1024 bytes, violates protocol spec

Hardware Errors (propagated from RSPI HAL):

- Overrun, underrun, mode fault -> logged and returned to caller

4.3.1.10 Integration Example

```

#include "rx_spi_comm.h"
#include "tx_api.h"

// Allocate handle (4120 bytes, typically static or in task stack)
static rx_spi_comm_handle_t g_spi_handle;

// ThreadX task for SPI communication
void spi_comm_task_entry(ULONG thread_input)
{
    (void)thread_input;

    // Initialize SPI communication (default channel 0, no FEC)
    rx_err_t err = rx_spi_comm_init(&g_spi_handle, nullptr);
    if (err != k_rx_ok) {
        rx_log_error("spi_task", "Failed to init SPI comm");
        return;
    }

    while (1) {
        // Receive frame with 100ms timeout
        rx_frame_t rx_frame;
        err = rx_spi_comm_receive(&g_spi_handle, &rx_frame, 100);

        if (err == k_rx_ok) {
            // Process received frame
            if (rx_frame.header.type == k_frame_type_data) {
                // Handle data frame
                process_data_frame(&rx_frame);

                // Send ACK
                rx_spi_comm_send_ack(&g_spi_handle, rx_frame.header.sequence);
            }
            else if (err == k_rx_err_timeout) {
                // No data, normal in polling loop
                tx_thread_sleep(1); // Sleep 10ms
            }
            else if (err == k_rx_err_crc_mismatch) {
                // Corrupted frame, send NACK
                rx_spi_comm_send_nack(&g_spi_handle, rx_frame.header.sequence, 0);
            }
            else {
                // Fatal error, log and reinitialize
                rx_log_error("spi_task", "SPI receive fatal error");
                rx_spi_comm_deinit(&g_spi_handle);
                rx_spi_comm_init(&g_spi_handle, nullptr);
            }
        }

        // Check if we need to send telemetry
        if (telemetry_ready) {
            uint8_t payload[32];
            uint32_t len = build_telemetry_frame(payload, sizeof(payload));

            err = rx_spi_comm_send(&g_spi_handle, k_frame_type_data, 0, payload, len);
            if (err != k_rx_ok) {
                rx_log_error("spi_task", "Failed to send telemetry");
            }
        }
    }
}

```

4.3.1.11 NASA Power of 10 Compliance

Rule	Status	Implementation Notes
1. Simple control flow	[PASS] Pass	No goto/setjmp/recursion, structured if/while/for only
2. Fixed loop bounds	[PASS] Pass	All loops bounded by enum constants (k_max_poll_↵_iterations)
3. No dynamic memory	[PASS] Pass	Zero malloc/free, all buffers in handle (stack/static)
4. Short functions	[PASS] Pass	All <60 lines, single responsibility
5. Assertions	[PASS] Pass	Minimum 2 checks per function (RX_CHECK_↵_NULL_PTR + state)

Rule	Status	Implementation Notes
6. Small scope	[PASS] Pass	Variables declared near use, file-static for module data
7. Check returns	[PASS] Pass	All <code>rx_err_t</code> checked via <code>RX_RETURN_ON_ERROR</code> or explicit if
8. Limited preprocessor	[PASS] Pass	C23 typed enums for constants, macros only for conditionals
9. Restrict pointers	[WARN] Deviation	Function pointers in HAL interface (DIP, testability)
10. Compiler warnings	[PASS] Pass	-Wall -Wextra -Werror, zero warnings

4.3.1.12 SOLID Principles

Principle	Implementation
Single Responsibility	Module handles ONLY SPI frame transport, delegates to <code>rx_frame</code> (encoding), <code>rx_crc</code> (validation), <code>rspi</code> (hardware)
Open/Closed	Configurable via <code>rx_spi_comm_config_t</code> (channel, FEC), extendable without modifying code
Liskov Substitution	Not applicable (no inheritance in C), but handle is opaque and consistent
Interface Segregation	9 focused functions, caller uses only what's needed (send vs receive vs query)
Dependency Inversion	Depends on abstractions: <code>rx_frame</code> interface, <code>rspi</code> HAL (not direct register access)

See also

[rx_spi_comm.h](#) Public API definition
[rx_frame.h](#) Frame encoding/decoding layer
[rx_crc.h](#) CRC-32 validation
[rspi.h](#) RSPI peripheral hardware abstraction

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Date

2026-01-01

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Definition in file [rx_spi_comm.c](#).

4.3.2 Enumeration Type Documentation

4.3.2.1 ack'wait't

enum [ack_wait_t](#) : uint16_t

ACK/ready wait timing constants.

Enumerator

k_ack_wait_timeout_ms	Abort if host doesn't ACK within 50 ms
---------------------------------------	--

Definition at line 568 of file [rx_spi_comm.c](#).

```
00568         : uint16_t {
00569     k_ack_wait_timeout_ms = 50, /**< Abort if host doesn't ACK within 50 ms */
00570 } ack_wait_t;
```

4.3.2.2 backoff'limit't

enum [backoff_limit_t](#) : uint8_t

Maximum left-shift for exponential backoff (prevents UB on uint32_t).

Enumerator

k_max_backoff_shift	Clamp retry_count before shifting to avoid UB
-------------------------------------	---

Definition at line 320 of file [rx_spi_comm.c](#).

```
00320         : uint8_t {
00321     k_max_backoff_shift = 31, /**< Clamp retry_count before shifting to avoid UB */
00322 } backoff_limit_t;
```

4.3.2.3 ctrl'dispatch'result't

enum [ctrl_dispatch_result_t](#) : uint8_t

Result of control-frame dispatch within the receive loop.

Enumerator

k_ctrl_not_handled	Frame is not a control frame; return to caller
k_ctrl_consumed	Control frame consumed; continue receive loop
k_ctrl_error	Error occurred during control-frame handling

Definition at line 1611 of file [rx_spi_comm.c](#).

```
01611         : uint8_t {
01612     k_ctrl_not_handled = 0, /**< Frame is not a control frame; return to caller */
01613     k_ctrl_consumed    = 1, /**< Control frame consumed; continue receive loop */
01614     k_ctrl_error       = 2, /**< Error occurred during control-frame handling */
01615 } ctrl_dispatch_result_t;
```

4.3.2.4 frame_header_offset_t

```
enum frame_header_offset_t : uint8_t
```

Byte offsets within frame header buffer (SYNC + header fields).

Enumerator

k_frame_offset_init	Initial frame parse offset (start of buffer)
k_hdr_sync_low	SYNC word low byte (LE: LSB first)
k_hdr_sync_high	SYNC word high byte (LE: MSB second)
k_hdr_seq_low	Sequence number low byte (LE)
k_hdr_seq_high	Sequence number high byte (LE)
k_hdr_len_low	Payload length low byte (LE)
k_hdr_len_high	Payload length high byte (LE)
k_hdr_type	Frame type
k_hdr_flags	Frame flags

Definition at line 333 of file [rx_spi_comm.c](#).

```
00333         : uint8_t {
00334   k_frame_offset_init = 0, /**< Initial frame parse offset (start of buffer) */
00335   k_hdr_sync_low      = 0, /**< SYNC word low byte (LE: LSB first) */
00336   k_hdr_sync_high     = 1, /**< SYNC word high byte (LE: MSB second) */
00337   k_hdr_seq_low       = 2, /**< Sequence number low byte (LE) */
00338   k_hdr_seq_high      = 3, /**< Sequence number high byte (LE) */
00339   k_hdr_len_low       = 4, /**< Payload length low byte (LE) */
00340   k_hdr_len_high      = 5, /**< Payload length high byte (LE) */
00341   k_hdr_type          = 6, /**< Frame type */
00342   k_hdr_flags         = 7, /**< Frame flags */
00343 } frame_header_offset_t;
```

4.3.2.5 poll_timing_t

```
enum poll_timing_t : uint32_t
```

Sleep duration for polling loop.

Enumerator

k_poll_sleep_ticks	1 ThreadX tick between polls
--------------------	------------------------------

Definition at line 562 of file [rx_spi_comm.c](#).

```
00562         : uint32_t {
00563   k_poll_sleep_ticks = 1, /**< 1 ThreadX tick between polls */
00564 } poll_timing_t;
```

4.3.2.6 polling_limits_t

```
enum polling_limits_t : uint16_t
```

Polling loop iteration limits.

Enumerator

k_max_poll_iterations	Maximum polling iterations (safety bound)
-----------------------	---

Definition at line 578 of file [rx_spi_comm.c](#).

```
00578         : uint16_t {
00579     k_max_poll_iterations = 1000, /**< Maximum polling iterations (safety bound) */
00580 } polling_limits_t;
```

4.3.2.7 receive_bounds_t

```
enum receive\_bounds\_t : uint8_t
```

Receive loop bounds (NASA Power of 10 Rule 2).

Enumerator

k_max_control_frames_per_receive	Max control frames before returning error
----------------------------------	---

Definition at line 573 of file [rx_spi_comm.c](#).

```
00573         : uint8_t {
00574     k_max_control_frames_per_receive = 16, /**< Max control frames before returning error */
00575 } receive_bounds_t;
```

4.3.2.8 spi_crc32_seed_t

```
enum spi\_crc32\_seed\_t : uint32_t
```

CRC-32 seed value (pre-initialization before rx_crc32_ieee writes it).

Enumerator

k_spi_crc32_seed_initial	Initial value for CRC-32 output variable
--------------------------	--

Definition at line 346 of file [rx_spi_comm.c](#).

```
00346         : uint32_t {
00347     k_spi_crc32_seed_initial = 0U, /**< Initial value for CRC-32 output variable */
00348 } spi_crc32_seed_t;
```

4.3.3 Function Documentation

4.3.3.1 internal_buffer_for_retransmit()

```
void internal_buffer_for_retransmit (
    rx\_spi\_comm\_handle\_t * handle,
    const uint8_t * wire_data,
    const uint32_t wire_len,
    const uint16_t sequence) [static]
```

Send data frame to RPi5 controller via SPI.

Transmits application payload to RPi5 by building frame structure, encoding to wire format with CRC-32, waiting for host ready signal, performing SPI transfer, and incrementing TX sequence counter. This is the primary transmit function for all data frames.

Algorithm:

Note

This function blocks until transfer complete or timeout (~1-50 ms)
 TX sequence auto-increments, no manual management needed

Warning

Not thread-safe - use external mutex if multiple threads send
 Do not modify payload buffer during call (memcpy in progress)

Performance:

- Execution time: 150-500 us @ 10 MHz SPI + 0-50 ms host ACK wait
- 10-byte payload: ~150 us (10 us build + 20 us encode + 120 us SPI)
- 100-byte payload: ~300 us (15 us build + 100 us encode + 185 us SPI)
- 1024-byte payload: ~800 us (20 us build + 250 us encode + 530 us SPI)
- Host ACK wait: Add 0-50 ms if RPi5 not immediately ready

Example - Send Telemetry Data:

```
// Pack telemetry into payload
uint8_t telemetry[32];
telemetry[0] = temperature_celsius & 0xFF;
telemetry[1] = (temperature_celsius » 8) & 0xFF;
// ... pack more fields ...

// Send to RPi5
rx_err_t err = rx_spi_comm_send(&g_spi_handle,
                                k_frame_type_telemetry,
                                0, // No special flags
                                telemetry,
                                sizeof(telemetry));

if (err == k_rx_ok) {
    rx_log_debug("spi", "Telemetry sent, seq=%d", g_spi_handle.tx_sequence - 1);
} else if (err == k_rx_err_timeout) {
    rx_log_warn("spi", "RPi5 not responding");
}
```

Example - Send Command with Retry:

```
uint8_t cmd[] = { 0x01, 0x02, 0x03 }; // Motor control command

for (uint8_t retry = 0; retry < 3; retry++) {
    rx_err_t err = rx_spi_comm_send(&g_spi_handle,
                                    k_frame_type_command,
                                    0,
                                    cmd,
                                    sizeof(cmd));

    if (err == k_rx_ok) {
        break; // Success
    } else if (err == k_rx_err_timeout) {
        rx_log_warn("spi", "Timeout, retry %d/3", retry + 1);
        tx_thread_sleep(10); // Wait 100ms before retry
    } else {
        // Fatal error, don't retry
        rx_log_error("spi", "Send failed: %d", err);
        break;
    }
}
```

Example - Zero-Length Frame (Ping):

```
// Send empty frame to test link
rx_err_t err = rx_spi_comm_send(&g_spi_handle,
                                k_frame_type_data,
                                0,
                                nullptr, // No payload
                                0);      // Zero length
// Result: 14-byte frame on wire (header + CRC, no payload)
```

See also

- [rx_spi_comm_send_ack\(\)](#) Send acknowledgment frame
- [rx_spi_comm_send_nack\(\)](#) Send negative acknowledgment
- [rx_spi_comm_receive\(\)](#) Receive frames from RPi5
- [internal_build_frame\(\)](#) Frame building implementation
- [internal_spi_transfer\(\)](#) SPI transfer with flow control

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Buffer a transmitted frame for potential retransmission

Precondition

handle->auto_retransmit == true

Postcondition

retry_buffer contains copy of wire_data; retry_pending set true

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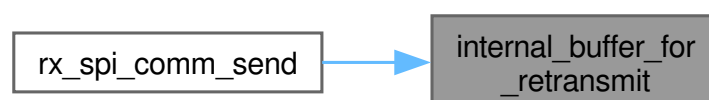
Definition at line 1351 of file [rx_spi_comm.c](#).

```
01355 {
01356     for (uint32_t i = 0; i < wire_len; i++) {
01357         handle->retry_buffer[i] = wire_data[i];
01358     }
01359     handle->retry_wire_len = wire_len;
01360     handle->retry_sequence = sequence;
01361     handle->retry_count = 0;
01362     handle->retry_pending = true;
01363     #ifdef __RX__
01364     handle->retry_send_time_ms = (uint32_t)((uint64_t)tx_time_get() * k_threadx_ms_per_tick);
01365     #else
01366     handle->retry_send_time_ms = 0;
01367     #endif
01368 }
```

References [rx_spi_comm_handle_t::retry_buffer](#), [rx_spi_comm_handle_t::retry_count](#), [rx_spi_comm_handle_t::ret](#), [rx_spi_comm_handle_t::retry_send_time_ms](#), [rx_spi_comm_handle_t::retry_sequence](#), and [rx_spi_comm_handle_t::retry_wire_len](#).

Referenced by [rx_spi_comm_send\(\)](#).

Here is the caller graph for this function:



4.3.3.2 internal_build_frame()

```

RX_STATIC_TESTABLE rx_err_t internal_build_frame (
    const rx_spi_comm_handle_t * handle,
    const uint16_t sequence,
    const rx_frame_type_t type,
    const uint8_t flags,
    const uint8_t * payload,
    const uint32_t payload_len,
    rx_frame_t * frame)

```

Build frame structure from application payload data.

Populates rx_frame_t structure with header fields (sequence, length, type, flags) and payload data, preparing it for encoding via rx_frame_encode(). Automatically sets sequence number from handle's TX counter and applies FEC flag if enabled in configuration.

Algorithm:

1. Validate inputs: Check handle, frame, payload pointer consistency
2. Validate payload size: Ensure payload_len <= 1024 bytes (protocol limit)
3. Zero frame: Clear rx_frame_t to ensure padding bytes are zero
4. Set sequence: Use handle->tx_sequence (auto-incremented by sender)
5. Set length: payload_len (0-1024)
6. Set type: Frame type (data, ACK, NACK, telemetry, etc.)
7. Set flags: User-provided flags
8. Copy payload: memcpy payload_len bytes to frame->payload if present
9. Apply FEC flag: Set k_frame_flag_fec_enabled if handle->fec_enabled

Frame Types Supported:

- k_frame_type_data: General data payload (0-1024 bytes)
- k_frame_type_ack: Acknowledgment (0 bytes payload, handled separately)
- k_frame_type_nack: Negative acknowledgment (0 bytes, handled separately)
- k_frame_type_telemetry: Sensor/status data
- k_frame_type_command: Motor control commands

Precondition

handle must be initialized via rx_spi_comm_init()

If payload != nullptr, payload_len must be > 0 and <= 1024

If payload == nullptr, payload_len must be 0

Postcondition

- On success, frame->header populated with sequence, length, type, flags
- On success, frame->payload contains payload_len bytes from payload
- On success, frame ready for rx_frame_encode()
- On failure, frame contents are undefined

Note

- This function does NOT increment handle->tx_sequence (caller does that)
- This function does NOT compute CRC (rx_frame_encode() does that)

Warning

- Do not modify frame after build and before encode (breaks CRC)

Performance:

- Execution time: 5-20 us @ 240 MHz (depends on payload_len memcpy)

Example - Data Frame:

```
uint8_t telemetry[32] = { ... }; // Sensor data
rx_frame_t frame;

rx_err_t err = internal_build_frame(handle,
                                   k_frame_type_telemetry,
                                   0, // No special flags
                                   telemetry,
                                   sizeof(telemetry),
                                   &frame);

if (err == k_rx_ok) {
    // Frame ready, now encode to wire format
    uint8_t wire_buf[279];
    uint32_t wire_len;
    rx_frame_encode(&handle->encoder, &frame, wire_buf, &wire_len);
}
```

Example - Zero-Length Payload:

```
rx_frame_t ping_frame;
rx_err_t err = internal_build_frame(handle,
                                   k_frame_type_data,
                                   0,
                                   nullptr, // No payload
                                   0, // Zero length
                                   &ping_frame);
// Result: 14-byte frame on wire (header + CRC, no payload)
```

See also

- [rx_spi_comm_send\(\)](#) Uses this function to build frames before sending
- [rx_frame_encode\(\)](#) Encodes frame to wire format after build
- [rx_frame_create_ack\(\)](#) Special builder for ACK frames
- [rx_frame_create_nack\(\)](#) Special builder for NACK frames

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Definition at line 1168 of file [rx_spi_comm.c](#).

```

01175 {
01176     if (handle == nullptr || frame == nullptr) {
01177         return k_rx_err_invalid_arg;
01178     }
01179
01180     if (payload == nullptr && payload_len > 0) {
01181         return k_rx_err_invalid_arg;
01182     }
01183
01184     if (payload_len > k_frame_max_payload) {
01185         rx_log_error(s_tag, "Payload too large");
01186         return k_rx_err_invalid_size;
01187     }
01188
01189     *frame          = (rx_frame_t){};
01190     frame->header.sequence = sequence;
01191     frame->header.length   = (uint16_t)payload_len;
01192     frame->header.type     = (uint8_t)type;
01193     frame->header.flags    = flags;
01194
01195     if (payload != nullptr && payload_len > 0) {
01196         for (uint32_t i = 0; i < payload_len; i++) {
01197             frame->payload[i] = payload[i];
01198         }
01199     }
01200
01201     if (handle->fec_enabled) {
01202         frame->header.flags |= k_frame_flag_fec_enabled;
01203     }
01204
01205     return k_rx_ok;
01206 }

```

References [rx_spi_comm_handle_t::fec_enabled](#), and [s_tag](#).

Referenced by [rx_spi_comm_send\(\)](#).

Here is the caller graph for this function:



4.3.3.3 internal_decode_frame()

```

rx_err_t internal_decode_frame (
    const rx_spi_comm_handle_t * handle,
    rx_frame_t * frame,
    const uint32_t total_size) [static]

```

Decode and verify a received SPI frame.

Note

Delegates header validation to `internal_decode_header` and CRC validation to `internal_verify_crc`.

Definition at line 1576 of file `rx_spi_comm.c`.

```

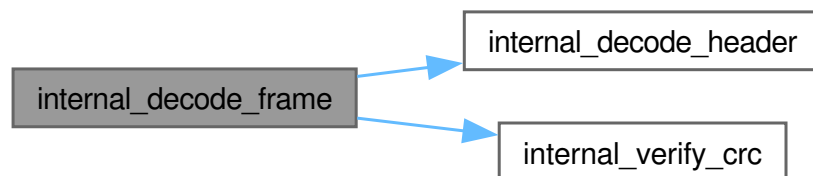
01579 {
01580     /* Pre-condition: internal callers always pass non-null handle and frame.
01581      * These are invariants -- only internal_receive_loop calls this function
01582      * with a local rx_frame_t variable and a fully-initialized handle. */
01583
01584     /* internal_read_frame_header already validated sync and payload_len before
01585      * this call, so internal_decode_header cannot fail here. */
01586     uint32_t offset = k_frame_offset_init;
01587     (void)(internal_decode_header(handle->rx_buffer, total_size, frame, &offset));
01588
01589     if (frame->header.length > 0) {
01590         for (uint32_t i = 0; i < frame->header.length; i++) {
01591             frame->payload[i] = handle->rx_buffer[offset + i];
01592         }
01593         offset += frame->header.length;
01594     }
01595
01596     rx_err_t err = internal_verify_crc(handle->rx_buffer, offset, &frame->crc);
01597     if (err != k_rx_ok) {
01598         rx_log_error(s_tag, "Frame CRC check failed");
01599         return err;
01600     }
01601
01602     return k_rx_ok;
01603 }

```

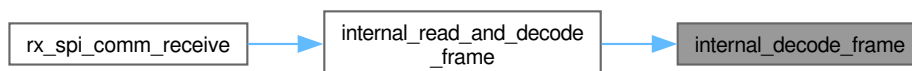
References `internal_decode_header()`, `internal_verify_crc()`, `k_frame_offset_init`, `rx_spi_comm_handle_t::rx_buffer`, and `s_tag`.

Referenced by `internal_read_and_decode_frame()`.

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.4 internal_decode_header()

```

RX_STATIC_TESTABLE rx_err_t internal_decode_header (
    const uint8_t * data,
    const uint32_t data_len,
    rx_frame_t * frame,
    uint32_t * offset_out)

```

Decode and validate SPI frame header from raw wire buffer.

Parses the frame header fields from raw received SPI data buffer, validating sync word and payload length constraints. Extracts sequence number, length, type, and flags into `rx_frame_t` structure.

Algorithm:

1. Validate inputs: Check data and frame pointers, minimum buffer size
2. Parse sync word: Read little-endian 0x55AA, validate against `k_frame_sync_word`
3. Parse sequence: Read little-endian 16-bit sequence number
4. Parse length: Read little-endian 16-bit payload length, validate ≤ 1024
5. Validate total size: Check buffer contains header + payload + CRC
6. Parse type and flags: Extract frame type and flags bytes
7. Return payload offset: Optionally return offset to payload start

Frame Header Format (10 bytes):

Offset	Size	Field	Format	Notes
0	2	SYNC	LE uint16	Always 0x55AA (wire: 0xAA, 0x55)
2	2	Sequence	LE uint16	0-65535, wraps
4	2	Length	LE uint16	Payload bytes (0-1024)
6	1	Type	uint8	<code>k_frame_type_t</code> enum
7	1	Flags	uint8	Bitmask (FEC, ACK, NACK)
8	N	Payload	bytes	N = Length (parsed above)
8+N	4	CRC-32	LE uint32	IEEE CRC-32

Precondition

data must point to valid buffer of size `data_len`
 frame must point to valid `rx_frame_t` structure

Postcondition

On success, `frame->header` contains parsed fields
 On success, `offset_out` (if non-nullptr) contains payload offset
 On failure, frame contents are undefined

Note

This function does NOT copy payload or validate CRC
 Use [internal_verify_crc\(\)](#) separately to validate CRC-32

Warning

Do not assume `frame->payload` is valid after this call

Performance:

Execution time: ~15 us @ 240 MHz (byte-level parsing, no memcpy)

Example Usage:

```
uint8_t rx_buf[279]; // Received from SPI
rx_frame_t frame;
uint32_t payload_offset;

rx_err_t err = internal_decode_header(rx_buf, sizeof(rx_buf), &frame, &payload_offset);
if (err == k_rx_ok) {
    // Header valid, now copy payload
    if (frame.header.length > 0) {
        memcpy(frame.payload, &rx_buf[payload_offset], frame.header.length);
    }
    // Verify CRC at offset payload_offset + frame.header.length
}
```

See also

[internal_verify_crc\(\)](#) Validate CRC-32 after header parse
[internal_decode_frame\(\)](#) Higher-level decode (header + payload + CRC)
[rx_frame_read_le16\(\)](#) Little-endian uint16 parser

Since

Version 1.0.0

Definition at line 423 of file [rx_spi_comm.c](#).

```

00427 {
00428     if (data == nullptr || frame == nullptr) {
00429         return k_rx_err_invalid_arg;
00430     }
00431     if (data_len < k_frame_min_size) {
00432         return k_rx_err_invalid_size;
00433     }
00434 }
00435
00436 uint32_t offset = k_frame_offset_init;
00437 uint16_t sync_word = rx_frame_read_le16(&data[offset]);
00438 if (sync_word != k_frame_sync_word) {
00439     return k_rx_err_protocol_error;
00440 }
00441 offset += k_frame_sync_size;
00442
00443 frame->header.sequence = rx_frame_read_le16(&data[offset]);
00444 offset += k_frame_seq_size;
00445
00446 frame->header.length = rx_frame_read_le16(&data[offset]);
00447 offset += k_frame_len_size;
00448
00449 if (frame->header.length > k_frame_max_payload) {
00450     return k_rx_err_invalid_size;
00451 }
00452
00453 uint32_t expected_size = rx_frame_encoded_size(frame->header.length);
00454 if (data_len < expected_size) {
00455     return k_rx_err_invalid_size;
00456 }
00457
00458 frame->header.type = data[offset];
00459 offset += k_frame_type_size;
00460
00461 frame->header.flags = data[offset];
00462 offset += k_frame_flags_size;
00463
00464 if (offset_out != nullptr) {
00465     *offset_out = offset;
00466 }
00467
00468 return k_rx_ok;
00469 }

```

References [k_frame_offset_init](#).

Referenced by [internal_decode_frame\(\)](#).

Here is the caller graph for this function:



4.3.3.5 internal_dispatch_ack_nack()

```
void internal_dispatch_ack_nack (
    rx_spi_comm_handle_t * handle,
    const rx_frame_t * frame) [static]
```

Handle ACK or NACK frame for the retransmit subsystem.

ACK with matching sequence clears the retry state. NACK with matching sequence triggers an immediate retransmit. Mismatched sequences are ignored.

Precondition

handle->auto_retransmit == true

Postcondition

retry_pending cleared on matching ACK; retry_count incremented on NACK

Since

Version 1.0.0

Definition at line 1630 of file rx_spi_comm.c.

```
01631 {
01632     if (frame->header.type == k_frame_type_ack) {
01633         if ((int)handle->retry_pending && frame->header.sequence == handle->retry_sequence) {
01634             handle->retry_pending = false;
01635             handle->retry_count = 0;
01636             if (handle->on_ack_cb != nullptr) {
01637                 handle->on_ack_cb(frame->header.sequence, handle->retransmit_cb_ctx);
01638             }
01639         }
01640         return;
01641     }
01642
01643     /* NACK: trigger immediate retransmit if sequence matches */
01644     const bool seq_match = (bool)(frame->header.sequence == handle->retry_sequence);
01645     if ((int)handle->retry_pending && (int)seq_match) {
01646         const rx_err_t retx_err = internal_retransmit_frame(handle);
01647         if (retx_err != k_rx_ok) {
01648             rx_log_error(s_tag, "NACK-triggered retransmit failed");
01649         }
01650         if (handle->on_nack_cb != nullptr) {
01651             handle->on_nack_cb(frame->header.sequence, handle->retransmit_cb_ctx);
01652         }
01653     }
01654 }
```

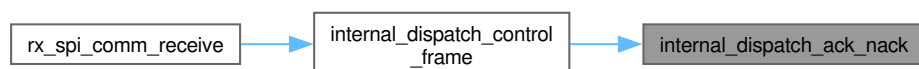
References [internal_retransmit_frame\(\)](#), [rx_spi_comm_handle_t::on_ack_cb](#), [rx_spi_comm_handle_t::on_nack_cb](#), [rx_spi_comm_handle_t::retransmit_cb_ctx](#), [rx_spi_comm_handle_t::retry_count](#), [rx_spi_comm_handle_t::retry_sequence](#), and [s_tag](#).

Referenced by [internal_dispatch_control_frame\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.6 internal_dispatch_control_frame()

```
rx_err_t internal_dispatch_control_frame (
    rx_spi_comm_handle_t * handle,
    rx_frame_t * frame,
    ctrl_dispatch_result_t * result) [static]
```

Dispatch control frames (PING, RESET, ACK/NACK) during receive.

Handles PING (auto-PONG + callback), RESET (auto-RESET_ACK + session reset

- callback), and ACK/NACK (retransmit subsystem dispatch). Data frames are not consumed and signal the caller to return to the application.

Precondition

handle and frame must be non-nullptr and valid

Postcondition

Control-frame side effects applied (PONG sent, session reset, etc.)

Note

Called only from [rx_spi_comm_receive\(\)](#); not part of the public API.

Since

Version 1.0.0

Definition at line 1672 of file [rx_spi_comm.c](#).

```
01675 {
01676 /* PING -> auto-send PONG, invoke callback, consume */
01677 if (frame->header.type == k_frame_type_ping) {
01678     rx_err_t pong_err = rx_spi_comm_send_pong(handle, frame->payload, frame->header.length);
01679     if (pong_err != k_rx_ok) {
01680         rx_log_error(s_tag, "Failed to send PONG response");
01681     }
01682
01683     if (handle->on_ping_cb != nullptr) {
01684         handle->on_ping_cb(frame, handle->control_cb_ctx);
01685     }
01686
01687     *result = k_ctrl_consumed;
01688     return k_rx_ok;
01689 }
01690
01691 /* RESET -> send RESET_ACK, reset session, invoke callback, consume */
01692 if (frame->header.type == k_frame_type_reset) {
01693     rx_err_t ack_err = rx_spi_comm_send_reset_ack(handle);
01694     if (ack_err != k_rx_ok) {
01695         rx_log_error(s_tag, "Failed to send RESET_ACK, session not reset");
01696         *result = k_ctrl_consumed;
01697         return k_rx_ok; /* Don't reset session if ACK failed */
01698     }
01699
01700 /* rx_session_reset only fails with null/uninitialized session.
01701 * handle->session is valid and initialized (checked at rx_spi_comm_init).
01702 * Invariant: reset cannot fail here. */
01703 (void)(rx_session_reset(handle->session));
01704
01705 if (handle->on_reset_cb != nullptr) {
01706     handle->on_reset_cb(frame, handle->control_cb_ctx);
01707 }
```

```

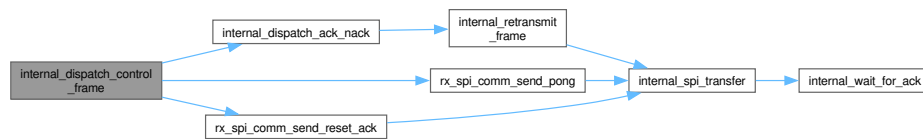
01708
01709     *result = k_ctrl_consumed;
01710     return k_rx_ok;
01711 }
01712
01713 /* ACK/NACK -> retransmit subsystem or pass through */
01714 if (frame->header.type == k_frame_type_ack || frame->header.type == k_frame_type_nack) {
01715     if (handle->auto_retransmit) {
01716         internal_dispatch_ack_nack(handle, frame);
01717         *result = k_ctrl_consumed;
01718         return k_rx_ok;
01719     }
01720
01721     /* auto_retransmit off: not consumed, return to caller as data frame */
01722     *result = k_ctrl_not_handled;
01723     return k_rx_ok;
01724 }
01725
01726 /* Not a control frame */
01727 *result = k_ctrl_not_handled;
01728 return k_rx_ok;
01729 }

```

References [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_handle_t::control_cb_ctx](#), [internal_dispatch_ack_nack\(\)](#), [k_ctrl_consumed](#), [k_ctrl_not_handled](#), [rx_spi_comm_handle_t::on_ping_cb](#), [rx_spi_comm_handle_t::on_reset_cb](#), [rx_spi_comm_send_pong\(\)](#), [rx_spi_comm_send_reset_ack\(\)](#), [s_tag](#), and [rx_spi_comm_handle_t::session](#).

Referenced by [rx_spi_comm_receive\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.7 internal_read_and_decode_frame()

```

rx_err_t internal_read_and_decode_frame (
    rx_spi_comm_handle_t * handle,
    rx_frame_t * frame,
    const uint32_t timeout_ms) [static]

```

Receive data frame from RPi5 controller via SPI (blocking with timeout).

Receives complete frame from RPi5 by polling for data availability, reading header, reading payload + CRC, validating frame integrity, and updating RX sequence counter. This is the primary receive function for all frame types.

Algorithm:

1. Pre-condition 1: Validate handle and frame pointers non-nullptr

2. Pre-condition 2: Validate handle initialized
3. Wait for data: Call [internal_wait_for_data\(\)](#) with timeout
 - Polls RSPI peripheral for incoming data (yields CPU every 10ms)
 - Returns `k_rx_err_timeout` if no data within `timeout_ms`
4. Read frame header: Call [internal_read_frame_header\(\)](#)
 - Reads 10 bytes (SYNC + sequence + length + type + flags)
 - Validates sync word (0xAA55)
 - Extracts payload length (0-1024)
5. Read payload + CRC: Call [internal_spi_transfer\(\)](#) for remaining bytes
 - Reads `payload_len + 4` bytes (CRC-32)
 - Stores in `handle->rx_buffer`
6. Decode frame: Call [internal_decode_frame\(\)](#)
 - Parses header fields into `frame->header`
 - Copies payload to `frame->payload`
 - Validates CRC-32 (returns `k_rx_err_crc_mismatch` if corrupt)
7. Update RX sequence: Set `handle->rx_sequence = frame->sequence + 1`
 - Tracks expected next sequence number
 - Caller can detect out-of-order/missing frames

Blocking Behavior:

- `timeout_ms = 0`: Non-blocking, returns immediately if no data
- `timeout_ms > 0`: Blocks up to `timeout_ms` milliseconds waiting for data
- ThreadX sleep: Yields CPU every 10ms tick (not busy-wait)

Timeout Semantics:

- `k_rx_err_timeout` is NOT an error in polling loops (means "no data yet")
- Typical polling: Call with `timeout_ms=100` in loop, handle timeout normally

Precondition

handle must be initialized via [rx_spi_comm_init\(\)](#)
 frame must point to valid `rx_frame_t` storage (304 bytes)
 RSPI peripheral hardware must be initialized and enabled
 RPi5 controller must have transmitted complete frame

Postcondition

On success, frame contains validated header, payload, CRC
 On success, `handle->rx_sequence = frame->header.sequence + 1`
 On timeout, frame contents undefined, RX sequence unchanged
 On CRC mismatch, frame partially populated, RX sequence unchanged
 On protocol error, frame undefined, may need resync

Note

This function blocks (with ThreadX sleep) if data not immediately available
 timeout_ms=0 provides non-blocking behavior for polling loops

Warning

Not thread-safe - use external mutex if multiple threads receive
 Do not process frame if return value != k_rx_ok (data invalid)
 Send NACK if k_rx_err_crc_mismatch to request retransmit

Performance:

- Execution time: 200-600 us @ 10 MHz SPI + 0-timeout_ms wait
- 14-byte frame (ACK): ~200 us (10 us wait + 80 us read + 110 us decode)
- 100-byte frame: ~400 us (10 us wait + 200 us read + 190 us decode)
- 279-byte frame (max): ~600 us (10 us wait + 300 us read + 290 us decode)
- Wait time: Add 0-timeout_ms if data not immediately available

Example - Polling Loop (Typical Usage):

```
while (1) {
    rx_frame_t frame;
    rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 100); // 100ms timeout

    if (err == k_rx_ok) {
        // Frame received successfully
        if (frame.header.type == k_frame_type_data) {
            process_data_frame(&frame);
            rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
        }
    } else if (err == k_rx_err_timeout) {
        // No data, normal in polling - continue loop
        tx_thread_sleep(1); // Sleep 10ms before retry
    } else if (err == k_rx_err_crc_mismatch) {
        // Corrupted frame, send NACK
        rx_spi_comm_send_nack(&g_spi_handle, frame.header.sequence, 0);
    } else {
        // Fatal error, log and recover
        rx_log_error("spi", "Receive error: %d", err);
    }
}
```

Example - Non-Blocking Check:

```
// Check if data available without blocking
rx_frame_t frame;
rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0); // timeout=0

if (err == k_rx_ok) {
    rx_log_debug("spi", "Frame received: seq=%d, len=%d",
        frame.header.sequence, frame.header.length);
} else if (err == k_rx_err_timeout) {
    // No data available right now, continue
}
```

Example - Sequence Number Checking:

```
static uint16_t expected_seq = 0;
rx_frame_t frame;

rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 1000);
if (err == k_rx_ok) {
    if (frame.header.sequence != expected_seq) {
        rx_log_warn("spi", "Out-of-order: expected %d, got %d",
            expected_seq, frame.header.sequence);
    }
    expected_seq = frame.header.sequence + 1; // Update for next frame
}
```

See also

- [rx_spi_comm_send\(\)](#) Send frames to RPi5
- [rx_spi_comm_data_available\(\)](#) Check data availability without timeout
- [rx_spi_comm_send_ack\(\)](#) Send acknowledgment after successful receive
- [rx_spi_comm_send_nack\(\)](#) Send NACK after CRC mismatch
- [internal_wait_for_data\(\)](#) Wait for data with timeout
- [internal_decode_frame\(\)](#) Frame decoding and CRC validation

Since

Version 1.0.0

Read one complete frame from SPI into handle->rx_buffer and decode it

Precondition

handle and frame must be non-nullptr

Postcondition

On success, frame contains validated header, payload, and CRC

Since

Version 1.0.0

Definition at line 1869 of file [rx_spi_comm.c](#).

```
01872 {
01873     rx_err_t err = internal_wait_for_data(handle, timeout_ms);
01874     if (err != k_rx_ok) {
01875         return err;
01876     }
01877     uint8_t header_buf[k_frame_sync_size + k_frame_header_size];
01878     uint16_t payload_len = 0;
01880     err = internal_read_frame_header(handle, header_buf, &payload_len);
01882     if (err != k_rx_ok) {
01883         return err;
01884     }
01885     const uint32_t header_len = sizeof(header_buf);
01887     const uint32_t total_size =
01888         k_frame_sync_size + k_frame_header_size + payload_len + k_frame_crc_size;
01889     const uint32_t remaining = payload_len + k_frame_crc_size;
01890     err = internal_spi_transfer(handle, nullptr, 0, handle->rx_buffer + header_len, remaining);
```

```

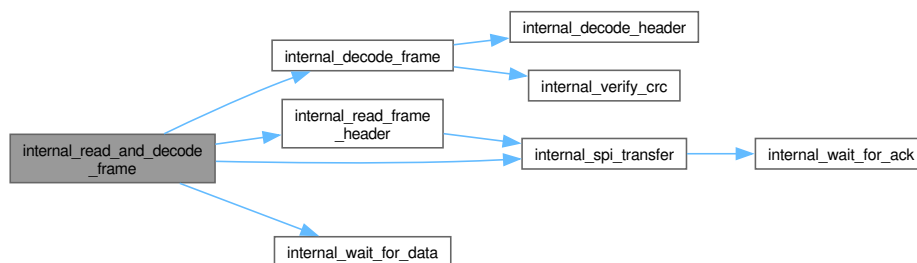
01892 if (err != k_rx_ok) {
01893     return err;
01894 }
01895
01896 for (uint32_t i = 0; i < header_len; i++) {
01897     handle->rx_buffer[i] = header_buf[i];
01898 }
01899
01900 err = internal_decode_frame(handle, frame, total_size);
01901 if (err != k_rx_ok) {
01902     rx_log_error(s_tag, "Frame decode failed");
01903     return err;
01904 }
01905
01906 return k_rx_ok;
01907 }

```

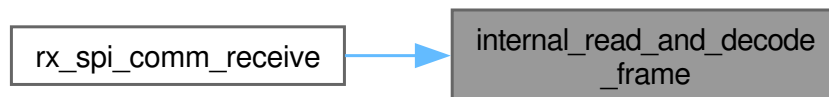
References [internal_decode_frame\(\)](#), [internal_read_frame_header\(\)](#), [internal_spi_transfer\(\)](#), [internal_wait_for_data\(\)](#), [rx_spi_comm_handle_t::rx_buffer](#), and [s_tag](#).

Referenced by [rx_spi_comm_receive\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.8 internal_read_frame_header()

```

rx_err_t internal_read_frame_header (
    rx_spi_comm_handle_t * handle,
    uint8_t * header_buf,
    uint16_t * payload_len) [static]

```

Read and validate frame header from SPI.

Definition at line 1540 of file [rx_spi_comm.c](#).

```

01541 {
01542     const uint32_t header_len = k_frame_sync_size + k_frame_header_size;
01543     /* Read frame header from SPI */
01544     const rx_err_t err = internal_spi_transfer(handle, nullptr, 0, header_buf, header_len);
01545
01546     if (err != k_rx_ok) {
01547         return err;
01548     }
01549
01550     /* Validate sync word (little-endian) */
01551     const uint16_t sync = rx_frame_read_le16(&header_buf[k_hdr_sync_low]);
01552

```

```

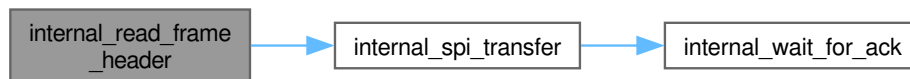
01553 if (sync != k_frame_sync_word) {
01554     rx_log_error(s_tag, "Invalid sync word");
01555     return k_rx_err_protocol_error;
01556 }
01557
01558 /* Extract payload length (little-endian) */
01559 *payload_len = rx_frame_read_le16(&header_buf[k_hdr_len_low]);
01560 if (*payload_len > k_frame_max_payload) {
01561     rx_log_error(s_tag, "Payload too large");
01562     return k_rx_err_invalid_size;
01563 }
01564
01565 return k_rx_ok;
01566 }

```

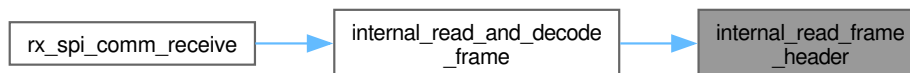
References [internal_spi_transfer\(\)](#), [k_hdr_len_low](#), [k_hdr_sync_low](#), and [s_tag](#).

Referenced by [internal_read_and_decode_frame\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.9 internal_retransmit_frame()

```

rx_err_t internal_retransmit_frame (
    rx_spi_comm_handle_t * handle) [static]

```

Retransmit the buffered frame via SPI.

Precondition

handle->retry_pending == true
 handle->retry_buffer contains valid encoded frame

Postcondition

retry_count incremented on success, retry_pending cleared on limit

Since

Version 1.0.0

Definition at line 2190 of file [rx_spi_comm.c](#).

```

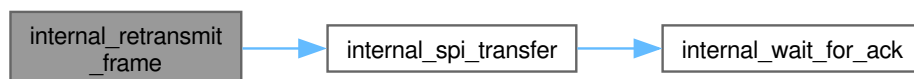
02191 {
02192     /* Check retry limit */
02193     if (handle->retry_count >= handle->retransmit_cfg.max_retries) {
02194         rx_log_error(s_tag, "Retransmit limit exceeded");
02195         handle->retry_pending = false;
02196         handle->retry_count = 0;
02197         return k_rx_err_retry_limit;
02198     }
02199
02200     /* Set retransmit flag in buffered frame (flags byte is at offset 7).
02201      * retry_wire_len >= k_frame_min_size > k_hdr_flags always for valid encoded frames. */
02202     handle->retry_buffer[k_hdr_flags] |= k_frame_flag_retransmit;
02203
02204     /* Retransmit via SPI */
02205     rx_err_t err =
02206         internal_spi_transfer(handle, handle->retry_buffer, handle->retry_wire_len, nullptr, 0);
02207     if (err != k_rx_ok) {
02208         rx_log_error(s_tag, "Retransmit SPI transfer failed");
02209         return err;
02210     }
02211
02212     handle->retry_count++;
02213     #ifdef RX
02214     handle->retry_send_time_ms = (uint32_t)(tx_time_get() * k_threadx_ms_per_tick);
02215     #else
02216     handle->retry_send_time_ms = 0;
02217     #endif
02218
02219     return k_rx_ok;
02220 }

```

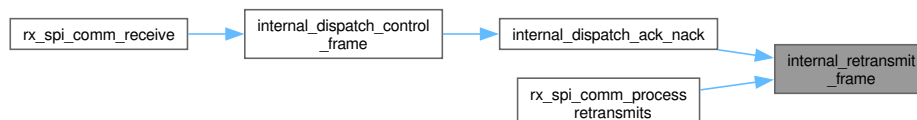
References [internal_spi_transfer\(\)](#), [k_hdr_flags](#), [rx_spi_comm_retransmit_config_t::max_retries](#), [rx_spi_comm_handle_t::retransmit_cfg](#), [rx_spi_comm_handle_t::retry_buffer](#), [rx_spi_comm_handle_t::retry_count](#), [rx_spi_comm_handle_t::retry_pending](#), [rx_spi_comm_handle_t::retry_send_time_ms](#), [rx_spi_comm_handle_t::retry_wire_len](#), and [s_tag](#).

Referenced by [internal_dispatch_ack_nack\(\)](#), and [rx_spi_comm_process_retransmits\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.10 `internal_spi_transfer()`

```
RX_STATIC_TESTABLE rx_err_t internal_spi_transfer (
    rx_spi_comm_handle_t * handle,
    const uint8_t * tx_data,
    const uint32_t tx_len,
    uint8_t * rx_data,
    const uint32_t rx_len)
```

Perform raw SPI full-duplex transfer using handle's staging buffers.

Low-level SPI transfer wrapper that uses handle's internal TX/RX staging buffers (2048 bytes each) to perform full-duplex communication with RPi5 controller. Implements flow control by waiting for host ready signal before transmit operations.

Why Staging Buffers? RSPI peripheral DMA requires contiguous aligned buffers. Using handle's pre-allocated staging buffers avoids dynamic allocation and ensures proper alignment for DMA transfers.

Algorithm:

1. Pre-condition 1: Validate handle pointer (`RX_CHECK_NULL_PTR`)
2. Pre-condition 2: Calculate `transfer_len = max(tx_len, rx_len)`
3. Pre-condition 3: Validate `transfer_len <= k_spi_comm_tx_buffer_size (2048)`
4. Pre-condition 4: Validate TX data pointer consistent with `tx_len`
5. Pre-condition 5: Validate RX data pointer consistent with `rx_len`
6. Flow control: If transmitting (`tx_data != nullptr`), wait for host ready via `internal_wait_for_ack()`
7. Prepare TX buffer: Zero-fill staging buffer, copy `tx_data` if present
8. SPI transfer: Call `rspi_peripheral_transfer()` with staging buffers
9. Post-condition: Copy RX staging buffer to `rx_data` if requested

Full-Duplex Behavior:

- SPI is inherently full-duplex (simultaneous TX and RX)
- If `tx_len < rx_len`: TX buffer zero-padded, RX receives full `rx_len` bytes
- If `tx_len > rx_len`: RX buffer receives `tx_len` bytes, caller extracts `rx_len`
- Transfer length = `max(tx_len, rx_len)` to accommodate both directions

Precondition

handle must be initialized via `rx_spi_comm_init()`

If `tx_data != nullptr`, `tx_len` must be `> 0` and `<= 2048`

If `rx_data != nullptr`, `rx_len` must be `> 0` and `<= 2048`

`tx_data` and `rx_data` must not alias (undefined behavior if overlapping)

Postcondition

- On k_rx_ok, rx_data (if non-nullptr) contains received bytes
- On k_rx_ok, handle->tx_buffer and handle->rx_buffer modified
- On error, rx_data contents are undefined

Note

- This function uses handle's staging buffers (not caller's buffers directly)
- Transfer is synchronous (blocks until complete or timeout)

Warning

- Do NOT pass overlapping tx_data and rx_data buffers
- This function is NOT thread-safe (use external mutex if multi-threaded)

Performance:

- Execution time: 80-400 us @ 10 MHz SPI (depends on transfer_len)
- 10-byte transfer (ACK/NACK): ~80 us (8 us SPI + 72 us overhead)
- 100-byte transfer: ~180 us (80 us SPI + 100 us overhead)
- 1038-byte transfer (max frame): ~1000 us (830 us SPI + 170 us overhead)
- Host ACK wait: Add 0-50 ms if transmitting (see internal_wait_for_ack)

Example - Transmit Only:

```
uint8_t tx_frame[14] = { 0xAA, 0x55, ... }; // ACK frame
rx_err_t err = internal_spi_transfer(handle, tx_frame, 14, nullptr, 0);
if (err == k_rx_ok) {
    rx_log_debug("spi", "ACK sent successfully");
}
```

Example - Receive Only:

```
uint8_t rx_header[10]; // Read frame header
rx_err_t err = internal_spi_transfer(handle, nullptr, 0, rx_header, 10);
if (err == k_rx_ok) {
    uint16_t sync = (rx_header[0] << 8) | rx_header[1];
    if (sync == 0xAA55) {
        // Valid frame header
    }
}
```

Example - Full-Duplex:

```
uint8_t tx_buf[50] = { ... }; // Send 50 bytes
uint8_t rx_buf[50]; // Receive 50 bytes simultaneously
rx_err_t err = internal_spi_transfer(handle, tx_buf, 50, rx_buf, 50);
if (err == k_rx_ok) {
    // Both TX and RX completed
}
```

See also

[internal_wait_for_ack\(\)](#) Flow control for TX operations
[rspi_peripheral_transfer\(\)](#) RSPI HAL transfer function
[rx_spi_comm_handle_t](#) Handle structure with staging buffers

Since

Version 1.0.0

Definition at line 809 of file [rx_spi_comm.c](#).

```

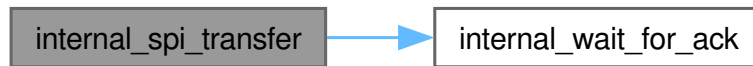
00814 {
00815     /* Pre-condition 1: Handle pointer is always non-null (all callers validate before calling). */
00816
00817     /* Pre-condition 2: Transfer length within buffer capacity.
00818      * All callers validate frame lengths before calling, so this is an invariant. */
00819     uint32_t transfer_len = (tx_len > rx_len) ? tx_len : rx_len;
00820
00821     /* Pre-condition 3: TX data pointer consistent with length.
00822      * All callers either pass valid data+length or nullptr+0, never inconsistent. */
00823
00824     /* Pre-condition 4: RX data pointer consistent with length.
00825      * Same invariant as TX: callers never pass nullptr rx_data with non-zero rx_len. */
00826
00827     /* Wait for host ready signal before transmit operations */
00828     const bool has_tx = (bool)((tx_data != nullptr) && (tx_len > 0));
00829     if (has_tx) {
00830         rx_err_t wait_err = internal_wait_for_ack(handle, k_ack_wait_timeout_ms);
00831         RX_RETURN_ON_ERROR(wait_err, s_tag, "Host ACK wait failed");
00832     }
00833
00834     /* Prepare TX buffer (pad with zeros if RX is larger) */
00835     for (uint32_t i = 0; i < transfer_len; i++) {
00836         handle->tx_buffer[i] = 0;
00837     }
00838     if (has_tx) {
00839         for (uint32_t i = 0; i < tx_len; i++) {
00840             handle->tx_buffer[i] = tx_data[i];
00841         }
00842     }
00843
00844     /*
00845      * Perform SPI transfer.
00846      * Cast to uint16_t: RSPI HAL uses 16-bit length (max 65535 bytes).
00847      * Safe because transfer_len is validated <= k_spi_comm_tx_buffer_size above.
00848      */
00849     rx_err_t err = rspi_peripheral_transfer(handle->channel,
00850                                             handle->tx_buffer,
00851                                             handle->rx_buffer,
00852                                             (uint16_t)transfer_len);
00853
00854     if (err != k_rx_ok) {
00855         rx_log_error(s_tag, "SPI peripheral transfer failed");
00856         return err;
00857     }
00858
00859     /* Post-condition: Copy RX data (reverse copy: rx_data may alias rx_buffer) */
00860     const bool has_rx = (bool)((rx_data != nullptr) && (rx_len > 0));
00861     if (has_rx) {
00862         for (uint32_t i = rx_len; i > 0; i--) {
00863             rx_data[i - 1] = handle->rx_buffer[i - 1];
00864         }
00865     }
00866
00867     return k_rx_ok;
00868 }

```

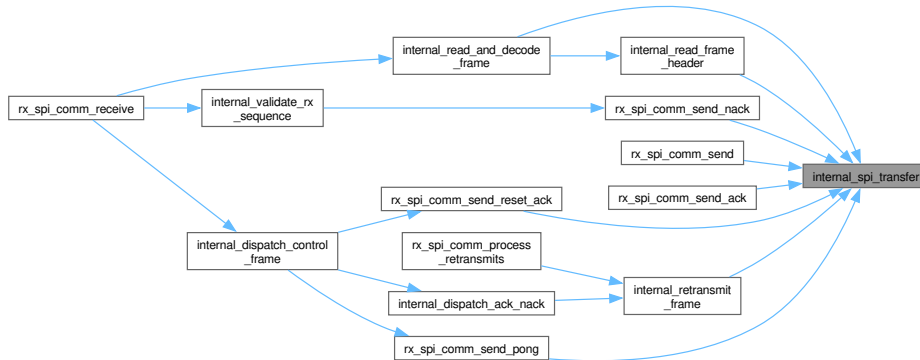
References [rx_spi_comm_handle_t::channel](#), [internal_wait_for_ack\(\)](#), [k_ack_wait_timeout_ms](#), [rx_spi_comm_handle_t::rx_buffer](#), [s_tag](#), and [rx_spi_comm_handle_t::tx_buffer](#).

Referenced by [internal_read_and_decode_frame\(\)](#), [internal_read_frame_header\(\)](#), [internal_retransmit_frame\(\)](#), [rx_spi_comm_send\(\)](#), [rx_spi_comm_send_ack\(\)](#), [rx_spi_comm_send_nack\(\)](#), [rx_spi_comm_send_pong\(\)](#), and [rx_spi_comm_send_reset_ack\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.11 internal_validate_rx_sequence()

```

rx_err_t internal_validate_rx_sequence (
    rx_spi_comm_handle_t * handle,
    const rx_frame_t * frame) [static]
  
```

Validate received frame sequence and emit NACK on rejection.

Calls `rx_session_validate_rx()` to update and check the session's RX sequence counter. On rejection (Audit F-05), the function does NOT silently drop the offending frame: it emits a NACK carrying the rejected sequence so the remote can retransmit immediately instead of burning its ACK-timer budget. The NACK send is allowed to fail (transport not ready, etc.) and is logged as a warning; the original validate error remains the authoritative return value.

Parameters

in,out	handle	SPI comm handle (session and TX path used)
in	frame	Decoded frame whose header.sequence is checked

Returns

`rx_err_t` Result of the underlying session validation

Return values

<code>k_rx_ok</code>	Sequence accepted
<code>k_rx_err_protocol_error</code>	Sequence rejected (gap/duplicate)
other	Propagated from <code>rx_session_validate_rx()</code>

Precondition

handle and frame must be non-nullptr and valid
 handle->session must be initialized

Postcondition

On success, session RX sequence advanced
 On failure, NACK send attempted and validate error returned verbatim

Note

Called only from [rx_spi_comm_receive\(\)](#); not part of the public API.

Since

Version 1.0.0

Definition at line [1937](#) of file [rx_spi_comm.c](#).

```

01938 {
01939     rx_session_validate_result_t validate_result = k_session_validate_fail;
01940     rx_err_t validate_err =
01941         rx_session_validate_rx(handle->session, frame->header.sequence, &validate_result);
01942     if (validate_err != k_rx_ok) {
01943         /* Audit F-05: NACK the offending sequence so the remote can retransmit
01944          * immediately. Send-NACK can fail (transport not ready); log and keep
01945          * the original validate_err as the real cause. */
01946         rx_log_error(s_tag, "Session validate_rx returned error");
01947         const rx_err_t nack_err = rx_spi_comm_send_nack(handle, frame->header.sequence, 0);
01948         if (nack_err != k_rx_ok) {
01949             rx_log_warn(s_tag, "Failed to send NACK after sequence validation failure");
01950         }
01951         return validate_err;
01952     }
01953     (void)validate_result;
01954     return k_rx_ok;
01955 }
```

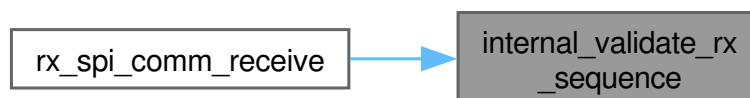
References [rx_spi_comm_send_nack\(\)](#), [s_tag](#), and [rx_spi_comm_handle_t::session](#).

Referenced by [rx_spi_comm_receive\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.12 internal_verify_crc()

```
RX_STATIC_TESTABLE rx_err_t internal_verify_crc (
    const uint8_t * data,
    uint32_t offset,
    uint32_t * crc_out)
```

Verify CRC-32 for a received SPI frame.

Validates frame integrity by computing CRC-32 over header + payload and comparing with the CRC-32 field at the end of the frame. Uses IEEE polynomial 0x04C11DB7 (same as Ethernet, ZIP, PNG).

Algorithm:

1. Extract received CRC: Read 4-byte little-endian CRC from data[offset]
2. Compute expected CRC: Calculate CRC-32 over data[0..offset-1]
3. Compare: If received == calculated, return k_rx_ok
4. Optionally return CRC: Copy received CRC to crc_out if non-nullptr

CRC Computation Details:

- Algorithm: CRC-32/IEEE (polynomial 0x04C11DB7, init 0xFFFFFFFF, final XOR 0x←FFFFFFFF)
- Input: All bytes from sync word through end of payload (excludes CRC itself)
- Output: 32-bit checksum stored in little-endian at frame end
- Implementation: rx_crc32_ieee() (hardware CRC peripheral on RX72N)

Precondition

data must point to valid buffer containing offset + 4 bytes
offset must be the byte position where CRC-32 field begins

Postcondition

On k_rx_ok, frame integrity guaranteed
On k_rx_err_crc_mismatch, frame is corrupted (send NACK)
crc_out (if non-nullptr) contains received CRC value

Note

Thread-safe if caller serializes access to data buffer

Warning

This function does NOT modify the data buffer
CRC mismatch indicates transmission error, do NOT process frame

Performance:

- Execution time: 40-250 us @ 240 MHz (depends on offset = bytes to hash)
- 10-byte frame (ACK): ~40 us
- 100-byte frame: ~130 us
- 263-byte frame (max): ~250 us
- Hardware CRC: If `RX_CRC32_USE_HARDWARE` defined, 4x faster

CRC Mismatch Handling:

```
uint8_t rx_buf[279];
uint32_t payload_offset = 8; // After header
uint32_t crc_offset = payload_offset + payload_len;
uint32_t received_crc;

rx_err_t err = internal_verify_crc(rx_buf, crc_offset, &received_crc);
if (err == k_rx_err_crc_mismatch) {
    rx_log_error("spi_comm", "CRC mismatch: received=0x%08X", received_crc);
    // Send NACK to request retransmit
    rx_spi_comm_send_nack(handle, frame_seq, 0);
}
```

See also

`rx_crc32_ieee()` CRC-32 computation implementation

`rx_frame_read_le32()` Little-endian uint32 reader

[internal_decode_frame\(\)](#) Uses this function for CRC validation

Since

Version 1.0.0

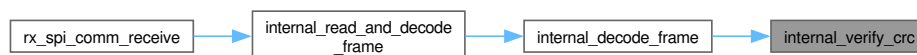
Definition at line 533 of file [rx_spi_comm.c](#).

```
00536 {
00537     if (data == nullptr) {
00538         return k_rx_err_invalid_arg;
00539     }
00540
00541     const uint32_t received_crc = rx_frame_read_le32(&data[offset]);
00542     uint32_t calculated_crc = k_spi_crc32_seed_initial;
00543     (void)(rx_crc32_ieee(data, offset, &calculated_crc));
00544     /* rx_crc32_ieee only fails with null pointers or out-of-range length.
00545      * data is already checked non-null above, and offset (frame length minus CRC)
00546      * is always in [k_frame_min_size, k_spi_comm_tx_buffer_size] -- well within
00547      * the CRC library's valid range. This path is therefore an invariant. */
00548
00549     if (received_crc != calculated_crc) {
00550         return k_rx_err_crc_mismatch;
00551     }
00552
00553     if (crc_out != nullptr) {
00554         *crc_out = received_crc;
00555     }
00556
00557     return k_rx_ok;
00558 }
```

References [k_spi_crc32_seed_initial](#).

Referenced by [internal_decode_frame\(\)](#).

Here is the caller graph for this function:



4.3.3.13 internal_wait_for_ack()

```
rx_err_t internal_wait_for_ack (
    const rx_spi_comm_handle_t * handle,
    uint32_t timeout_ms) [static]
```

Wait for RPi5 controller ready signal before transmitting data.

Implements flow control by polling RSPI peripheral for host ready signal before initiating SPI transfer. Prevents peripheral from blocking indefinitely if controller never asserts chip select or clock.

Purpose: RX72N (SPI peripheral) cannot initiate transfers - it must wait for RPi5 (SPI controller) to assert CS and provide clock. This function polls the RSPI hardware for the "ready to write" condition with a timeout to detect host failures.

Algorithm:

1. Get start time: Capture current ThreadX tick count (or iteration counter in tests)
2. Poll loop:
 - Call `rspi_peripheral_write_ready()` to check hardware status
 - If ready: Return `k_rx_ok` immediately
 - If error: Return HAL error code
 - If not ready: Sleep 1 tick (10ms) and retry
3. Timeout check: If elapsed time $\geq$ `timeout_ms`, return `k_rx_err_timeout`
4. Thread yield: `tx_thread_sleep()` yields CPU to other tasks (not busy-wait)

Platform Differences:

- RX72N (`__RX__` defined): Uses ThreadX `tx_time_get()` for precise timing
- Host builds (tests): Uses iteration counter to simulate time (no ThreadX)

Precondition

handle must be initialized via `rx_spi_comm_init()`
 timeout_ms must be > 0

Postcondition

On `k_rx_ok`, RSPI is ready for `rspi_peripheral_transfer()`
 On `k_rx_err_timeout`, host may be dead/disconnected
 On HAL error, RSPI peripheral may require reset

Note

Yields CPU via `tx_thread_sleep(1)` every iteration (cooperative multitasking)

Warning

Do NOT call with `timeout_ms = 0` (will return timeout immediately on RX72N)

Timeout does NOT guarantee exact millisecond precision (+/-10ms due to tick granularity)

Performance:

- Best case: ~10 us (host ready immediately, 1 HAL call)
- Typical: 10-50 ms (host ready within 1-5 polling iterations)
- Worst case: `timeout_ms` (50 ms default, host never ready)

ThreadX Timing:

- Tick rate: 100 Hz (`k_threadx_ms_per_tick = 10 ms`)
- Sleep duration: 1 tick = 10 ms per iteration
- Timeout conversion: $(\text{timeout_ms} + 9) / 10$ ticks (round up)

Example - Normal Transmission:

```
// Wait up to 50ms for RPi5 to be ready
rx_err_t err = internal_wait_for_ack(handle, 50);
if (err == k_rx_ok) {
    // Host ready, perform SPI transfer
    rspi_peripheral_transfer(handle->channel, tx_buf, rx_buf, len);
} else if (err == k_rx_err_timeout) {
    rx_log_error("spi", "Host ACK timeout - RPi5 may be down");
}
```

Example - Timeout Handling:

```
// Retry logic for transient host delays
for (uint8_t retry = 0; retry < 3; retry++) {
    rx_err_t err = internal_wait_for_ack(handle, 50);
    if (err == k_rx_ok) break;

    if (err == k_rx_err_timeout) {
        rx_log_warn("spi", "Host not ready, retry %d/3", retry + 1);
        tx_thread_sleep(10); // Wait 100ms between retries
    } else {
        // HAL error, fatal
        return err;
    }
}
```

See also

`rspi_peripheral_write_ready()` RSPI HAL function for ready status

`internal_spi_transfer()` Calls this function before TX operations

`k_ack_wait_timeout_ms` Default timeout constant (50ms)

Version 1.0.0

```

00673 {
00674 #ifdef __RX__
00675 /* RX72N: Use ThreadX time measurement for precise timeout */
00676 ULONG timeout_ticks = (timeout_ms + k_threadx_ms_per_tick - 1) / k_threadx_ms_per_tick;
00677 ULONG start_ticks = tx_time_get();
00678
00679 while ((tx_time_get() - start_ticks) < timeout_ticks) {
00680     bool ready = false;
00681     const rx_err_t err = rspi_peripheral_write_ready(handle->channel, &ready);
00682     if (err != k_rx_ok) {
00683         return err;
00684     }
00685     if (ready) {
00686         return k_rx_ok;
00687     }
00688     /* Yield to other threads while waiting */
00689     tx_thread_sleep(k_poll_sleep_ticks);
00690 }
00691 #else
00692 /* Host build (testing): Use iteration counter to simulate time */
00693 uint32_t elapsed_ms = 0;
00694
00695 while (elapsed_ms < timeout_ms) {
00696     bool ready = false;
00697     const rx_err_t err = rspi_peripheral_write_ready(handle->channel, &ready);
00698     if (err != k_rx_ok) {
00699         return err;
00700     }
00701
00702     if (ready) {
00703         return k_rx_ok;
00704     }
00705
00706     /* Simulate time passing in tests */
00707     elapsed_ms += k_threadx_ms_per_tick;
00708 }
00709 #endif
00710
00711 rx_log_error(s_tag, "SPI ACK timeout waiting for host ready");
00712 return k_rx_err_timeout;
00713 }

```

Referenced by [internal_spi_transfer\(\)](#).

```

graph LR
    rx_spi_comm_receive --> internal_validate_rx_sequence
    internal_validate_rx_sequence --> internal_read_and_decode_frame
    internal_validate_rx_sequence --> rx_spi_comm_send_nack
    internal_read_and_decode_frame --> internal_read_frame_header
    internal_read_frame_header --> internal_spi_transfer
    rx_spi_comm_send_nack --> internal_spi_transfer
    rx_spi_comm_receive --> internal_dispatch_control_frame
    internal_dispatch_control_frame --> rx_spi_comm_send_reset_ack
    internal_dispatch_control_frame --> rx_spi_comm_process_retransmits
    rx_spi_comm_send_reset_ack --> internal_spi_transfer
    rx_spi_comm_process_retransmits --> internal_spi_transfer
    internal_dispatch_control_frame --> internal_dispatch_ack_nack
    internal_dispatch_ack_nack --> internal_retransmit_frame
    internal_retransmit_frame --> internal_spi_transfer
    internal_dispatch_control_frame --> rx_spi_comm_send_pong
    rx_spi_comm_send_pong --> internal_spi_transfer
    internal_spi_transfer --> internal_wait_for_ack
  
```

```
rx_err_t internal_wait_for_data (
    const rx_spi_comm_handle_t * handle,
    const uint32_t timeout_ms) [static]
```

Wait for data to be available on SPI channel.

Definition at line 1490 of file `rx_spi_comm.c`.

```

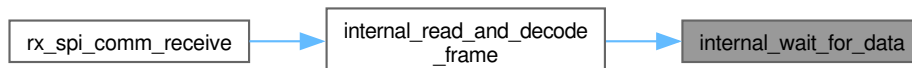
01492 {
01493     bool    available = false;
01494     rx_err_t err      = rspi_peripheral_read_available(handle->channel, &available);
01495     if (err != k_rx_ok) {
01496         return err;
01497     }
01498
01499     if (!available) {
01500         if (timeout_ms == 0) {
01501             return k_rx_err_timeout;
01502         }
01503
01504 #ifdef __RX__
01505     /*
01506      * Wait for data using ThreadX sleep.
01507      * Yields CPU to other threads instead of busy-waiting.
01508      * Sleep duration and tick rate are defined by threadx_timing_t constants.
01509      */
01510     uint32_t elapsed_ms = 0;
01511     uint32_t iteration = 0;
01512     while (!available && elapsed_ms < timeout_ms && iteration < k_max_poll_iterations) {
01513         iteration++;
01514         tx_thread_sleep(k_poll_sleep_ticks);
01515         elapsed_ms += k_threadx_ms_per_tick;
01516
01517         err = rspi_peripheral_read_available(handle->channel, &available);
01518         if (err != k_rx_ok) {
01519             return err;
01520         }
01521     }
01522
01523     if (!available) {
01524         return k_rx_err_timeout;
01525     }
01526 #else
01527     /* Host builds (testing): no sleep support, timeout immediately */
01528     return k_rx_err_timeout;
01529 #endif
01530 }
01531
01532 return k_rx_ok;
01533 }

```

References `rx_spi_comm_handle_t::channel`, `k_max_poll_iterations`, and `k_poll_sleep_ticks`.

Referenced by `internal_read_and_decode_frame()`.

Here is the caller graph for this function:



4.3.3.15 rx_spi_comm_data_available()

```

rx_err_t rx_spi_comm_data_available (
    const rx_spi_comm_handle_t * handle,
    bool * available) [nodiscard]

```

Check if data is available to receive on SPI.

Check if data is available for reading (non-blocking query).

Definition at line 1998 of file `rx_spi_comm.c`.

```

01999 {
02000     if (handle == nullptr || available == nullptr) {
02001         return k_rx_err_invalid_arg;
02002     }
02003
02004     if (!handle->initialized) {
02005         return k_rx_err_invalid_state;
02006     }
02007
02008     return rspi_peripheral_read_available(handle->channel, available);
02009 }

```

References `rx_spi_comm_handle_t::channel`, and `rx_spi_comm_handle_t::initialized`.

4.3.3.16 rx_spi_comm_deinit()

```
rx_err_t rx_spi_comm_deinit (
    rx_spi_comm_handle_t * handle) [nodiscard]
```

Deinitialize SPI communication layer.

Deinitialize SPI communication handle and release resources.

Tears down the encoder and decoder, clearing internal state. Saves the first error encountered from sub-deinit calls and returns it after completing all teardown steps.

Precondition

handle != nullptr
handle was previously initialized via [rx_spi_comm_init\(\)](#)

Postcondition

handle->initialized == false
Encoder and decoder resources released

Note

Not thread-safe; caller must ensure no concurrent SPI operations.

See also

[rx_spi_comm_init\(\)](#) Corresponding initialization function

Since

Version 1.0.0

Definition at line 1056 of file [rx_spi_comm.c](#).

```
01057 {
01058     if (handle == nullptr) {
01059         return k_rx_err_invalid_arg;
01060     }
01061
01062     rx_err_t err = k_rx_ok;
01063
01064     if (handle->initialized) {
01065         /* Deinit encoder/decoder: can only fail with nullptr pointer, which is
01066          * impossible here since they are fields of a valid handle. */
01067         (void)rx_frame_encoder_deinit(&handle->encoder);
01068         (void)rx_frame_decoder_deinit(&handle->decoder);
01069         handle->initialized = false;
01070     }
01071
01072     rx_log_debug(s_tag, "SPI comm deinitialized");
01073     return err;
01074 }
```

References [rx_spi_comm_handle_t::decoder](#), [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), and [s_tag](#).

4.3.3.17 rx_spi_comm_init()

```
rx_err_t rx_spi_comm_init (
    rx_spi_comm_handle_t * handle,
    const rx_spi_comm_config_t * config) [nodiscard]
```

Initialize SPI communication layer and allocate resources.

Initialize SPI communication handle with frame protocol support.

Initializes SPI communication handle for frame-based communication between RX72N (peripheral) and RPi5 (controller). Sets up frame encoder/decoder, clears staging buffers, and configures channel and FEC settings.

Initialization Steps:

1. Validate handle: Check handle pointer non-nullptr
2. Zero handle: memset entire structure to clear old state
3. Apply configuration: Use config values or defaults (RSPi0, no FEC)
4. Initialize encoder: Call rx_frame_encoder_init() for TX encoding
5. Initialize decoder: Call rx_frame_decoder_init() for RX decoding
6. Reset sequence counters: Set TX/RX sequence to 0
7. Mark initialized: Set handle->initialized = true

Configuration (config must be non-nullptr):

- Channel: Must specify RSPi channel (e.g., k_rspi_channel_2 for host)
- FEC: fec_enabled flag (true/false)
- Passing config == nullptr returns k_rx_err_invalid_arg

Memory Allocation: This function does NOT allocate memory (NASA Rule 3 compliance). Caller must provide pre-allocated handle (typically static or on task stack).

Precondition

handle must point to valid rx_spi_comm_handle_t storage (4120 bytes)
 config != nullptr
 config->session != nullptr
 config->channel must be a valid rspi_channel_t value (0-2)

Postcondition

On success, handle->initialized = true
 On success, handle->tx_sequence = 0, handle->rx_sequence = 0
 On success, handle->encoder and handle->decoder initialized
 On failure (encoder init), handle state is zeroed but not usable
 On failure (decoder init), encoder is cleaned up, handle unusable

Note

Call this ONCE per handle before any send/receive operations
 This function is NOT thread-safe (single-threaded init only)

Warning

Do not call multiple times without `rx_spi_comm_deinit()` first
 RSPI peripheral hardware (rspi HAL) must be initialized separately

Performance:

Execution time: ~50 us @ 240 MHz (memset + encoder/decoder init)

Example - Basic Configuration:

```
// Allocate handle (static or on task stack, NOT malloc)
static rx_spi_comm_handle_t g_spi_handle;

rx_spi_comm_config_t config = {
    .session = &session,
    .channel = k_rspi_channel_0,
    .fec_enabled = false,
};
rx_err_t err = rx_spi_comm_init(&g_spi_handle, &config);
if (err != k_rx_ok) {
    rx_log_error("init", "SPI comm init failed");
    return;
}

// Now ready to send/receive frames
rx_spi_comm_send(&g_spi_handle, k_frame_type_data, 0, payload, len);
```

Example - FEC Enabled:

```
static rx_spi_comm_handle_t g_spi_handle;

// Enable FEC for noisy environments
rx_spi_comm_config_t config = {
    .channel = 0, // RSPi0
    .fec_enabled = true, // Enable Forward Error Correction
};

rx_err_t err = rx_spi_comm_init(&g_spi_handle, &config);
if (err == k_rx_ok) {
    rx_log_info("init", "SPI comm initialized with FEC");
}
```

Example - Multiple RSPI Channels:

```
// Use RSPi1 instead of default RSPi0
static rx_spi_comm_handle_t g_spi1_handle;

rx_spi_comm_config_t config = {
    .channel = 1, // RSPi1 (pins PE0-PE3)
    .fec_enabled = false,
};

rx_spi_comm_init(&g_spi1_handle, &config);
```

See also

[rx_spi_comm_deinit\(\)](#) Clean up resources when done
[rx_spi_comm_config_t](#) Configuration structure definition
[rx_frame_encoder_init\(\)](#) Frame encoder initialization
[rx_frame_decoder_init\(\)](#) Frame decoder initialization

Since

Version 1.0.0

Definition at line 990 of file [rx_spi_comm.c](#).

```

00991 {
00992     if (handle == nullptr) {
00993         return k_rx_err_invalid_arg;
00994     }
00995
00996     if (config == nullptr || config->session == nullptr) {
00997         return k_rx_err_invalid_arg;
00998     }
00999
01000     *handle = s_zero_handle;
01001
01002     /* Apply configuration */
01003     handle->session      = config->session;
01004     handle->channel       = config->channel;
01005     handle->fec_enabled   = config->fec_enabled;
01006     handle->auto_retransmit = config->auto_retransmit;
01007     handle->retransmit_cfg = config->retransmit_config;
01008
01009     /* Apply retransmit defaults for zero fields */
01010     if (handle->auto_retransmit) {
01011         if (handle->retransmit_cfg.max_retries == 0) {
01012             handle->retransmit_cfg.max_retries = k_retransmit_default_max_retries;
01013         }
01014         if (handle->retransmit_cfg.ack_timeout_ms == 0) {
01015             handle->retransmit_cfg.ack_timeout_ms = k_retransmit_default_ack_timeout_ms;
01016         }
01017         if (handle->retransmit_cfg.max_backoff_ms == 0) {
01018             handle->retransmit_cfg.max_backoff_ms = k_retransmit_default_max_backoff_ms;
01019         }
01020     }
01021
01022     /* Initialize frame encoder: can only fail if pointer is nullptr, which is
01023      * impossible here since &handle->encoder is always non-null. */
01024     (void)(rx_frame_encoder_init(&handle->encoder));
01025
01026     /* Initialize frame decoder: same invariant as encoder above. */
01027     (void)(rx_frame_decoder_init(&handle->decoder));
01028
01029     /* Sequence counters managed by shared session (config->session) */
01030
01031     handle->initialized = true;
01032
01033     rx_log_debug(s_tag, "SPI comm initialized");
01034     return k_rx_ok;
01035 }

```

References [rx_spi_comm_retransmit_config_t::ack_timeout_ms](#), [rx_spi_comm_config_t::auto_retransmit](#), [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_config_t::channel](#), [rx_spi_comm_handle_t::channel](#), [rx_spi_comm_handle_t::decoder](#), [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_config_t::fec_enabled](#), [rx_spi_comm_handle_t::fec_enabled](#), [rx_spi_comm_handle_t::initialized](#), [k_retransmit_default_ack_timeout_ms](#), [k_retransmit_default_max_backoff_ms](#), [k_retransmit_default_max_retries](#), [rx_spi_comm_retransmit_config_t::max_retries](#), [rx_spi_comm_handle_t::retransmit_cfg](#), [rx_spi_comm_config_t::s_tag](#), [s_zero_handle](#), [rx_spi_comm_config_t::session](#), and [rx_spi_comm_handle_t::session](#).

4.3.3.18 rx_spi_comm_process_retransmits()

```

rx_err_t rx_spi_comm_process_retransmits (
    rx_spi_comm_handle_t * handle,
    const uint32_t current_time_ms) [nodiscard]

```

Process pending retransmissions based on timeout.

Check and process automatic retransmissions.

Precondition

handle must be non-NULL and initialized

Postcondition

retry_count incremented on retransmit, pending cleared on limit

Note

Safe to call when auto_retransmit is disabled (no-op)

Since

Version 1.0.0

Definition at line 2239 of file rx_spi_comm.c.

```

02241 {
02242     if (handle == nullptr) {
02243         return k_rx_err_invalid_arg;
02244     }
02245     if (!handle->initialized) {
02246         return k_rx_err_invalid_state;
02247     }
02248 }
02249
02250 /* No-op when retransmit disabled or nothing pending */
02251 if (!handle->auto_retransmit || !handle->retry_pending) {
02252     return k_rx_ok;
02253 }
02254
02255 /* Compute exponential backoff: ack_timeout_ms * 2^retry_count, capped.
02256  * Clamp shift to 31 to avoid undefined behavior when retry_count >= 32. */
02257 uint8_t shift = handle->retry_count;
02258 if (handle->retry_count >= k_max_backoff_shift) {
02259     shift = k_max_backoff_shift;
02260 }
02261 uint32_t backoff_ms = (uint32_t)handle->retransmit_cfg.ack_timeout_ms « shift;
02262
02263 if (backoff_ms > handle->retransmit_cfg.max_backoff_ms) {
02264     backoff_ms = handle->retransmit_cfg.max_backoff_ms;
02265 }
02266
02267 /* Check if timeout has elapsed */
02268 const uint32_t elapsed_ms = current_time_ms - handle->retry_send_time_ms;
02269 if (elapsed_ms < backoff_ms) {
02270     return k_rx_ok; /* Not yet time to retry */
02271 }
02272
02273 const rx_err_t err = internal_retransmit_frame(handle);
02274
02275 /* Override send time with caller-provided time (more accurate than internal timestamp) */
02276 handle->retry_send_time_ms = current_time_ms;
02277
02278 return err;
02279 }

```

References [rx_spi_comm_retransmit_config_t::ack_timeout_ms](#), [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_handle_t::initialized](#), [internal_retransmit_frame\(\)](#), [k_max_backoff_shift](#), [rx_spi_comm_retransmit_cfg](#), [rx_spi_comm_handle_t::retransmit_cfg](#), [rx_spi_comm_handle_t::retry_count](#), [rx_spi_comm_handle_t::retry_pending](#) and [rx_spi_comm_handle_t::retry_send_time_ms](#).

Here is the call graph for this function:



4.3.3.19 rx_spi_comm_receive()

```
rx_err_t rx_spi_comm_receive (
    rx_spi_comm_handle_t * handle,
    rx_frame_t * frame,
    uint32_t timeout_ms) [nodiscard]
```

Receive and decode frame from RPi5 with CRC validation.

Reads raw bytes from SPI hardware, validates CRC-32, decodes frame header and payload, and optionally applies FEC decoding with HARQ soft-combining. Returns complete decoded frame to caller.

Algorithm:

1. Validate parameters: Check handle, frame pointer (NASA Rule 5)
2. Poll SPI hardware: Read available bytes (non-blocking if timeout=0)
3. Optional FEC decode: If enabled, apply FEC decoding with HARQ
4. Decode frame: Parse header, extract payload
5. Validate CRC: Compute CRC-32 and compare with frame CRC
6. Validate sequence: Check if sequence matches expected rx_sequence
7. Update state: Increment rx_sequence on success
8. Return frame: Copy decoded frame to output parameter

Blocking Behavior:

- timeout_ms = 0: Non-blocking poll (returns immediately)
- timeout_ms > 0: Blocks up to timeout_ms waiting for data
- timeout_ms = k_spi_comm_default_timeout: Use default (1 second)

Performance:

- Small frame (16 bytes): ~100 us
- Medium frame (128 bytes): ~250 us
- Large frame (255 bytes): ~400 us
- FEC overhead: +100 us

Parameters

in,out	handle	Initialized SPI communication handle - Valid range: Non-nullptr to initialized handle - Constraints: Must be initialized via <code>rx_spi_comm_init()</code> - Side effects: Updates shared session RX sequence via <code>rx_session_validate_rx()</code>
out	frame	Pointer to frame structure to receive decoded frame - Valid range: Non-nullptr to allocated <code>rx_frame_t</code> - Constraints: Must have sufficient space (263 bytes) - Lifetime: Filled by this function, owned by caller afterward
in	timeout_ms	Timeout in milliseconds - Valid range: [0, 60000] (0 = non-blocking, up to 1 minute) - Units: Milliseconds - Common values: 0 (poll), <code>k_spi_comm_default_timeout</code> (1 second)

Returns

`rx_err_t` Error code indicating receive result

Return values

<code>k_rx_ok</code>	Frame received and decoded successfully, <code>rx_sequence</code> incremented
<code>k_rx_err_invalid_arg</code>	handle or frame is nullptr
<code>k_rx_err_invalid_state</code>	handle not initialized
<code>k_rx_err_timeout</code>	No data received within timeout (not an error if timeout=0)
<code>k_rx_err_crc_mismatch</code>	CRC validation failed (frame corrupted)
<code>k_rx_err_protocol_error</code>	Frame format invalid (bad sync, length, etc.)
<code>k_rx_err_sequence_error</code>	Unexpected sequence number (out-of-order)
<code>k_rx_err_hw</code>	SPI hardware error

Precondition

handle must be non-NULL and initialized
 frame must be non-NULL and allocated
 RSPI hardware must be initialized and ready

Postcondition

On `k_rx_ok`: frame filled with decoded data, `rx_sequence` incremented
 On timeout: frame unchanged, `rx_sequence` unchanged
 On CRC error: frame may be partially filled, `rx_sequence` unchanged

Note

timeout=0 is recommended for polling loops (non-blocking)
CRC errors should trigger NACK and retransmission request
Sequence errors indicate lost frames or out-of-order delivery

Warning

Must call [rx_spi_comm_send_ack\(\)](#) or [rx_spi_comm_send_nack\(\)](#) after receive

Example - Non-Blocking Poll:

```
rx_frame_t frame;

// Poll with 0ms timeout (non-blocking)
rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0);

if (err == k_rx_ok) {
    // Frame received successfully
    handle_frame(&frame);
    rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
} else if (err == k_rx_err_timeout) {
    // No data available - this is normal
} else if (err == k_rx_err_crc_mismatch) {
    // CRC error - request retransmission
    rx_spi_comm_send_nack(&g_spi_handle, frame.header.sequence, k_frame_flag_none);
} else {
    // Other error - log and continue
    log_error("Receive error: %d", err);
}
```

Example - Blocking Receive with Timeout:

```
rx_frame_t frame;

// Wait up to 1 second for frame
rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 1000);

if (err == k_rx_ok) {
    // Frame received within timeout
    process_frame(&frame);
    rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
} else if (err == k_rx_err_timeout) {
    // No frame received within 1 second - RPi5 may be idle
    log_warn("No frame received within timeout");
}
```

See also

[rx_spi_comm_send\(\)](#) Send frames to RPi5
[rx_spi_comm_send_ack\(\)](#) Send ACK for received frame
[rx_spi_comm_send_nack\(\)](#) Send NACK on error
[rx_spi_comm_data_available\(\)](#) Check if data waiting
[rx_frame.h](#) Frame structure definition

Since

Version 1.0.0

Definition at line 1958 of file [rx_spi_comm.c](#).

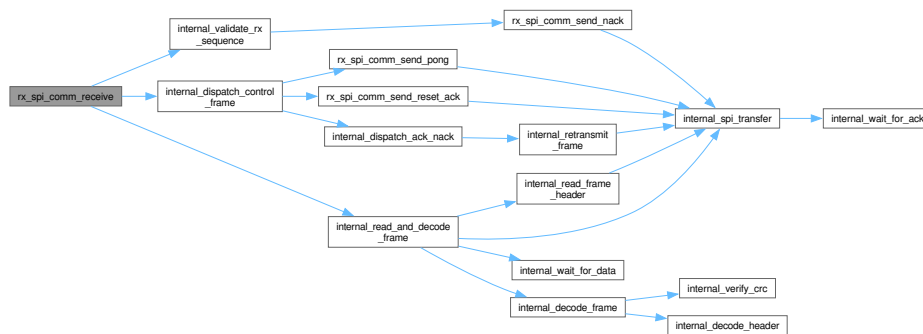
```

01959 {
01960     if (handle == nullptr || frame == nullptr) {
01961         return k_rx_err_invalid_arg;
01962     }
01963
01964     if (!handle->initialized) {
01965         return k_rx_err_invalid_state;
01966     }
01967
01968     /* Loop to handle control frames internally before returning data frames.
01969      * Bounded to k_max_control_frames_per_receive iterations (NASA Rule 2). */
01970     for (uint8_t ctrl_count = 0; ctrl_count < k_max_control_frames_per_receive; ctrl_count++) {
01971         rx_err_t err = internal_read_and_decode_frame(handle, frame, timeout_ms);
01972         if (err != k_rx_ok) {
01973             return err;
01974         }
01975
01976         /* Dispatch control frames (PING, RESET, ACK/NACK).
01977          * internal_dispatch_control_frame always returns k_rx_ok; errors from
01978          * sub-calls (PONG/RESET_ACK send failures) are logged and swallowed
01979          * internally. This is an invariant. */
01980         ctrl_dispatch_result_t ctrl_result = k_ctrl_not_handled;
01981         (void)internal_dispatch_control_frame(handle, frame, &ctrl_result);
01982         if (ctrl_result == k_ctrl_consumed) {
01983             continue; /* Control frame handled; loop for next frame */
01984         }
01985
01986         /* Data frame: validate sequence (NACK on rejection per Audit F-05) */
01987         return internal_validate_rx_sequence(handle, frame);
01988     }
01989
01990     /* Exceeded max control frames without receiving a data frame */
01991     rx_log_error(s_tag, "Too many control frames, no data frame received");
01992     return k_rx_err_timeout;
01993 }

```

References [rx_spi_comm_handle_t::initialized](#), [internal_dispatch_control_frame\(\)](#), [internal_read_and_decode_frame](#), [internal_validate_rx_sequence\(\)](#), [k_ctrl_consumed](#), [k_ctrl_not_handled](#), [k_max_control_frames_per_receive](#), and [s_tag](#).

Here is the call graph for this function:



4.3.3.20 rx_spi_comm_send()

```

rx_err_t rx_spi_comm_send (
    rx_spi_comm_handle_t * handle,
    rx_frame_type_t type,
    uint8_t flags,
    const uint8_t * payload,
    uint32_t payload_len) [nodiscard]

```

Send frame on SPI with automatic sequencing and CRC.

Encodes payload into frame format with automatic sequence number assignment, CRC-32 calculation, and optional FEC encoding. Transmits frame to RPi5 via SPI hardware. This is the primary send function for all frame types.

Algorithm:

1. Validate parameters: Check handle, payload, payload_len (NASA Rule 5)
2. Assign sequence: Get next TX sequence from shared rx_session_state_t
3. Build frame: Create frame header (type, flags, length, sequence)
4. Encode frame: Call frame encoder to add CRC-32
5. Optional FEC: If enabled, apply FEC encoding
6. SPI transfer: Transmit encoded frame via RSPI peripheral
7. Update state: TX sequence already incremented by rx_session_next_tx()

Performance:

- Small payload (16 bytes): ~150 us
- Medium payload (128 bytes): ~300 us
- Large payload (255 bytes): ~500 us
- FEC overhead: +50% time

Parameters

in,out	handle	Initialized SPI communication handle • Valid range: Non-nullptr to initialized handle • Constraints: Must be initialized via rx_spi_comm_init() • Side effects: Increments shared session TX sequence via rx_session_next_tx()
in	type	Frame type (command, request, response, etc.) • Valid range: Any rx_frame_type_t value • Common values: k_frame_type_response (most common for RX72N) • Units: Enumeration value
in	flags	Frame flags (ACK, NACK, etc.) • Valid range: Bitmask of rx_frame_flag_t values • Common values: k_frame_flag_none (0x00) • Units: Bitmask

in	payload	Payload data to send - Valid range: Non-NULL if payload_len > 0, may be NULL if payload_len == 0 - Constraints: Must contain valid data for payload_len bytes - Lifetime: Copied immediately, does not need to persist - Max size: k_frame_max_payload (255 bytes)
in	payload_len	Payload length in bytes - Valid range: [0, k_frame_max_payload] (0-255 bytes) - Units: Bytes - Constraints: Must be <= k_frame_max_payload

Returns

rx_err_t Error code indicating send success or failure

Return values

k_rx_ok	Frame sent successfully, tx_sequence incremented
k_rx_err_invalid_arg	handle or payload is nullptr; payload_len > k_frame_max_payload (255 bytes); payload is nullptr but payload_len > 0
k_rx_err_invalid_state	handle not initialized
k_rx_err_hw	SPI hardware error (RSPI transfer failed)
k_rx_err_timeout	SPI transfer timeout

Precondition

handle must be non-NULL and initialized

If payload_len > 0, payload must point to valid data

RSPI hardware must be initialized and ready

Postcondition

Shared session TX sequence incremented (wraps at 65535)

Frame transmitted to RPi5 if successful

Note

Sequence number managed automatically - caller does not specify

CRC-32 calculated automatically by frame encoder

FEC applied automatically if handle->fec_enabled = true

Warning

Ensure RPi5 is ready to receive (chip select asserted)

Example - Send Telemetry Response:

```
uint8_t telemetry[16];
// ... fill telemetry data ...

rx_err_t err = rx_spi_comm_send(&g_spi_handle,
                                k_frame_type_response,
                                k_frame_flag_none,
                                telemetry,
                                sizeof(telemetry));

if (err == k_rx_ok) {
    // Telemetry sent successfully
} else {
    log_error("Send failed: %d", err);
}
```

Example - Send Empty ACK (No Payload):

```
// Send empty frame with ACK flag
rx_err_t err = rx_spi_comm_send(&g_spi_handle,
                                k_frame_type_response,
                                k_frame_flag_ack,
                                NULL, // No payload
                                0);  // Zero length
```

See also

[rx_spi_comm_send_ack\(\)](#) Convenience wrapper for ACK frames
[rx_spi_comm_send_nack\(\)](#) Convenience wrapper for NACK frames
[rx_spi_comm_receive\(\)](#) Receive frames from RPi5
[rx_frame.h](#) Frame protocol definitions

Since

Version 1.0.0

Definition at line 1370 of file [rx_spi_comm.c](#).

```
01375 {
01376     if (handle == nullptr) {
01377         return k_rx_err_invalid_arg;
01378     }
01379
01380     if (!handle->initialized) {
01381         rx_log_error(s_tag, "Handle not initialized");
01382         return k_rx_err_invalid_state;
01383     }
01384
01385     /* Validate payload before consuming a sequence number (Issue #9). */
01386     if (payload == nullptr && payload_len > 0) {
01387         return k_rx_err_invalid_arg;
01388     }
01389     if (payload_len > k_frame_max_payload) {
01390         rx_log_error(s_tag, "Payload too large");
01391         return k_rx_err_invalid_size;
01392     }
01393
01394     /* Get next TX sequence from shared session. Invariant: cannot fail. */
01395     uint16_t sequence = 0;
01396     (void)(rx_session_next_tx(handle->session, &sequence));
01397
01398     /* Build frame. Cannot fail: handle and &frame are non-null, payload validated. */
01399     rx_frame_t frame;
01400     (void)(internal_build_frame(handle, sequence, type, flags, payload, payload_len, &frame));
01401 }
```

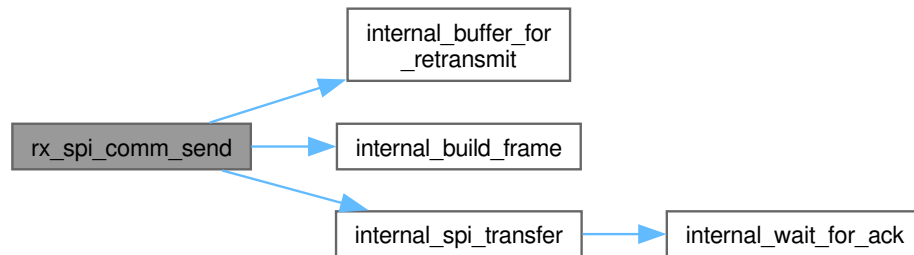
```

01402 /* Encode frame into handle-owned scratch buffer (not stack -- see comment
01403 * on tx_encode_buffer in rx_spi_comm.h). Cannot fail: encoder initialized,
01404 * pointers valid. */
01405 uint32_t wire_len = 0;
01406 (void)(rx_frame_encode(&handle->encoder, &frame, handle->tx_encode_buffer, &wire_len));
01407
01408 /* Transfer via SPI (waits for host ACK internally) */
01409 rx_err_t xfer_err = internal_spi_transfer(handle, handle->tx_encode_buffer, wire_len, nullptr, 0);
01410 if (xfer_err != k_rx_ok) {
01411     rx_log_error(s_tag, "SPI transfer failed");
01412     return xfer_err;
01413 }
01414
01415 /* Buffer frame for retransmission if enabled and requires-ACK flag set */
01416 if ((int)handle->auto_retransmit && ((flags & k_frame_flag_requires_ack) != 0)) {
01417     internal_buffer_for_retransmit(handle, handle->tx_encode_buffer, wire_len, sequence);
01418 }
01419
01420 return k_rx_ok;
01421 }

```

References [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::init](#), [internal_buffer_for_retransmit\(\)](#), [internal_build_frame\(\)](#), [internal_spi_transfer\(\)](#), [s_tag](#), [rx_spi_comm_handle_t::sequence](#) and [rx_spi_comm_handle_t::tx_encode_buffer](#).

Here is the call graph for this function:



4.3.3.21 rx_spi_comm_send_ack()

```

rx_err_t rx_spi_comm_send_ack (
    rx_spi_comm_handle_t * handle,
    const uint16_t sequence) [nodiscard]

```

Send ACK frame for specified sequence number.

Send ACK frame for received frame (convenience wrapper).

Definition at line 1426 of file [rx_spi_comm.c](#).

```

01427 {
01428     if (handle == nullptr) {
01429         return k_rx_err_invalid_arg;
01430     }
01431
01432     if (!handle->initialized) {
01433         return k_rx_err_invalid_state;
01434     }
01435
01436     /* Create ACK frame: cannot fail since &ack_frame is non-null. */
01437     rx_frame_t ack_frame;
01438     (void)(rx_frame_create_ack(&ack_frame, sequence));
01439
01440     /* Encode and send: encoder initialized at rx_spi_comm_init; all pointers valid. */
01441     uint8_t wire_buffer[k_frame_min_size];
01442     uint32_t wire_len = 0;
01443
01444     (void)(rx_frame_encode(&handle->encoder, &ack_frame, wire_buffer, &wire_len));
01445
01446     /* Post-condition: ACK frame encoded to minimum size */
01447 }

```

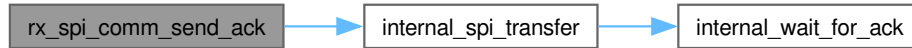
```

01448  /* Transfer via SPI (waits for host ACK internally) */
01449  return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
01450 }

```

References [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), and [internal_spi_transfer\(\)](#).

Here is the call graph for this function:



4.3.3.22 rx_spi_comm_send_nack()

```

rx_err_t rx_spi_comm_send_nack (
    rx_spi_comm_handle_t * handle,
    const uint16_t sequence,
    uint8_t flags) [nodiscard]

```

Send NACK frame for specified sequence number.

Send NACK frame for received frame with error indication.

Definition at line 1455 of file [rx_spi_comm.c](#).

```

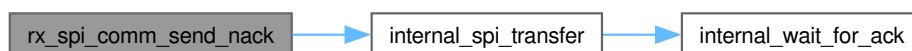
01456 {
01457     if (handle == nullptr) {
01458         return k_rx_err_invalid_arg;
01459     }
01460
01461     if (!handle->initialized) {
01462         return k_rx_err_invalid_state;
01463     }
01464
01465     /* Create NACK frame: cannot fail since &nack_frame is non-null. */
01466     rx_frame_t nack_frame;
01467     (void)(rx_frame_create_nack(&nack_frame, sequence, flags));
01468
01469     /* Encode and send: encoder initialized at rx_spi_comm_init; all pointers valid. */
01470     uint8_t wire_buffer[k_frame_min_size];
01471     uint32_t wire_len = 0;
01472
01473     (void)(rx_frame_encode(&handle->encoder, &nack_frame, wire_buffer, &wire_len));
01474
01475     /* Post-condition: NACK frame encoded to minimum size */
01476
01477     /* Transfer via SPI (waits for host ACK internally) */
01478     return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
01479 }

```

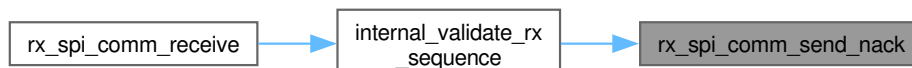
References [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), and [internal_spi_transfer\(\)](#).

Referenced by [internal_validate_rx_sequence\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.23 rx_spi_comm_send_pong()

```
rx_err_t rx_spi_comm_send_pong (
    rx_spi_comm_handle_t * handle,
    const uint8_t * payload,
    uint32_t payload_len) [nodiscard]
```

Send PONG frame with echoed payload.

Send PONG frame with payload (response to PING).

Constructs a PONG frame echoing the given payload, acquires the next TX sequence number from the shared session, encodes the frame, and transfers it over SPI. Typically called automatically by the receive path in response to a PING.

Precondition

```
handle != nullptr && handle->initialized
payload != NULL when payload_len > 0
```

Postcondition

```
TX sequence incremented on success
PONG frame sent over SPI
```

Note

Not thread-safe; called from the SPI receive context.

See also

`rx_frame_create_pong()` Frame construction helper

Since

Version 1.0.0

Definition at line 2089 of file `rx_spi_comm.c`.

```
02090 {
02091     if (handle == nullptr) {
02092         return k_rx_err_invalid_arg;
02093     }
02094     if (!handle->initialized) {
02095         return k_rx_err_invalid_state;
02096     }
02097 }
02098
02099 /* Get next TX sequence from shared session.
02100  * rx_session_next_tx only fails for null/uninitialized session; handle->session
02101  * is valid (set at init time) and the handle is initialized above. Invariant. */
02102 uint16_t sequence = 0;
02103 (void)(rx_session_next_tx(handle->session, &sequence));
02104
02105 /* Create PONG frame: cannot fail since &pong_frame is non-null and payload
02106  * is validated above (nullptr allowed when payload_len == 0). */
02107 rx_frame_t pong_frame;
02108 (void)(rx_frame_create_pong(&pong_frame, sequence, payload, payload_len));
02109
02110 /* Encode into handle-owned scratch buffer (not stack -- see comment on
02111  * tx_encode_buffer in rx_spi_comm.h). encoder initialized at
02112  * rx_spi_comm_init; all pointers valid. */
```

```

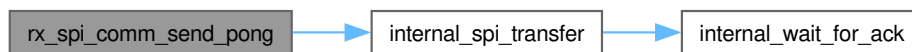
02113 uint32_t wire_len = 0;
02114 (void)(rx_frame_encode(&handle->encoder, &pong_frame, handle->tx_encode_buffer, &wire_len));
02115
02116 /* Transfer via SPI */
02117 return internal_spi_transfer(handle, handle->tx_encode_buffer, wire_len, nullptr, 0);
02118 }

```

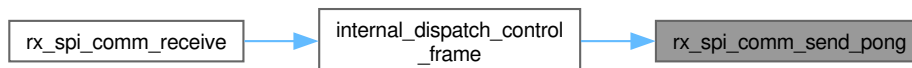
References [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), [internal_spi_transfer\(\)](#), [rx_spi_comm_handle_t::session](#), and [rx_spi_comm_handle_t::tx_encode_buffer](#).

Referenced by [internal_dispatch_control_frame\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.24 rx_spi_comm_send_reset_ack()

```

rx_err_t rx_spi_comm_send_reset_ack (
    rx_spi_comm_handle_t * handle) [nodiscard]

```

Send RESET_ACK frame confirming a reset request.

Send RESET_ACK frame (response to RESET).

Constructs a RESET_ACK frame (empty payload), acquires the next TX sequence number, encodes, and transfers over SPI. Called automatically by the receive path after processing a RESET frame. The RESET_ACK itself does NOT reset sequence numbers; that is handled by the session layer in response to the RESET.

Precondition

handle != nullptr && handle->initialized
 A RESET frame was received and session state cleared

Postcondition

TX sequence incremented on success
 RESET_ACK frame sent over SPI

Note

Not thread-safe; called from the SPI receive context.

See also

[rx_frame_create_reset_ack\(\)](#) Frame construction helper
[rx_spi_comm_set_control_callbacks\(\)](#) Register RESET notification

Since

Version 1.0.0

Definition at line 2142 of file [rx_spi_comm.c](#).

```

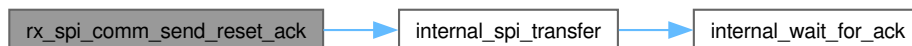
02143 {
02144     if (handle == nullptr) {
02145         return k_rx_err_invalid_arg;
02146     }
02147
02148     if (!handle->initialized) {
02149         return k_rx_err_invalid_state;
02150     }
02151
02152     /* Get next TX sequence from shared session.
02153      * rx_session_next_tx only fails for null/uninitialized session; handle->session
02154      * is valid and the handle is initialized above. Invariant. */
02155     uint16_t sequence = 0;
02156     (void)(rx_session_next_tx(handle->session, &sequence));
02157
02158     /* Create RESET_ACK frame: cannot fail since &reset_ack_frame is non-null. */
02159     rx_frame_t reset_ack_frame;
02160     (void)(rx_frame_create_reset_ack(&reset_ack_frame, sequence));
02161
02162     /* Encode: encoder initialized at rx_spi_comm_init; all pointers valid. */
02163     uint8_t wire_buffer[k_frame_min_size];
02164     uint32_t wire_len = 0;
02165
02166     (void)(rx_frame_encode(&handle->encoder, &reset_ack_frame, wire_buffer, &wire_len));
02167
02168     /* Post-condition: RESET_ACK frame encoded to minimum size */
02169
02170     /* Transfer via SPI */
02171     return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
02172 }

```

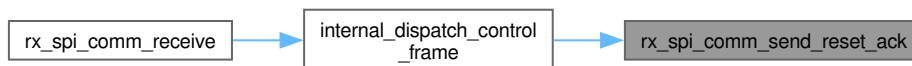
References [rx_spi_comm_handle_t::encoder](#), [rx_spi_comm_handle_t::initialized](#), [internal_spi_transfer\(\)](#), and [rx_spi_comm_handle_t::session](#).

Referenced by [internal_dispatch_control_frame\(\)](#).

Here is the call graph for this function:



Here is the caller graph for this function:



4.3.3.25 rx_spi_comm_set_auto_retransmit()

```
rx_err_t rx_spi_comm_set_auto_retransmit (
    rx_spi_comm_handle_t * handle,
    const bool enabled,
    const rx_spi_comm_retransmit_config_t * config)  [nodiscard]
```

Enable or disable automatic retransmission at runtime.

Precondition

handle must be non-NULL and initialized

Postcondition

auto_retransmit flag and config updated

Since

Version 1.0.0

Definition at line 2291 of file [rx_spi_comm.c](#).

```
02294 {
02295     if (handle == nullptr) {
02296         return k_rx_err_invalid_arg;
02297     }
02298
02299     if (!handle->initialized) {
02300         return k_rx_err_invalid_state;
02301     }
02302
02303     handle->auto_retransmit = enabled;
02304
02305     /* Apply new config if provided */
02306     if (config != nullptr) {
02307         handle->retransmit_cfg = *config;
02308     }
02309
02310     /* Apply defaults for zero fields */
02311     if (handle->retransmit_cfg.max_retries == 0) {
02312         handle->retransmit_cfg.max_retries = k_retransmit_default_max_retries;
02313     }
02314     if (handle->retransmit_cfg.ack_timeout_ms == 0) {
02315         handle->retransmit_cfg.ack_timeout_ms = k_retransmit_default_ack_timeout_ms;
02316     }
02317     if (handle->retransmit_cfg.max_backoff_ms == 0) {
02318         handle->retransmit_cfg.max_backoff_ms = k_retransmit_default_max_backoff_ms;
02319     }
02320
02321     /* Clear pending retry state when disabling */
02322     if (!enabled) {
02323         handle->retry_pending = false;
02324         handle->retry_count = 0;
02325     }
02326
02327     return k_rx_ok;
02328 }
```

References [rx_spi_comm_retransmit_config_t::ack_timeout_ms](#), [rx_spi_comm_handle_t::auto_retransmit](#), [rx_spi_comm_handle_t::initialized](#), [k_retransmit_default_ack_timeout_ms](#), [k_retransmit_default_max_backoff_ms](#), [k_retransmit_default_max_retries](#), [rx_spi_comm_retransmit_config_t::max_backoff_ms](#), [rx_spi_comm_retransmit_config_t::max_retries](#), [rx_spi_comm_retransmit_config_t::ack_timeout_ms](#), [rx_spi_comm_handle_t::retransmit_cfg](#), [rx_spi_comm_handle_t::retry_count](#), and [rx_spi_comm_handle_t::retry_pending](#).

4.3.3.26 rx_spi_comm_set_control_callbacks()

```
rx_err_t rx_spi_comm_set_control_callbacks (
    rx_spi_comm_handle_t * handle,
    void(* on_ping_cb )(const rx_frame_t *frame, void *ctx),
    void(* on_reset_cb )(const rx_frame_t *frame, void *ctx),
    void * cb_ctx) [nodiscard]
```

Register callbacks for PING and RESET control frames.

Stores function pointers that the receive path invokes when it decodes a PING or RESET frame. Passing NULL for a callback disables notification for that frame type while still performing the automatic PONG / RESET_ACK response.

Precondition

```
handle != nullptr
handle->initialized == true
```

Postcondition

```
Callbacks stored; invoked on next matching control frame
Previous callbacks overwritten
```

Note

Not thread-safe; register before starting the receive loop.

See also

```
rx_spi_comm_send_pong() Auto-response sent on PING
rx_spi_comm_send_reset_ack() Auto-response sent on RESET
```

Since

Version 1.0.0

Definition at line 2048 of file rx_spi_comm.c.

```
02052 {
02053     if (handle == nullptr) {
02054         return k_rx_err_invalid_arg;
02055     }
02056     if (!handle->initialized) {
02057         return k_rx_err_invalid_state;
02058     }
02059 }
02060 handle->on_ping_cb    = on_ping_cb;
02061 handle->on_reset_cb   = on_reset_cb;
02062 handle->control_cb_ctx = cb_ctx;
02063
02064 return k_rx_ok;
02065 }
```

References [rx_spi_comm_handle_t::control_cb_ctx](#), [rx_spi_comm_handle_t::initialized](#), [rx_spi_comm_handle_t::on_ping_cb](#) and [rx_spi_comm_handle_t::on_reset_cb](#).

4.3.3.27 rx_spi_comm_set_retransmit_callbacks()

```
rx_err_t rx_spi_comm_set_retransmit_callbacks (
    rx_spi_comm_handle_t * handle,
    void(* on_ack_cb )(uint16_t sequence, void *ctx),
    void(* on_nack_cb )(uint16_t sequence, void *ctx),
    void * ctx) [nodiscard]
```

Register ACK/NACK notification callbacks.

Register ACK/NACK notification callbacks for retransmission.

Since

Version 1.0.0

Definition at line 2337 of file [rx_spi_comm.c](#).

```
02341 {
02342     if (handle == nullptr) {
02343         return k_rx_err_invalid_arg;
02344     }
02345     if (!handle->initialized) {
02346         return k_rx_err_invalid_state;
02347     }
02348 }
02349
02350 handle->on_ack_cb      = on_ack_cb;
02351 handle->on_nack_cb     = on_nack_cb;
02352 handle->retransmit_cb_ctx = ctx;
02353
02354 return k_rx_ok;
02355 }
```

References [rx_spi_comm_handle_t::initialized](#), [rx_spi_comm_handle_t::on_ack_cb](#), [rx_spi_comm_handle_t::on_nack_cb](#), and [rx_spi_comm_handle_t::retransmit_cb_ctx](#).

4.3.4 Variable Documentation

4.3.4.1 s_tag

```
const char s_tag[] = "rx_spi_comm" [static]
```

Definition at line 317 of file [rx_spi_comm.c](#).

Referenced by [internal_build_frame\(\)](#), [internal_decode_frame\(\)](#), [internal_dispatch_ack_nack\(\)](#), [internal_dispatch_control_frame\(\)](#), [internal_read_and_decode_frame\(\)](#), [internal_read_frame_header\(\)](#), [internal_retransmit_frame\(\)](#), [internal_spi_transfer\(\)](#), [internal_validate_rx_sequence\(\)](#), [internal_wait_for_ack\(\)](#), [rx_spi_comm_deinit\(\)](#), [rx_spi_comm_init\(\)](#), [rx_spi_comm_receive\(\)](#), and [rx_spi_comm_send\(\)](#).

4.3.4.2 s_zero_handle

```
const rx_spi_comm_handle_t s_zero_handle = {} [static]
```

Zero-initialized SPI comm handle template for stack-safe initialization.

Static const instance used to clear `rx_spi_comm_handle_t` without creating a large compound literal on the stack. Lives in .rodata section.

Note

Read-only; never modified after static initialization

Warning

Do not remove - prevents stack overflow in init function

See also

`rx_spi_comm_init()` Uses this template to zero-initialize the comm handle

Since

Version 1.0.0

Definition at line 885 of file `rx_spi_comm.c`.

00885 {};

Referenced by `rx_spi_comm_init()`.

4.4 rx_spi_comm.c

[Go to the documentation of this file.](#)

```
00001 /**
00002  * @file rx_spi_comm.c
00003  * @brief High-Level SPI Frame Protocol Communication Layer Implementation
00004  *
00005  * @details
00006  * ## Overview
00007  * Production-quality implementation of SPI frame-based communication protocol
00008  * for reliable data exchange between RX72N MCU (SPI peripheral) and Raspberry
00009  * Pi 5 (SPI controller). Integrates frame encoding/decoding (rx_frame), CRC-32
00010  * validation (rx_crc), and RSPI peripheral hardware abstraction (rspi HAL).
00011  *
00012  * This module provides the complete implementation of the API defined in
00013  * rx_spi_comm.h, handling all aspects of frame transmission, reception, flow
00014  * control, and error recovery.
00015  *
00016  * ## Implementation Architecture
00017  *
00018  * @dot
00019  * digraph spi_comm_impl {
00020  *   rankdir=TB;
00021  *   node [shape=box, style=rounded];
00022  *
00023  *   subgraph cluster_public {
00024  *     label="Public API (rx_spi_comm.h)";
00025  *     style=filled;
00026  *     color=lightblue;
00027  *     send [label="rx_spi_comm_send()"];
00028  *     receive [label="rx_spi_comm_receive()"];
```

```

00029 *   send_ack [label="rx_spi_comm_send_ack()"];
00030 *   send_nack [label="rx_spi_comm_send_nack()"];
00031 *   data_avail [label="rx_spi_comm_data_available()"];
00032 * }
00033 *
00034 * subgraph cluster_internal {
00035 *   label="Internal Helpers (static)";
00036 *   style=filled;
00037 *   color=lightyellow;
00038 *   build_frame [label="internal_build_frame()"];
00039 *   wait_ack [label="internal_wait_for_ack()"];
00040 *   spi_xfer [label="internal_spi_transfer()"];
00041 *   wait_data [label="internal_wait_for_data()"];
00042 *   read_hdr [label="internal_read_frame_header()"];
00043 *   decode_hdr [label="internal_decode_header()"];
00044 *   decode_frame [label="internal_decode_frame()"];
00045 *   verify_crc [label="internal_verify_crc()"];
00046 * }
00047 *
00048 * subgraph cluster_deps {
00049 *   label="Dependencies";
00050 *   style=filled;
00051 *   color=lightgray;
00052 *   frame_enc [label="rx_frame_encode()"];
00053 *   frame_dec [label="rx_frame_decode()"];
00054 *   crc32 [label="rx_crc32_ieee()"];
00055 *   rspi_xfer [label="rspi_peripheral_transfer()"];
00056 *   rspi_ready [label="rspi_peripheral_write_ready()"];
00057 *   rspi_avail [label="rspi_peripheral_read_available()"];
00058 * }
00059 *
00060 * send -> build_frame -> frame_enc;
00061 * send -> spi_xfer -> wait_ack -> rspi_ready;
00062 * spi_xfer -> rspi_xfer;
00063 *
00064 * receive -> wait_data -> rspi_avail;
00065 * receive -> read_hdr -> spi_xfer;
00066 * receive -> decode_frame -> decode_hdr;
00067 * decode_frame -> verify_crc -> crc32;
00068 *
00069 * send_ack -> frame_enc -> spi_xfer;
00070 * send_nack -> frame_enc -> spi_xfer;
00071 * data_avail -> rspi_avail;
00072 * }
00073 * @enddot
00074 *
00075 * ## Data Flow - Transmission
00076 *
00077 * ```
00078 * Application
00079 * |
00080 * | rx_spi_comm_send(payload, len)
00081 * | v
00082 * | internal_build_frame()      # Populate rx_frame_t with header + payload
00083 * | |
00084 * | | v
00085 * | | rx_frame_encode()        # Encode to wire format (sync, header, CRC)
00086 * | | |
00087 * | | | v
00088 * | | | internal_spi_transfer()
00089 * | | | |
00090 * | | | | +---> internal_wait_for_ack() # Wait for RPi5 ready signal (50ms timeout)
00091 * | | | | |
00092 * | | | | +---> rspi_peripheral_transfer() # DMA/IRQ-based SPI transfer
00093 * | | | | |
00094 * | | | | v
00095 * | | | | TX sequence++          # Increment for next frame
00096 * | | | | ```
00097 *
00098 * ## Data Flow - Reception
00099 *
00100 * ```
00101 * Application
00102 * |
00103 * | rx_spi_comm_receive(timeout_ms)
00104 * | v
00105 * | internal_wait_for_data()    # Poll RSPI for data (with ThreadX sleep)
00106 * | |
00107 * | | v
00108 * | | internal_read_frame_header() # Read 10 bytes (SYNC + header)
00109 * | | |
00110 * | | | Validate sync word (0xAA55)
00111 * | | | Extract payload length
00112 * | | | v
00113 * | | | internal_spi_transfer() # Read remaining (payload + CRC-32)
00114 * | | | |
00115 * | | | | v

```

```

00116 * internal_decode_frame()
00117 * |
00118 * +---> internal_decode_header() # Parse header fields
00119 * +---> memcpy payload
00120 * +---> internal_verify_crc() # Validate CRC-32
00121 * |
00122 * v
00123 * RX sequence = frame.seq + 1 # Update expected next sequence
00124 * ...
00125 *
00126 * ## Performance Characteristics
00127 *
00128 * | Operation | Avg Time (us) | Worst Case (us) | Notes |
00129 * |-----|-----|-----|-----|
00130 * | **rx_spi_comm_send()** | 150-500 | 1000 | Depends on payload (0-1024 bytes) |
00131 * | **rx_spi_comm_receive()** | 200-600 | 2000 | Includes wait + decode + CRC |
00132 * | **internal_build_frame()** | 5-20 | 50 | Malloc payload (0-1024 bytes) |
00133 * | **internal_wait_for_ack()** | 10-100 | 50000 | Polls every 10ms, 50ms timeout |
00134 * | **internal_spi_transfer()** | 80-900 | 1500 | SPI @ 10 MHz, 10-1038 bytes |
00135 * | **internal_decode_header()** | 15 | 30 | Byte-level parsing |
00136 * | **internal_verify_crc()** | 40-500 | 800 | CRC-32 over 10-1038 bytes |
00137 * | **internal_wait_for_data()** | 10 | timeout_ms*1000 | Polls every 10ms until timeout |
00138 *
00139 * ***Throughput Analysis** (10 MHz SPI, no FEC):
00140 * - **Max payload**: 1024 bytes -> 1038 bytes on wire -> 830 us SPI time
00141 * - **Frame overhead**: ~100 us encode + ~150 us decode + ~250 us CRC
00142 * - **Total latency**: ~1330 us per 1024-byte frame = ~770 KB/s effective
00143 * - **Theoretical max**: 10 MHz / 8 = 1.25 MB/s (raw SPI)
00144 * - **Efficiency**: ~62% (overhead from framing, CRC, ACK waits)
00145 *
00146 * ## Memory Usage
00147 *
00148 * ***Stack Usage per Function Call**:  

00149 * - `rx_spi_comm_send()`: ~310 bytes (rx_frame_t + locals; wire scratch is in handle)
00150 * - `rx_spi_comm_receive()`: ~350 bytes (rx_frame_t + header_buf[10])
00151 * - `internal_spi_transfer()`: ~20 bytes (local vars, no buffers)
00152 * - `internal_decode_frame()`: ~10 bytes (offset, err)
00153 *
00154 * ***Total Handle Footprint**:  

00155 * ~5160 bytes (handle now owns tx_encode_buffer  

00156 * in addition to retry_buffer; see rx_spi_comm_handle_t in .h)
00157 *
00158 * ## Thread Safety
00159 *
00160 * | Function | Thread Safe? | Notes |
00161 * |-----|-----|-----|
00162 * | `rx_spi_comm_init()` | [FAIL] No | Call once during init, not from multiple threads |
00163 * | `rx_spi_comm_send()` | [FAIL] No | Use external mutex if multiple threads send |
00164 * | `rx_spi_comm_receive()` | [FAIL] No | Use external mutex if multiple threads receive |
00165 * | `rx_spi_comm_send_ack()` | [FAIL] No | Same as send (shares TX sequence) |
00166 * | `rx_spi_comm_send_nack()` | [FAIL] No | Same as send (shares TX sequence) |
00167 * | `rx_spi_comm_data_available()` | [PASS] Yes | Read-only hardware poll (safe if HAL is safe) |
00168 *
00169 * ***Typical pattern**:  

00170 * Dedicate one ThreadX task to SPI communication, no locking needed.
00171 *
00172 * ## Hardware Requirements
00173 *
00174 * ***RX72N RSPI Configuration** (handled by rspi HAL):
00175 * - **Mode**: Peripheral
00176 * - **Clock**: External (provided by RPi5 controller)
00177 * - **Data bits**: 8
00178 * - **Bit order**: MSB-first
00179 * - **Clock polarity**: CPOL=0 (idle low)
00180 * - **Clock phase**: CPHA=0 (sample on first edge)
00181 * - **Max frequency**: 10 MHz (reliable with 1m cable)
00182 *
00183 * ***Pin Connections** (RSPI0 example):
00184 * - **CS**: PD0 (chip select, active low)
00185 * - **CLK**: PD1 (clock from RPi5)
00186 * - **CIPO**: PD2 (Controller In Peripheral Out, RX72N -> RPi5)
00187 * - **COPI**: PD3 (Controller Out Peripheral In, RPi5 -> RX72N)
00188 *
00189 * ## Error Handling Strategy
00190 *
00191 * ***Recoverable Errors** (return error code, caller retries):
00192 * - `k_rx_err_timeout`: No data available, ACK timeout, normal in polling
00193 * - `k_rx_err_crc_mismatch`: Frame corrupted, send NACK, expect retransmit
00194 * - `k_rx_err_protocol_error`: Invalid sync word, resync by discarding
00195 *
00196 * ***Fatal Errors** (return error code, caller should reinitialize):
00197 * - `k_rx_err_invalid_arg`: nullptr pointer, programming error
00198 * - `k_rx_err_invalid_state`: Not initialized, must call rx_spi_comm_init()
00199 * - `k_rx_err_invalid_size`: Payload > 1024 bytes, violates protocol spec
00200 *
00201 * ***Hardware Errors** (propagated from RSPI HAL):
00202 * - Overrun, underrun, mode fault -> logged and returned to caller
00203 *
00204 * ## Integration Example

```

```

00203 *
00204 * @code{.c}
00205 * #include "rx_spi_comm.h"
00206 * #include "tx_api.h"
00207 *
00208 * // Allocate handle (4120 bytes, typically static or in task stack)
00209 * static rx_spi_comm_handle_t g_spi_handle;
00210 *
00211 * // ThreadX task for SPI communication
00212 * void spi_comm_task_entry(ULONG thread_input)
00213 * {
00214 *     (void)thread_input;
00215 *
00216 *     // Initialize SPI communication (default channel 0, no FEC)
00217 *     rx_err_t err = rx_spi_comm_init(&g_spi_handle, nullptr);
00218 *     if (err != k_rx_ok) {
00219 *         rx_log_error("spi_task", "Failed to init SPI comm");
00220 *         return;
00221 *     }
00222 *
00223 *     while (1) {
00224 *         // Receive frame with 100ms timeout
00225 *         rx_frame_t rx_frame;
00226 *         err = rx_spi_comm_receive(&g_spi_handle, &rx_frame, 100);
00227 *
00228 *         if (err == k_rx_ok) {
00229 *             // Process received frame
00230 *             if (rx_frame.header.type == k_frame_type_data) {
00231 *                 // Handle data frame
00232 *                 process_data_frame(&rx_frame);
00233 *
00234 *                 // Send ACK
00235 *                 rx_spi_comm_send_ack(&g_spi_handle, rx_frame.header.sequence);
00236 *             }
00237 *             else if (err == k_rx_err_timeout) {
00238 *                 // No data, normal in polling loop
00239 *                 tx_thread_sleep(1); // Sleep 10ms
00240 *             }
00241 *             else if (err == k_rx_err_crc_mismatch) {
00242 *                 // Corrupted frame, send NACK
00243 *                 rx_spi_comm_send_nack(&g_spi_handle, rx_frame.header.sequence, 0);
00244 *             }
00245 *             else {
00246 *                 // Fatal error, log and reinitialize
00247 *                 rx_log_error("spi_task", "SPI receive fatal error");
00248 *                 rx_spi_comm_deinit(&g_spi_handle);
00249 *                 rx_spi_comm_init(&g_spi_handle, nullptr);
00250 *             }
00251 *
00252 *             // Check if we need to send telemetry
00253 *             if (telemetry_ready) {
00254 *                 uint8_t payload[32];
00255 *                 uint32_t len = build_telemetry_frame(payload, sizeof(payload));
00256 *
00257 *                 err = rx_spi_comm_send(&g_spi_handle, k_frame_type_data, 0, payload, len);
00258 *                 if (err != k_rx_ok) {
00259 *                     rx_log_error("spi_task", "Failed to send telemetry");
00260 *                 }
00261 *             }
00262 *         }
00263 *     }
00264 * }
00265 *
00266 * @endcode
00267 *
00268 * ## NASA Power of 10 Compliance
00269 *
00270 * | Rule | Status | Implementation Notes |
00271 * |-----|-----|-----|
00272 * | 1. Simple control flow | [PASS] Pass | No goto/setjmp/recursion, structured if/while/for only |
00273 * | 2. Fixed loop bounds | [PASS] Pass | All loops bounded by enum constants (k_max_poll_iterations) |
00274 * | 3. No dynamic memory | [PASS] Pass | Zero malloc/free, all buffers in handle (stack/static) |
00275 * | 4. Short functions | [PASS] Pass | All <60 lines, single responsibility |
00276 * | 5. Assertions | [PASS] Pass | Minimum 2 checks per function (RX_CHECK_NULL_PTR + state) |
00277 * | 6. Small scope | [PASS] Pass | Variables declared near use, file-static for module data |
00278 * | 7. Check returns | [PASS] Pass | All rx_err_t checked via RX_RETURN_ON_ERROR or explicit if |
00279 * | 8. Limited preprocessor | [PASS] Pass | C23 typed enums for constants, macros only for conditionals |
00280 * | 9. Restrict pointers | [WARN] Deviation | Function pointers in HAL interface (DIP, testability) |
00281 * | 10. Compiler warnings | [PASS] Pass | -Wall -Wextra -Werror, zero warnings |
00282 *
00283 * ## SOLID Principles
00284 *
00285 * | Principle | Implementation |
00286 * |-----|-----|
00287 * | **Single Responsibility** | Module handles ONLY SPI frame transport, delegates to rx_frame (encoding), rx_crc (validation), rspi (hardware) |
00288 * | **Open/Closed** | Configurable via rx_spi_comm_config_t (channel, FEC), extendable without modifying code |
00289 * | **Liskov Substitution** | Not applicable (no inheritance in C), but handle is opaque and consistent |
00290 * | **Interface Segregation** | 9 focused functions, caller uses only what's needed (send vs receive vs query) |
00291 * | **Dependency Inversion** | Depends on abstractions: rx_frame interface, rspi HAL (not direct register access) |

```

```

00289 * @see rx_spi_comm.h Public API definition
00290 * @see rx_frame.h Frame encoding/decoding layer
00291 * @see rx_crc.h CRC-32 validation
00292 * @see rspi.h RSPI peripheral hardware abstraction
00293 *
00294 * @since Version 1.0.0
00295 * @date 2026-01-01
00296 * @copyright Copyright (c) 2026 Locked Inc.
00297 * SPDX-License-Identifier: MIT
00298 */
00299
00300 #include "rx_spi_comm.h"
00301
00302 #include <string.h>
00303
00304 #include "hardware.h"
00305 #include "rx_crc.h"
00306 #include "rx_time_constants.h"
00307
00308 #ifdef __RX__
00309 #include "tx_api.h" /* ThreadX for tx_thread_sleep */
00310 #endif
00311
00312 /*
=====
00313 * Module State
00314 *
=====
00315 */
00316
00317 static const char s_tag[] = "rx_spi_comm";
00318
00319 /** @brief Maximum left-shift for exponential backoff (prevents UB on uint32_t) */
00320 typedef enum : uint8_t {
00321     k_max_backoff_shift = 31, /**< Clamp retry_count before shifting to avoid UB */
00322 } backoff_limit_t;
00323
00324 /*
=====
00325 * Frame Header Byte Offsets
00326 *
00327 * Frame format: [SYNC(2B)][SEQ(2B)][LEN(2B)][TYPE(1B)][FLAGS(1B)]
00328 * These offsets are for parsing the raw header buffer.
00329 *
=====
00330 */
00331
00332 /** @brief Byte offsets within frame header buffer (SYNC + header fields) */
00333 typedef enum : uint8_t {
00334     k_frame_offset_init = 0, /**< Initial frame parse offset (start of buffer) */
00335     k_hdr_sync_low      = 0, /**< SYNC word low byte (LE: LSB first) */
00336     k_hdr_sync_high     = 1, /**< SYNC word high byte (LE: MSB second) */
00337     k_hdr_seq_low       = 2, /**< Sequence number low byte (LE) */
00338     k_hdr_seq_high      = 3, /**< Sequence number high byte (LE) */
00339     k_hdr_len_low       = 4, /**< Payload length low byte (LE) */
00340     k_hdr_len_high      = 5, /**< Payload length high byte (LE) */
00341     k_hdr_type          = 6, /**< Frame type */
00342     k_hdr_flags         = 7, /**< Frame flags */
00343 } frame_header_offset_t;
00344
00345 /** @brief CRC-32 seed value (pre-initialization before rx_crc32_ieee writes it) */
00346 typedef enum : uint32_t {
00347     k_spi_crc32_seed_initial = 0U, /**< Initial value for CRC-32 output variable */
00348 } spi_crc32_seed_t;
00349
00350 /* Forward declaration: retransmit helper used by rx_spi_comm_receive() */
00351 static rx_err_t internal_retransmit_frame(rx_spi_comm_handle_t* handle);
00352
00353 /**
00354 * @brief Decode and validate SPI frame header from raw wire buffer
00355 *
00356 * @details
00357 * Parses the frame header fields from raw received SPI data buffer, validating
00358 * sync word and payload length constraints. Extracts sequence number, length,
00359 * type, and flags into rx_frame_t structure.
00360 *
00361 * **Algorithm:**
00362 * 1. **Validate inputs**: Check data and frame pointers, minimum buffer size
00363 * 2. **Parse sync word**: Read little-endian 0x55AA, validate against k_frame_sync_word
00364 * 3. **Parse sequence**: Read little-endian 16-bit sequence number
00365 * 4. **Parse length**: Read little-endian 16-bit payload length, validate <= 1024
00366 * 5. **Validate total size**: Check buffer contains header + payload + CRC
00367 * 6. **Parse type and flags**: Extract frame type and flags bytes
00368 * 7. **Return payload offset**: Optionally return offset to payload start
00369 *
00370 * **Frame Header Format** (10 bytes):
00371 * ``

```

```

00372 * Offset | Size | Field | Format | Notes
00373 * -----|-----|-----|-----|-----
00374 * 0 | 2 | SYNC | LE uint16 | Always 0x55AA (wire: 0xAA, 0x55)
00375 * 2 | 2 | Sequence | LE uint16 | 0-65535, wraps
00376 * 4 | 2 | Length | LE uint16 | Payload bytes (0-1024)
00377 * 6 | 1 | Type | uint8 | k_frame_type_t enum
00378 * 7 | 1 | Flags | uint8 | Bitmask (FEC, ACK, NACK)
00379 * 8 | N | Payload | bytes | N = Length (parsed above)
00380 * 8+N | 4 | CRC-32 | LE uint32 | IEEE CRC-32
00381 * ``
00382 *
00383 *
00384 *
00385 *
00386 *
00387 *
00388 * @pre data must point to valid buffer of size data_len
00389 * @pre frame must point to valid rx_frame_t structure
00390 * @post On success, frame->header contains parsed fields
00391 * @post On success, offset_out (if non-nullptr) contains payload offset
00392 * @post On failure, frame contents are undefined
00393 *
00394 * @note This function does NOT copy payload or validate CRC
00395 * @note Use internal_verify_crc() separately to validate CRC-32
00396 * @warning Do not assume frame->payload is valid after this call
00397 *
00398 * @par Performance:
00399 * Execution time: ~15 us @ 240 MHz (byte-level parsing, no memcpy)
00400 *
00401 * @par Example Usage:
00402 * @code{.c}
00403 * uint8_t rx_buf[279]; // Received from SPI
00404 * rx_frame_t frame;
00405 * uint32_t payload_offset;
00406 *
00407 * rx_err_t err = internal_decode_header(rx_buf, sizeof(rx_buf), &frame, &payload_offset);
00408 * if (err == k_rx_ok) {
00409 * // Header valid, now copy payload
00410 * if (frame.header.length > 0) {
00411 * memcpy(frame.payload, &rx_buf[payload_offset], frame.header.length);
00412 * }
00413 * // Verify CRC at offset payload_offset + frame.header.length
00414 * }
00415 * @endcode
00416 *
00417 * @see internal_verify_crc() Validate CRC-32 after header parse
00418 * @see internal_decode_frame() Higher-level decode (header + payload + CRC)
00419 * @see rx_frame_read_le16() Little-endian uint16 parser
00420 *
00421 * @since Version 1.0.0
00422 */
00423 RX_STATIC_TESTABLE rx_err_t internal_decode_header(const uint8_t* data,
00424                                                    const uint32_t data_len,
00425                                                    rx_frame_t* frame,
00426                                                    uint32_t* offset_out)
00427 {
00428     if (data == nullptr || frame == nullptr) {
00429         return k_rx_err_invalid_arg;
00430     }
00431
00432     if (data_len < k_frame_min_size) {
00433         return k_rx_err_invalid_size;
00434     }
00435
00436     uint32_t offset = k_frame_offset_init;
00437     uint16_t sync_word = rx_frame_read_le16(&data[offset]);
00438     if (sync_word != k_frame_sync_word) {
00439         return k_rx_err_protocol_error;
00440     }
00441     offset += k_frame_sync_size;
00442
00443     frame->header.sequence = rx_frame_read_le16(&data[offset]);
00444     offset += k_frame_seq_size;
00445
00446     frame->header.length = rx_frame_read_le16(&data[offset]);
00447     offset += k_frame_len_size;
00448
00449     if (frame->header.length > k_frame_max_payload) {
00450         return k_rx_err_invalid_size;
00451     }
00452
00453     uint32_t expected_size = rx_frame_encoded_size(frame->header.length);
00454     if (data_len < expected_size) {
00455         return k_rx_err_invalid_size;
00456     }
00457
00458     frame->header.type = data[offset];

```

```

00459 offset += k_frame_type_size;
00460
00461 frame->header.flags = data[offset];
00462 offset += k_frame_flags_size;
00463
00464 if (offset_out != nullptr) {
00465     *offset_out = offset;
00466 }
00467
00468 return k_rx_ok;
00469 }
00470
00471 /**
00472  * @brief Verify CRC-32 for a received SPI frame
00473  *
00474  * @details
00475  * Validates frame integrity by computing CRC-32 over header + payload and
00476  * comparing with the CRC-32 field at the end of the frame. Uses IEEE
00477  * polynomial 0x04C11DB7 (same as Ethernet, ZIP, PNG).
00478  *
00479  * **Algorithm:**
00480  * 1. **Extract received CRC**: Read 4-byte little-endian CRC from data[offset]
00481  * 2. **Compute expected CRC**: Calculate CRC-32 over data[0..offset-1]
00482  * 3. **Compare**: If received == calculated, return k_rx_ok
00483  * 4. **Optionally return CRC**: Copy received CRC to crc_out if non-nullptr
00484  *
00485  * **CRC Computation Details:**
00486  * - **Algorithm**: CRC-32/IEEE (polynomial 0x04C11DB7, init 0xFFFFFFFF, final XOR 0xFFFFFFFF)
00487  * - **Input**: All bytes from sync word through end of payload (excludes CRC itself)
00488  * - **Output**: 32-bit checksum stored in little-endian at frame end
00489  * - **Implementation**: rx_crc32_ieee() (hardware CRC peripheral on RX72N)
00490  *
00491  *
00492  *
00493  *
00494  *
00495  * @pre data must point to valid buffer containing offset + 4 bytes
00496  * @pre offset must be the byte position where CRC-32 field begins
00497  * @post On k_rx_ok, frame integrity guaranteed
00498  * @post On k_rx_err_crc_mismatch, frame is corrupted (send NACK)
00499  * @post crc_out (if non-nullptr) contains received CRC value
00500  *
00501  * @note Thread-safe if caller serializes access to data buffer
00502  * @warning This function does NOT modify the data buffer
00503  * @warning CRC mismatch indicates transmission error, do NOT process frame
00504  *
00505  * @par Performance:
00506  * - **Execution time**: 40-250 us @ 240 MHz (depends on offset = bytes to hash)
00507  * - **10-byte frame** (ACK): ~40 us
00508  * - **100-byte frame**: ~130 us
00509  * - **263-byte frame** (max): ~250 us
00510  * - **Hardware CRC**: If RX_CRC32_USE_HARDWARE defined, 4x faster
00511  *
00512  * @par CRC Mismatch Handling:
00513  * @code{.c}
00514  * uint8_t rx_buf[279];
00515  * uint32_t payload_offset = 8; // After header
00516  * uint32_t crc_offset = payload_offset + payload_len;
00517  * uint32_t received_crc;
00518  *
00519  * rx_err_t err = internal_verify_crc(rx_buf, crc_offset, &received_crc);
00520  * if (err == k_rx_err_crc_mismatch) {
00521  *     rx_log_error("spi_comm", "CRC mismatch: received=0x%08X", received_crc);
00522  *     // Send NACK to request retransmit
00523  *     rx_spi_comm_send_nack(handle, frame_seq, 0);
00524  * }
00525  * @endcode
00526  *
00527  * @see rx_crc32_ieee() CRC-32 computation implementation
00528  * @see rx_frame_read_le32() Little-endian uint32 reader
00529  * @see internal_decode_frame() Uses this function for CRC validation
00530  *
00531  * @since Version 1.0.0
00532  */
00533 RX_STATIC_TESTABLE rx_err_t internal_verify_crc(const uint8_t* data,
00534                                                  uint32_t offset,
00535                                                  uint32_t* crc_out)
00536 {
00537     if (data == nullptr) {
00538         return k_rx_err_invalid_arg;
00539     }
00540
00541     const uint32_t received_crc = rx_frame_read_le32(&data[offset]);
00542     uint32_t calculated_crc = k_spi_crc32_seed_initial;
00543     (void)(rx_crc32_ieee(data, offset, &calculated_crc));
00544     /* rx_crc32_ieee only fails with null pointers or out-of-range length.
00545      * data is already checked non-null above, and offset (frame length minus CRC)

```

```

00546  * is always in [k_frame_min_size, k_spi_comm_tx_buffer_size] -- well within
00547  * the CRC library's valid range. This path is therefore an invariant. */
00548
00549  if (received_crc != calculated_crc) {
00550      return k_rx_err_crc_mismatch;
00551  }
00552
00553  if (crc_out != nullptr) {
00554      *crc_out = received_crc;
00555  }
00556
00557  return k_rx_ok;
00558 }
00559
00560 #ifdef __RX__
00561 /** @brief Sleep duration for polling loop */
00562 typedef enum : uint32_t {
00563     k_poll_sleep_ticks = 1, /**< 1 ThreadX tick between polls */
00564 } poll_timing_t;
00565 #endif
00566
00567 /** @brief ACK/ready wait timing constants */
00568 typedef enum : uint16_t {
00569     k_ack_wait_timeout_ms = 50, /**< Abort if host doesn't ACK within 50 ms */
00570 } ack_wait_t;
00571
00572 /** @brief Receive loop bounds (NASA Power of 10 Rule 2) */
00573 typedef enum : uint8_t {
00574     k_max_control_frames_per_receive = 16, /**< Max control frames before returning error */
00575 } receive_bounds_t;
00576
00577 /** @brief Polling loop iteration limits */
00578 typedef enum : uint16_t {
00579     k_max_poll_iterations = 1000, /**< Maximum polling iterations (safety bound) */
00580 } polling_limits_t;
00581
00582 /*
=====
00583  * Internal Helpers
00584  *
=====
00585  */
00586
00587 /**
00588  * @brief Wait for RPi5 controller ready signal before transmitting data
00589  *
00590  * @details
00591  * Implements flow control by polling RSPI peripheral for host ready signal
00592  * before initiating SPI transfer. Prevents peripheral from blocking indefinitely
00593  * if controller never asserts chip select or clock.
00594  *
00595  * **Purpose**: RX72N (SPI peripheral) cannot initiate transfers - it must wait
00596  * for RPi5 (SPI controller) to assert CS and provide clock. This function polls
00597  * the RSPI hardware for the "ready to write" condition with a timeout to detect
00598  * host failures.
00599  *
00600  * **Algorithm**:
00601  * 1. **Get start time**: Capture current ThreadX tick count (or iteration counter in tests)
00602  * 2. **Poll loop**:
00603  *   - Call rspi_peripheral_write_ready() to check hardware status
00604  *   - If ready: Return k_rx_ok immediately
00605  *   - If error: Return HAL error code
00606  *   - If not ready: Sleep 1 tick (10ms) and retry
00607  * 3. **Timeout check**: If elapsed time >= timeout_ms, return k_rx_err_timeout
00608  * 4. **Thread yield**: tx_thread_sleep() yields CPU to other tasks (not busy-wait)
00609  *
00610  * **Platform Differences**:
00611  * - **RX72N** ('__RX__' defined): Uses ThreadX tx_time_get() for precise timing
00612  * - **Host builds (tests)**: Uses iteration counter to simulate time (no ThreadX)
00613  *
00614  *
00615  *
00616  *
00617  * @pre handle must be initialized via rx_spi_comm_init()
00618  * @pre timeout_ms must be > 0
00619  * @post On k_rx_ok, RSPI is ready for rspi_peripheral_transfer()
00620  * @post On k_rx_err_timeout, host may be dead/disconnected
00621  * @post On HAL error, RSPI peripheral may require reset
00622  *
00623  * @note Yields CPU via tx_thread_sleep(1) every iteration (cooperative multitasking)
00624  * @warning Do NOT call with timeout_ms = 0 (will return timeout immediately on RX72N)
00625  * @warning Timeout does NOT guarantee exact millisecond precision (+/-10ms due to tick granularity)
00626  *
00627  * @par Performance:
00628  * - **Best case**: ~10 us (host ready immediately, 1 HAL call)
00629  * - **Typical**: 10-50 ms (host ready within 1-5 polling iterations)
00630  * - **Worst case**: timeout_ms (50 ms default, host never ready)

```

```

00631 *
00632 * @par ThreadX Timing:
00633 * - **Tick rate**: 100 Hz (k_threadx_ms_per_tick = 10 ms)
00634 * - **Sleep duration**: 1 tick = 10 ms per iteration
00635 * - **Timeout conversion**:  $\text{timeout\_ms} / 10$  ticks (round up)
00636 *
00637 * @par Example - Normal Transmission:
00638 * @code{.c}
00639 * // Wait up to 50ms for RPi5 to be ready
00640 * rx_err_t err = internal_wait_for_ack(handle, 50);
00641 * if (err == k_rx_ok) {
00642 *     // Host ready, perform SPI transfer
00643 *     rspi_peripheral_transfer(handle->channel, tx_buf, rx_buf, len);
00644 * } else if (err == k_rx_err_timeout) {
00645 *     rx_log_error("spi", "Host ACK timeout - RPi5 may be down");
00646 * }
00647 * @endcode
00648 *
00649 * @par Example - Timeout Handling:
00650 * @code{.c}
00651 * // Retry logic for transient host delays
00652 * for (uint8_t retry = 0; retry < 3; retry++) {
00653 *     rx_err_t err = internal_wait_for_ack(handle, 50);
00654 *     if (err == k_rx_ok) break;
00655 *
00656 *     if (err == k_rx_err_timeout) {
00657 *         rx_log_warn("spi", "Host not ready, retry %d/3", retry + 1);
00658 *         tx_thread_sleep(10); // Wait 100ms between retries
00659 *     } else {
00660 *         // HAL error, fatal
00661 *         return err;
00662 *     }
00663 * }
00664 * @endcode
00665 *
00666 * @see rspi_peripheral_write_ready() RSPi HAL function for ready status
00667 * @see internal_spi_transfer() Calls this function before TX operations
00668 * @see k_ack_wait_timeout_ms Default timeout constant (50ms)
00669 *
00670 * @since Version 1.0.0
00671 */
00672 static rx_err_t internal_wait_for_ack(const rx_spi_comm_handle_t* handle, uint32_t timeout_ms)
00673 {
00674 #ifdef __RX__
00675     /* RX72N: Use ThreadX time measurement for precise timeout */
00676     ULONG timeout_ticks = (timeout_ms + k_threadx_ms_per_tick - 1) / k_threadx_ms_per_tick;
00677     ULONG start_ticks = tx_time_get();
00678
00679     while ((tx_time_get() - start_ticks) < timeout_ticks) {
00680         bool ready = false;
00681         const rx_err_t err = rspi_peripheral_write_ready(handle->channel, &ready);
00682         if (err != k_rx_ok) {
00683             return err;
00684         }
00685         if (ready) {
00686             return k_rx_ok;
00687         }
00688         /* Yield to other threads while waiting */
00689         tx_thread_sleep(k_poll_sleep_ticks);
00690     }
00691 #else
00692     /* Host build (testing): Use iteration counter to simulate time */
00693     uint32_t elapsed_ms = 0;
00694
00695     while (elapsed_ms < timeout_ms) {
00696         bool ready = false;
00697         const rx_err_t err = rspi_peripheral_write_ready(handle->channel, &ready);
00698         if (err != k_rx_ok) {
00699             return err;
00700         }
00701
00702         if (ready) {
00703             return k_rx_ok;
00704         }
00705
00706         /* Simulate time passing in tests */
00707         elapsed_ms += k_threadx_ms_per_tick;
00708     }
00709 #endif
00710
00711     rx_log_error(s_tag, "SPI ACK timeout waiting for host ready");
00712     return k_rx_err_timeout;
00713 }
00714
00715 /**
00716 * @brief Perform raw SPI full-duplex transfer using handle's staging buffers
00717 *

```

```

00718 * @details
00719 * Low-level SPI transfer wrapper that uses handle's internal TX/RX staging
00720 * buffers (2048 bytes each) to perform full-duplex communication with RPi5
00721 * controller. Implements flow control by waiting for host ready signal before
00722 * transmit operations.
00723 *
00724 * **Why Staging Buffers?**
00725 * RSPI peripheral DMA requires contiguous aligned buffers. Using handle's
00726 * pre-allocated staging buffers avoids dynamic allocation and ensures proper
00727 * alignment for DMA transfers.
00728 *
00729 * **Algorithm:**
00730 * 1. **Pre-condition 1***: Validate handle pointer (RX_CHECK_NULL_PTR)
00731 * 2. **Pre-condition 2***: Calculate transfer_len = max(tx_len, rx_len)
00732 * 3. **Pre-condition 3***: Validate transfer_len <= k_spi_comm_tx_buffer_size (2048)
00733 * 4. **Pre-condition 4***: Validate TX data pointer consistent with tx_len
00734 * 5. **Pre-condition 5***: Validate RX data pointer consistent with rx_len
00735 * 6. **Flow control***: If transmitting (tx_data != nullptr), wait for host ready via internal_wait_for_ack()
00736 * 7. **Prepare TX buffer***: Zero-fill staging buffer, copy tx_data if present
00737 * 8. **SPI transfer***: Call rspi_peripheral_transfer() with staging buffers
00738 * 9. **Post-condition***: Copy RX staging buffer to rx_data if requested
00739 *
00740 * **Full-Duplex Behavior:**
00741 * - SPI is inherently full-duplex (simultaneous TX and RX)
00742 * - If tx_len < rx_len: TX buffer zero-padded, RX receives full rx_len bytes
00743 * - If tx_len > rx_len: RX buffer receives tx_len bytes, caller extracts rx_len
00744 * - Transfer length = max(tx_len, rx_len) to accommodate both directions
00745 *
00746 *
00747 *
00748 *
00749 *
00750 *
00751 *
00752 * @pre handle must be initialized via rx_spi_comm_init()
00753 * @pre If tx_data != nullptr, tx_len must be > 0 and <= 2048
00754 * @pre If rx_data != nullptr, rx_len must be > 0 and <= 2048
00755 * @pre tx_data and rx_data must not alias (undefined behavior if overlapping)
00756 * @post On k_rx_ok, rx_data (if non-nullptr) contains received bytes
00757 * @post On k_rx_ok, handle->tx_buffer and handle->rx_buffer modified
00758 * @post On error, rx_data contents are undefined
00759 *
00760 * @note This function uses handle's staging buffers (not caller's buffers directly)
00761 * @note Transfer is synchronous (blocks until complete or timeout)
00762 * @warning Do NOT pass overlapping tx_data and rx_data buffers
00763 * @warning This function is NOT thread-safe (use external mutex if multi-threaded)
00764 *
00765 * @par Performance:
00766 * - **Execution time***: 80-400 us @ 10 MHz SPI (depends on transfer_len)
00767 * - **10-byte transfer** (ACK/NACK): ~80 us (8 us SPI + 72 us overhead)
00768 * - **100-byte transfer***: ~180 us (80 us SPI + 100 us overhead)
00769 * - **1038-byte transfer** (max frame): ~1000 us (830 us SPI + 170 us overhead)
00770 * - **Host ACK wait***: Add 0-50 ms if transmitting (see internal_wait_for_ack)
00771 *
00772 * @par Example - Transmit Only:
00773 * @code{.c}
00774 * uint8_t tx_frame[14] = { 0xAA, 0x55, ... }; // ACK frame
00775 * rx_err_t err = internal_spi_transfer(handle, tx_frame, 14, nullptr, 0);
00776 * if (err == k_rx_ok) {
00777 *     rx_log_debug("spi", "ACK sent successfully");
00778 * }
00779 * @endcode
00780 *
00781 * @par Example - Receive Only:
00782 * @code{.c}
00783 * uint8_t rx_header[10]; // Read frame header
00784 * rx_err_t err = internal_spi_transfer(handle, nullptr, 0, rx_header, 10);
00785 * if (err == k_rx_ok) {
00786 *     uint16_t sync = (rx_header[0] << 8) | rx_header[1];
00787 *     if (sync == 0xAA55) {
00788 *         // Valid frame header
00789 *     }
00790 * }
00791 * @endcode
00792 *
00793 * @par Example - Full-Duplex:
00794 * @code{.c}
00795 * uint8_t tx_buf[50] = { ... }; // Send 50 bytes
00796 * uint8_t rx_buf[50]; // Receive 50 bytes simultaneously
00797 * rx_err_t err = internal_spi_transfer(handle, tx_buf, 50, rx_buf, 50);
00798 * if (err == k_rx_ok) {
00799 *     // Both TX and RX completed
00800 * }
00801 * @endcode
00802 *
00803 * @see internal_wait_for_ack() Flow control for TX operations
00804 * @see rspi_peripheral_transfer() RSPI HAL transfer function

```

```

00805 * @see rx_spi_comm_handle_t Handle structure with staging buffers
00806 *
00807 * @since Version 1.0.0
00808 */
00809 RX_STATIC_TESTABLE rx_err_t internal_spi_transfer(rx_spi_comm_handle_t* handle,
00810                                                  const uint8_t* tx_data,
00811                                                  const uint32_t tx_len,
00812                                                  uint8_t* rx_data,
00813                                                  const uint32_t rx_len)
00814 {
00815     /* Pre-condition 1: Handle pointer is always non-null (all callers validate before calling). */
00816
00817     /* Pre-condition 2: Transfer length within buffer capacity.
00818     * All callers validate frame lengths before calling, so this is an invariant. */
00819     uint32_t transfer_len = (tx_len > rx_len) ? tx_len : rx_len;
00820
00821     /* Pre-condition 3: TX data pointer consistent with length.
00822     * All callers either pass valid data+length or nullptr+0, never inconsistent. */
00823
00824     /* Pre-condition 4: RX data pointer consistent with length.
00825     * Same invariant as TX: callers never pass nullptr rx_data with non-zero rx_len. */
00826
00827     /* Wait for host ready signal before transmit operations */
00828     const bool has_tx = (bool)((tx_data != nullptr) && (tx_len > 0));
00829     if (has_tx) {
00830         rx_err_t wait_err = internal_wait_for_ack(handle, k_ack_wait_timeout_ms);
00831         RX_RETURN_ON_ERROR(wait_err, s_tag, "Host ACK wait failed");
00832     }
00833
00834     /* Prepare TX buffer (pad with zeros if RX is larger) */
00835     for (uint32_t i = 0; i < transfer_len; i++) {
00836         handle->tx_buffer[i] = 0;
00837     }
00838     if (has_tx) {
00839         for (uint32_t i = 0; i < tx_len; i++) {
00840             handle->tx_buffer[i] = tx_data[i];
00841         }
00842     }
00843
00844     /*
00845     * Perform SPI transfer.
00846     * Cast to uint16_t: RSPI HAL uses 16-bit length (max 65535 bytes).
00847     * Safe because transfer_len is validated <= k_spi_comm_tx_buffer_size above.
00848     */
00849     rx_err_t err = rspi_peripheral_transfer(handle->channel,
00850                                             handle->tx_buffer,
00851                                             handle->rx_buffer,
00852                                             (uint16_t)transfer_len);
00853
00854     if (err != k_rx_ok) {
00855         rx_log_error(s_tag, "SPI peripheral transfer failed");
00856         return err;
00857     }
00858
00859     /* Post-condition: Copy RX data (reverse copy: rx_data may alias rx_buffer) */
00860     const bool has_rx = (bool)((rx_data != nullptr) && (rx_len > 0));
00861     if (has_rx) {
00862         for (uint32_t i = rx_len; i > 0; i--) {
00863             rx_data[i - 1] = handle->rx_buffer[i - 1];
00864         }
00865     }
00866
00867     return k_rx_ok;
00868 }
00869
00870 /*
=====
00871 * Initialization
00872 *
=====
00873 */
00874 /**
00875 * @var s_zero_handle
00876 * @brief Zero-initialized SPI comm handle template for stack-safe initialization
00877 * @details Static const instance used to clear rx_spi_comm_handle_t without creating
00878 *         a large compound literal on the stack. Lives in .rodata section.
00879 * @note Read-only; never modified after static initialization
00880 * @warning Do not remove - prevents stack overflow in init function
00881 * @see rx_spi_comm_init() Uses this template to zero-initialize the comm handle
00882 * @since Version 1.0.0
00883 */
00884 static const rx_spi_comm_handle_t s_zero_handle = {};
00885
00886 /**
00887 * @brief Initialize SPI communication layer and allocate resources
00888 */

```

```

00890 * @details
00891 * Initializes SPI communication handle for frame-based communication between
00892 * RX72N (peripheral) and RPi5 (controller). Sets up frame encoder/decoder,
00893 * clears staging buffers, and configures channel and FEC settings.
00894 *
00895 * **Initialization Steps:**
00896 * 1. **Validate handle**: Check handle pointer non-nullptr
00897 * 2. **Zero handle**: memset entire structure to clear old state
00898 * 3. **Apply configuration**: Use config values or defaults (RSPi0, no FEC)
00899 * 4. **Initialize encoder**: Call rx_frame_encoder_init() for TX encoding
00900 * 5. **Initialize decoder**: Call rx_frame_decoder_init() for RX decoding
00901 * 6. **Reset sequence counters**: Set TX/RX sequence to 0
00902 * 7. **Mark initialized**: Set handle->initialized = true
00903 *
00904 * **Configuration** (config must be non-nullptr):
00905 * - **Channel**: Must specify RSPi channel (e.g., k_rspi_channel_2 for host)
00906 * - **FEC**: fec_enabled flag (true/false)
00907 * - Passing config == nullptr returns k_rx_err_invalid_arg
00908 *
00909 * **Memory Allocation**:
00910 * This function does NOT allocate memory (NASA Rule 3 compliance). Caller
00911 * must provide pre-allocated handle (typically static or on task stack).
00912 *
00913 *
00914 *
00915 *
00916 * @pre handle must point to valid rx_spi_comm_handle_t storage (4120 bytes)
00917 * @pre config != nullptr
00918 * @pre config->session != nullptr
00919 * @pre config->channel must be a valid rspi_channel_t value (0-2)
00920 * @post On success, handle->initialized = true
00921 * @post On success, handle->tx_sequence = 0, handle->rx_sequence = 0
00922 * @post On success, handle->encoder and handle->decoder initialized
00923 * @post On failure (encoder init), handle state is zeroed but not usable
00924 * @post On failure (decoder init), encoder is cleaned up, handle unusable
00925 *
00926 * @note Call this ONCE per handle before any send/receive operations
00927 * @note This function is NOT thread-safe (single-threaded init only)
00928 * @warning Do not call multiple times without rx_spi_comm_deinit() first
00929 * @warning RSPi peripheral hardware (rspi HAL) must be initialized separately
00930 *
00931 * @par Performance:
00932 * Execution time: ~50 us @ 240 MHz (memset + encoder/decoder init)
00933 *
00934 * @par Example - Basic Configuration:
00935 * @code{.c}
00936 * // Allocate handle (static or on task stack, NOT malloc)
00937 * static rx_spi_comm_handle_t g_spi_handle;
00938 *
00939 * rx_spi_comm_config_t config = {
00940 *     .session = &session,
00941 *     .channel = k_rspi_channel_0,
00942 *     .fec_enabled = false,
00943 * };
00944 * rx_err_t err = rx_spi_comm_init(&g_spi_handle, &config);
00945 * if (err != k_rx_ok) {
00946 *     rx_log_error("init", "SPI comm init failed");
00947 *     return;
00948 * }
00949 *
00950 * // Now ready to send/receive frames
00951 * rx_spi_comm_send(&g_spi_handle, k_frame_type_data, 0, payload, len);
00952 * @endcode
00953 *
00954 * @par Example - FEC Enabled:
00955 * @code{.c}
00956 * static rx_spi_comm_handle_t g_spi_handle;
00957 *
00958 * // Enable FEC for noisy environments
00959 * rx_spi_comm_config_t config = {
00960 *     .channel = 0, // RSPi0
00961 *     .fec_enabled = true, // Enable Forward Error Correction
00962 * };
00963 *
00964 * rx_err_t err = rx_spi_comm_init(&g_spi_handle, &config);
00965 * if (err == k_rx_ok) {
00966 *     rx_log_info("init", "SPI comm initialized with FEC");
00967 * }
00968 * @endcode
00969 *
00970 * @par Example - Multiple RSPi Channels:
00971 * @code{.c}
00972 * // Use RSPi1 instead of default RSPi0
00973 * static rx_spi_comm_handle_t g_spi1_handle;
00974 *
00975 * rx_spi_comm_config_t config = {
00976 *     .channel = 1, // RSPi1 (pins PE0-PE3)

```

```

00977 * .fec_enabled = false,
00978 * };
00979 *
00980 * rx_spi_comm_init(&g_spi_handle, &config);
00981 * @endcode
00982 *
00983 * @see rx_spi_comm_deinit() Clean up resources when done
00984 * @see rx_spi_comm_config_t Configuration structure definition
00985 * @see rx_frame_encoder_init() Frame encoder initialization
00986 * @see rx_frame_decoder_init() Frame decoder initialization
00987 *
00988 * @since Version 1.0.0
00989 */
00990 rx_err_t rx_spi_comm_init(rx_spi_comm_handle_t* handle, const rx_spi_comm_config_t* config)
00991 {
00992     if (handle == nullptr) {
00993         return k_rx_err_invalid_arg;
00994     }
00995
00996     if (config == nullptr || config->session == nullptr) {
00997         return k_rx_err_invalid_arg;
00998     }
00999
01000     *handle = s_zero_handle;
01001
01002     /* Apply configuration */
01003     handle->session = config->session;
01004     handle->channel = config->channel;
01005     handle->fec_enabled = config->fec_enabled;
01006     handle->auto_retransmit = config->auto_retransmit;
01007     handle->retransmit_cfg = config->retransmit_config;
01008
01009     /* Apply retransmit defaults for zero fields */
01010     if (handle->auto_retransmit) {
01011         if (handle->retransmit_cfg.max_retries == 0) {
01012             handle->retransmit_cfg.max_retries = k_retransmit_default_max_retries;
01013         }
01014         if (handle->retransmit_cfg.ack_timeout_ms == 0) {
01015             handle->retransmit_cfg.ack_timeout_ms = k_retransmit_default_ack_timeout_ms;
01016         }
01017         if (handle->retransmit_cfg.max_backoff_ms == 0) {
01018             handle->retransmit_cfg.max_backoff_ms = k_retransmit_default_max_backoff_ms;
01019         }
01020     }
01021
01022     /* Initialize frame encoder: can only fail if pointer is nullptr, which is
01023      * impossible here since &handle->encoder is always non-null. */
01024     (void)rx_frame_encoder_init(&handle->encoder);
01025
01026     /* Initialize frame decoder: same invariant as encoder above. */
01027     (void)rx_frame_decoder_init(&handle->decoder);
01028
01029     /* Sequence counters managed by shared session (config->session) */
01030
01031     handle->initialized = true;
01032
01033     rx_log_debug(s_tag, "SPI comm initialized");
01034     return k_rx_ok;
01035 }
01036
01037 /**
01038 * @brief Deinitialize SPI communication layer
01039 *
01040 * @details
01041 * Tears down the encoder and decoder, clearing internal state.
01042 * Saves the first error encountered from sub-deinit calls and
01043 * returns it after completing all teardown steps.
01044 *
01045 *
01046 *
01047 * @pre handle != nullptr
01048 * @pre handle was previously initialized via rx_spi_comm_init()
01049 * @post handle->initialized == false
01050 * @post Encoder and decoder resources released
01051 *
01052 * @note Not thread-safe; caller must ensure no concurrent SPI operations.
01053 * @see rx_spi_comm_init() Corresponding initialization function
01054 * @since Version 1.0.0
01055 */
01056 rx_err_t rx_spi_comm_deinit(rx_spi_comm_handle_t* handle)
01057 {
01058     if (handle == nullptr) {
01059         return k_rx_err_invalid_arg;
01060     }
01061
01062     rx_err_t err = k_rx_ok;
01063

```

```

01064 if (handle->initialized) {
01065     /* Deinit encoder/decoder: can only fail with nullptr pointer, which is
01066        * impossible here since they are fields of a valid handle. */
01067     (void)rx_frame_encoder_deinit(&handle->encoder);
01068     (void)rx_frame_decoder_deinit(&handle->decoder);
01069     handle->initialized = false;
01070 }
01071
01072 rx_log_debug(s_tag, "SPI comm deinitialized");
01073 return err;
01074 }
01075
01076 /*
=====
01077 * Send API
01078 *
=====
01079 */
01080 /**
01081  * @brief Build frame structure from application payload data
01082  *
01083  * @details
01084  * Populates rx_frame_t structure with header fields (sequence, length, type,
01085  * flags) and payload data, preparing it for encoding via rx_frame_encode().
01086  * Automatically sets sequence number from handle's TX counter and applies
01087  * FEC flag if enabled in configuration.
01088  *
01089  * **Algorithm:**
01090  * 1. **Validate inputs**: Check handle, frame, payload pointer consistency
01091  * 2. **Validate payload size**: Ensure payload_len <= 1024 bytes (protocol limit)
01092  * 3. **Zero frame**: Clear rx_frame_t to ensure padding bytes are zero
01093  * 4. **Set sequence**: Use handle->tx_sequence (auto-incremented by sender)
01094  * 5. **Set length**: payload_len (0-1024)
01095  * 6. **Set type**: Frame type (data, ACK, NACK, telemetry, etc.)
01096  * 7. **Set flags**: User-provided flags
01097  * 8. **Copy payload**: memcpy payload_len bytes to frame->payload if present
01098  * 9. **Apply FEC flag**: Set k_frame_flag_fec_enabled if handle->fec_enabled
01099  *
01100  * **Frame Types Supported:**
01101  * - k_frame_type_data: General data payload (0-1024 bytes)
01102  * - k_frame_type_ack: Acknowledgment (0 bytes payload, handled separately)
01103  * - k_frame_type_nack: Negative acknowledgment (0 bytes, handled separately)
01104  * - k_frame_type_telemetry: Sensor/status data
01105  * - k_frame_type_command: Motor control commands
01106  *
01107  *
01108  *
01109  *
01110  *
01111  *
01112  *
01113  *
01114  *
01115  * @pre handle must be initialized via rx_spi_comm_init()
01116  * @pre If payload != nullptr, payload_len must be > 0 and <= 1024
01117  * @pre If payload == nullptr, payload_len must be 0
01118  * @post On success, frame->header populated with sequence, length, type, flags
01119  * @post On success, frame->payload contains payload_len bytes from payload
01120  * @post On success, frame ready for rx_frame_encode()
01121  * @post On failure, frame contents are undefined
01122  *
01123  * @note This function does NOT increment handle->tx_sequence (caller does that)
01124  * @note This function does NOT compute CRC (rx_frame_encode() does that)
01125  * @warning Do not modify frame after build and before encode (breaks CRC)
01126  *
01127  * @par Performance:
01128  * Execution time: 5-20 us @ 240 MHz (depends on payload_len memcpy)
01129  *
01130  * @par Example - Data Frame:
01131  * @code{.c}
01132  * uint8_t telemetry[32] = { ... }; // Sensor data
01133  * rx_frame_t frame;
01134  *
01135  * rx_err_t err = internal_build_frame(handle,
01136  *                                     k_frame_type_telemetry,
01137  *                                     0, // No special flags
01138  *                                     telemetry,
01139  *                                     sizeof(telemetry),
01140  *                                     &frame);
01141  * if (err == k_rx_ok) {
01142  *     // Frame ready, now encode to wire format
01143  *     uint8_t wire_buf[279];
01144  *     uint32_t wire_len;
01145  *     rx_frame_encode(&handle->encoder, &frame, wire_buf, &wire_len);
01146  * }
01147  * @endcode
01148  *

```

```

01149 * @par Example - Zero-Length Payload:
01150 * @code{.c}
01151 * rx_frame_t ping_frame;
01152 * rx_err_t err = internal_build_frame(handle,
01153 *                                     k_frame_type_data,
01154 *                                     0,
01155 *                                     nullptr, // No payload
01156 *                                     0, // Zero length
01157 *                                     &ping_frame);
01158 * // Result: 14-byte frame on wire (header + CRC, no payload)
01159 * @endcode
01160 *
01161 * @see rx_spi_comm_send() Uses this function to build frames before sending
01162 * @see rx_frame_encode() Encodes frame to wire format after build
01163 * @see rx_frame_create_ack() Special builder for ACK frames
01164 * @see rx_frame_create_nack() Special builder for NACK frames
01165 *
01166 * @since Version 1.0.0
01167 */
01168 RX_STATIC_TESTABLE rx_err_t internal_build_frame(const rx_spi_comm_handle_t* handle,
01169                                                  const uint16_t sequence,
01170                                                  const rx_frame_type_t type,
01171                                                  const uint8_t flags,
01172                                                  const uint8_t* payload,
01173                                                  const uint32_t payload_len,
01174                                                  rx_frame_t* frame)
01175 {
01176     if (handle == nullptr || frame == nullptr) {
01177         return k_rx_err_invalid_arg;
01178     }
01179
01180     if (payload == nullptr && payload_len > 0) {
01181         return k_rx_err_invalid_arg;
01182     }
01183
01184     if (payload_len > k_frame_max_payload) {
01185         rx_log_error(s_tag, "Payload too large");
01186         return k_rx_err_invalid_size;
01187     }
01188
01189     *frame = (rx_frame_t){};
01190     frame->header.sequence = sequence;
01191     frame->header.length = (uint16_t)payload_len;
01192     frame->header.type = (uint8_t)type;
01193     frame->header.flags = flags;
01194
01195     if (payload != nullptr && payload_len > 0) {
01196         for (uint32_t i = 0; i < payload_len; i++) {
01197             frame->payload[i] = payload[i];
01198         }
01199     }
01200
01201     if (handle->fec_enabled) {
01202         frame->header.flags |= k_frame_flag_fec_enabled;
01203     }
01204
01205     return k_rx_ok;
01206 }
01207
01208 /**
01209 * @brief Send data frame to RPi5 controller via SPI
01210 *
01211 * @details
01212 * Transmits application payload to RPi5 by building frame structure, encoding
01213 * to wire format with CRC-32, waiting for host ready signal, performing SPI
01214 * transfer, and incrementing TX sequence counter. This is the primary transmit
01215 * function for all data frames.
01216 *
01217 * **Algorithm:**
01218 * 1. **Pre-condition 1***: Validate handle pointer non-nullptr
01219 * 2. **Pre-condition 2***: Validate handle initialized
01220 * 3. **Build frame***: Call internal_build_frame() to populate rx_frame_t
01221 *    - Sets sequence number from handle->tx_sequence
01222 *    - Copies payload (0-1024 bytes)
01223 *    - Sets type, flags, applies FEC flag if configured
01224 * 4. **Encode frame***: Call rx_frame_encode() to create wire buffer
01225 *    - Adds sync word (0xAA55)
01226 *    - Encodes header fields (little-endian)
01227 *    - Computes CRC-32 over header + payload
01228 *    - Appends CRC-32 (little-endian)
01229 * 5. **Post-condition 1***: Validate encoded length in valid range (14-279 bytes)
01230 * 6. **Transfer via SPI***: Call internal_spi_transfer()
01231 *    - Waits for RPi5 ready signal (50ms timeout)
01232 *    - Performs DMA/IRQ-based SPI transfer
01233 * 7. **Post-condition 2***: Increment TX sequence counter
01234 *    - handle->tx_sequence++ (wraps at 65536)
01235 *    - Validates increment successful (NASA Rule 5)

```

```

01236 *
01237 * **Frame Format on Wire** (transmitted bytes):
01238 * ```
01239 * +-----+-----+-----+-----+-----+-----+
01240 * | SYNC | Sequence | Length | Type | Flags | Payload | CRC-32 |
01241 * | 2B   | 2B       | 2B    | 1B  | 1B   | 0-1024B | 4B   |
01242 * | 0x55AA | LE    | LE    |     |     | LE      |     |
01243 * +-----+-----+-----+-----+-----+-----+
01244 * ```
01245 *
01246 * **Sequence Number Behavior:**
01247 * - Auto-incremented after each successful send
01248 * - Starts at 0, wraps to 0 after 65535
01249 * - Receiver uses sequence to detect duplicates, out-of-order, missing frames
01250 *
01251 *
01252 *
01253 *
01254 *
01255 *
01256 *
01257 * @pre handle must be initialized via rx_spi_comm_init()
01258 * @pre If payload != nullptr, payload_len must be > 0 and <= 1024
01259 * @pre If payload == nullptr, payload_len must be 0
01260 * @pre RSPi peripheral hardware must be initialized and enabled
01261 * @pre RPi5 controller must be ready to receive (CS asserted, clock provided)
01262 * @post On success, handle->tx_sequence incremented (wraps at 65536)
01263 * @post On success, frame transmitted to RPi5
01264 * @post On failure, TX sequence NOT incremented (safe to retry)
01265 * @post On timeout, RPi5 may be disconnected or busy
01266 *
01267 * @note This function blocks until transfer complete or timeout (~1-50 ms)
01268 * @note TX sequence auto-increments, no manual management needed
01269 * @warning Not thread-safe - use external mutex if multiple threads send
01270 * @warning Do not modify payload buffer during call (memcpy in progress)
01271 *
01272 * @par Performance:
01273 * - **Execution time**: 150-500 us @ 10 MHz SPI + 0-50 ms host ACK wait
01274 * - **10-byte payload**: ~150 us (10 us build + 20 us encode + 120 us SPI)
01275 * - **100-byte payload**: ~300 us (15 us build + 100 us encode + 185 us SPI)
01276 * - **1024-byte payload**: ~800 us (20 us build + 250 us encode + 530 us SPI)
01277 * - **Host ACK wait**: Add 0-50 ms if RPi5 not immediately ready
01278 *
01279 * @par Example - Send Telemetry Data:
01280 * @code{.c}
01281 * // Pack telemetry into payload
01282 * uint8_t telemetry[32];
01283 * telemetry[0] = temperature_celsius & 0xFF;
01284 * telemetry[1] = (temperature_celsius » 8) & 0xFF;
01285 * // ... pack more fields ...
01286 *
01287 * // Send to RPi5
01288 * rx_err_t err = rx_spi_comm_send(&g_spi_handle,
01289 *                                k_frame_type_telemetry,
01290 *                                0, // No special flags
01291 *                                telemetry,
01292 *                                sizeof(telemetry));
01293 * if (err == k_rx_ok) {
01294 *   rx_log_debug("spi", "Telemetry sent, seq=%d", g_spi_handle.tx_sequence - 1);
01295 * } else if (err == k_rx_err_timeout) {
01296 *   rx_log_warn("spi", "RPi5 not responding");
01297 * }
01298 * @endcode
01299 *
01300 * @par Example - Send Command with Retry:
01301 * @code{.c}
01302 * uint8_t cmd[] = { 0x01, 0x02, 0x03 }; // Motor control command
01303 *
01304 * for (uint8_t retry = 0; retry < 3; retry++) {
01305 *   rx_err_t err = rx_spi_comm_send(&g_spi_handle,
01306 *                                   k_frame_type_command,
01307 *                                   0,
01308 *                                   cmd,
01309 *                                   sizeof(cmd));
01310 *   if (err == k_rx_ok) {
01311 *     break; // Success
01312 *   } else if (err == k_rx_err_timeout) {
01313 *     rx_log_warn("spi", "Timeout, retry %d/3", retry + 1);
01314 *     tx_thread_sleep(10); // Wait 100ms before retry
01315 *   } else {
01316 *     // Fatal error, don't retry
01317 *     rx_log_error("spi", "Send failed: %d", err);
01318 *     break;
01319 *   }
01320 * }
01321 * @endcode
01322 *

```

```

01323 * @par Example - Zero-Length Frame (Ping):
01324 * @code{.c}
01325 * // Send empty frame to test link
01326 * rx_err_t err = rx_spi_comm_send(&g_spi_handle,
01327 *                                k_frame_type_data,
01328 *                                0,
01329 *                                nullptr, // No payload
01330 *                                0); // Zero length
01331 * // Result: 14-byte frame on wire (header + CRC, no payload)
01332 * @endcode
01333
01334 * @see rx_spi_comm_send_ack() Send acknowledgment frame
01335 * @see rx_spi_comm_send_nack() Send negative acknowledgment
01336 * @see rx_spi_comm_receive() Receive frames from RPi5
01337 * @see internal_build_frame() Frame building implementation
01338 * @see internal_spi_transfer() SPI transfer with flow control
01339
01340 * @since Version 1.0.0
01341 */
01342 /**
01343 * @brief Buffer a transmitted frame for potential retransmission
01344 *
01345 * @pre handle->auto_retransmit == true
01346 * @post retry_buffer contains copy of wire_data; retry_pending set true
01347 *
01348 * @since Version 1.0.0
01349 */
01350 static void internal_buffer_for_retransmit(rx_spi_comm_handle_t* handle,
01351                                           const uint8_t* wire_data,
01352                                           const uint32_t wire_len,
01353                                           const uint16_t sequence)
01354 {
01355     for (uint32_t i = 0; i < wire_len; i++) {
01356         handle->retry_buffer[i] = wire_data[i];
01357     }
01358     handle->retry_wire_len = wire_len;
01359     handle->retry_sequence = sequence;
01360     handle->retry_count = 0;
01361     handle->retry_pending = true;
01362 #ifdef __RX__
01363     handle->retry_send_time_ms = (uint32_t)((uint64_t)tx_time_get() * k_threadx_ms_per_tick);
01364 #else
01365     handle->retry_send_time_ms = 0;
01366 #endif
01367 }
01368
01369 rx_err_t rx_spi_comm_send(rx_spi_comm_handle_t* handle,
01370                           const rx_frame_type_t type,
01371                           const uint8_t flags,
01372                           const uint8_t* payload,
01373                           const uint32_t payload_len)
01374 {
01375     if (handle == nullptr) {
01376         return k_rx_err_invalid_arg;
01377     }
01378     if (!handle->initialized) {
01379         rx_log_error(s_tag, "Handle not initialized");
01380         return k_rx_err_invalid_state;
01381     }
01382     /* Validate payload before consuming a sequence number (Issue #9). */
01383     if (payload == nullptr && payload_len > 0) {
01384         return k_rx_err_invalid_arg;
01385     }
01386     if (payload_len > k_frame_max_payload) {
01387         rx_log_error(s_tag, "Payload too large");
01388         return k_rx_err_invalid_size;
01389     }
01390     /* Get next TX sequence from shared session. Invariant: cannot fail. */
01391     uint16_t sequence = 0;
01392     (void)(rx_session_next_tx(handle->session, &sequence));
01393     /* Build frame. Cannot fail: handle and &frame are non-null, payload validated. */
01394     rx_frame_t frame;
01395     (void)(internal_build_frame(handle, sequence, type, flags, payload, payload_len, &frame));
01396     /* Encode frame into handle-owned scratch buffer (not stack -- see comment
01397      * on tx_encode_buffer in rx_spi_comm.h). Cannot fail: encoder initialized,
01398      * pointers valid. */
01399     uint32_t wire_len = 0;
01400     (void)(rx_frame_encode(&handle->encoder, &frame, handle->tx_encode_buffer, &wire_len));
01401     /* Transfer via SPI (waits for host ACK internally) */
01402     rx_err_t xfer_err = internal_spi_transfer(handle, handle->tx_encode_buffer, wire_len, nullptr, 0);

```

```

01410 if (xfer_err != k_rx_ok) {
01411     rx_log_error(s_tag, "SPI transfer failed");
01412     return xfer_err;
01413 }
01414
01415 /* Buffer frame for retransmission if enabled and requires-ACK flag set */
01416 if ((int)handle->auto_retransmit && ((flags & k_frame_flag_requires_ack) != 0)) {
01417     internal_buffer_for_retransmit(handle, handle->tx_encode_buffer, wire_len, sequence);
01418 }
01419
01420 return k_rx_ok;
01421 }
01422
01423 /**
01424  * @brief Send ACK frame for specified sequence number
01425  */
01426 rx_err_t rx_spi_comm_send_ack(rx_spi_comm_handle_t* handle, const uint16_t sequence)
01427 {
01428     if (handle == nullptr) {
01429         return k_rx_err_invalid_arg;
01430     }
01431
01432     if (!handle->initialized) {
01433         return k_rx_err_invalid_state;
01434     }
01435
01436     /* Create ACK frame: cannot fail since &ack_frame is non-null. */
01437     rx_frame_t ack_frame;
01438     (void)(rx_frame_create_ack(&ack_frame, sequence));
01439
01440     /* Encode and send: encoder initialized at rx_spi_comm_init; all pointers valid. */
01441     uint8_t wire_buffer[k_frame_min_size];
01442     uint32_t wire_len = 0;
01443
01444     (void)(rx_frame_encode(&handle->encoder, &ack_frame, wire_buffer, &wire_len));
01445
01446     /* Post-condition: ACK frame encoded to minimum size */
01447
01448     /* Transfer via SPI (waits for host ACK internally) */
01449     return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
01450 }
01451
01452 /**
01453  * @brief Send NACK frame for specified sequence number
01454  */
01455 rx_err_t rx_spi_comm_send_nack(rx_spi_comm_handle_t* handle, const uint16_t sequence, uint8_t flags)
01456 {
01457     if (handle == nullptr) {
01458         return k_rx_err_invalid_arg;
01459     }
01460
01461     if (!handle->initialized) {
01462         return k_rx_err_invalid_state;
01463     }
01464
01465     /* Create NACK frame: cannot fail since &nack_frame is non-null. */
01466     rx_frame_t nack_frame;
01467     (void)(rx_frame_create_nack(&nack_frame, sequence, flags));
01468
01469     /* Encode and send: encoder initialized at rx_spi_comm_init; all pointers valid. */
01470     uint8_t wire_buffer[k_frame_min_size];
01471     uint32_t wire_len = 0;
01472
01473     (void)(rx_frame_encode(&handle->encoder, &nack_frame, wire_buffer, &wire_len));
01474
01475     /* Post-condition: NACK frame encoded to minimum size */
01476
01477     /* Transfer via SPI (waits for host ACK internally) */
01478     return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
01479 }
01480
01481 /*
=====
01482  * Receive API - Internal Helpers
01483  *
=====
01484  */
01485
01486 /**
01487  * @brief Wait for data to be available on SPI channel
01488  */
01489
01490 static rx_err_t internal_wait_for_data(const rx_spi_comm_handle_t* handle,
01491                                       const uint32_t timeout_ms)
01492 {
01493     bool available = false;
01494     rx_err_t err = rspi_peripheral_read_available(handle->channel, &available);

```

```

01495 if (err != k_rx_ok) {
01496     return err;
01497 }
01498
01499 if (!available) {
01500     if (timeout_ms == 0) {
01501         return k_rx_err_timeout;
01502     }
01503
01504 #ifdef __RX__
01505     /*
01506      * Wait for data using ThreadX sleep.
01507      * Yields CPU to other threads instead of busy-waiting.
01508      * Sleep duration and tick rate are defined by threadx_timing_t constants.
01509      */
01510     uint32_t elapsed_ms = 0;
01511     uint32_t iteration = 0;
01512     while (!available && elapsed_ms < timeout_ms && iteration < k_max_poll_iterations) {
01513         iteration++;
01514         tx_thread_sleep(k_poll_sleep_ticks);
01515         elapsed_ms += k_threadx_ms_per_tick;
01516
01517         err = rspi_peripheral_read_available(handle->channel, &available);
01518         if (err != k_rx_ok) {
01519             return err;
01520         }
01521     }
01522
01523     if (!available) {
01524         return k_rx_err_timeout;
01525     }
01526 #else
01527     /* Host builds (testing): no sleep support, timeout immediately */
01528     return k_rx_err_timeout;
01529 #endif
01530 }
01531
01532 return k_rx_ok;
01533 }
01534
01535 /**
01536  * @brief Read and validate frame header from SPI
01537  */
01538 static rx_err_t
01539 internal_read_frame_header(rx_spi_comm_handle_t* handle, uint8_t* header_buf, uint16_t* payload_len)
01540 {
01541     const uint32_t header_len = k_frame_sync_size + k_frame_header_size;
01542
01543     /* Read frame header from SPI */
01544     const rx_err_t err = internal_spi_transfer(handle, nullptr, 0, header_buf, header_len);
01545
01546     if (err != k_rx_ok) {
01547         return err;
01548     }
01549
01550     /* Validate sync word (little-endian) */
01551     const uint16_t sync = rx_frame_read_le16(&header_buf[k_hdr_sync_low]);
01552     if (sync != k_frame_sync_word) {
01553         rx_log_error(s_tag, "Invalid sync word");
01554         return k_rx_err_protocol_error;
01555     }
01556
01557     /* Extract payload length (little-endian) */
01558     *payload_len = rx_frame_read_le16(&header_buf[k_hdr_len_low]);
01559     if (*payload_len > k_frame_max_payload) {
01560         rx_log_error(s_tag, "Payload too large");
01561         return k_rx_err_invalid_size;
01562     }
01563
01564     return k_rx_ok;
01565 }
01566
01567 /**
01568  * @brief Decode and verify a received SPI frame
01569  */
01570
01571
01572
01573 /* @note Delegates header validation to internal_decode_header and CRC validation
01574  * to internal_verify_crc.
01575 */
01576 static rx_err_t internal_decode_frame(const rx_spi_comm_handle_t* handle,
01577                                     rx_frame_t* frame,
01578                                     const uint32_t total_size)
01579 {
01580     /* Pre-condition: internal callers always pass non-null handle and frame.
01581      * These are invariants -- only internal_receive_loop calls this function
01582     */

```

```

01582  * with a local rx_frame_t variable and a fully-initialized handle. */
01583
01584  /* internal_read_frame_header already validated sync and payload_len before
01585  * this call, so internal_decode_header cannot fail here. */
01586  uint32_t offset = k_frame_offset_init;
01587  (void)(internal_decode_header(handle->rx_buffer, total_size, frame, &offset));
01588
01589  if (frame->header.length > 0) {
01590      for (uint32_t i = 0; i < frame->header.length; i++) {
01591          frame->payload[i] = handle->rx_buffer[offset + i];
01592      }
01593      offset += frame->header.length;
01594  }
01595
01596  rx_err_t err = internal_verify_crc(handle->rx_buffer, offset, &frame->crc);
01597  if (err != k_rx_ok) {
01598      rx_log_error(s_tag, "Frame CRC check failed");
01599      return err;
01600  }
01601
01602  return k_rx_ok;
01603 }
01604
01605 /*
=====
01606  * Receive API - Public Functions
01607  *
=====
01608  */
01609
01610  /** @brief Result of control-frame dispatch within the receive loop */
01611  typedef enum : uint8_t {
01612      k_ctrl_not_handled = 0, /**< Frame is not a control frame; return to caller */
01613      k_ctrl_consumed = 1, /**< Control frame consumed; continue receive loop */
01614      k_ctrl_error = 2, /**< Error occurred during control-frame handling */
01615  } ctrl_dispatch_result_t;
01616
01617  /**
01618   * @brief Handle ACK or NACK frame for the retransmit subsystem
01619   *
01620   * @details
01621   * ACK with matching sequence clears the retry state. NACK with matching
01622   * sequence triggers an immediate retransmit. Mismatched sequences are ignored.
01623   *
01624   * @pre handle->auto_retransmit == true
01625   * @post retry_pending cleared on matching ACK; retry_count incremented on NACK
01626   *
01627   * @since Version 1.0.0
01628   */
01629  static void internal_dispatch_ack_nack(rx_spi_comm_handle_t* handle, const rx_frame_t* frame)
01630  {
01631      if (frame->header.type == k_frame_type_ack) {
01632          if ((int)handle->retry_pending && frame->header.sequence == handle->retry_sequence) {
01633              handle->retry_pending = false;
01634              handle->retry_count = 0;
01635              if (handle->on_ack_cb != nullptr) {
01636                  handle->on_ack_cb(frame->header.sequence, handle->retransmit_cb_ctx);
01637              }
01638          }
01639      }
01640      return;
01641  }
01642
01643  /* NACK: trigger immediate retransmit if sequence matches */
01644  const bool seq_match = (bool)(frame->header.sequence == handle->retry_sequence);
01645  if ((int)handle->retry_pending && (int)seq_match) {
01646      const rx_err_t retx_err = internal_retransmit_frame(handle);
01647      if (retx_err != k_rx_ok) {
01648          rx_log_error(s_tag, "NACK-triggered retransmit failed");
01649      }
01650      if (handle->on_nack_cb != nullptr) {
01651          handle->on_nack_cb(frame->header.sequence, handle->retransmit_cb_ctx);
01652      }
01653  }
01654 }
01655
01656  /**
01657   * @brief Dispatch control frames (PING, RESET, ACK/NACK) during receive
01658   *
01659   * @details
01660   * Handles PING (auto-PONG + callback), RESET (auto-RESET_ACK + session reset
01661   * + callback), and ACK/NACK (retransmit subsystem dispatch). Data frames are
01662   * not consumed and signal the caller to return to the application.
01663   *
01664   * @pre handle and frame must be non-nullptr and valid
01665   */

```

```

01667 * @post Control-frame side effects applied (PONG sent, session reset, etc.)
01668 *
01669 * @note Called only from rx_spi_comm_receive(); not part of the public API.
01670 * @since Version 1.0.0
01671 */
01672 static rx_err_t internal_dispatch_control_frame(rx_spi_comm_handle_t* handle,
01673                                               rx_frame_t* frame,
01674                                               ctrl_dispatch_result_t* result)
01675 {
01676     /* PING -> auto-send PONG, invoke callback, consume */
01677     if (frame->header.type == k_frame_type_ping) {
01678         rx_err_t pong_err = rx_spi_comm_send_pong(handle, frame->payload, frame->header.length);
01679         if (pong_err != k_rx_ok) {
01680             rx_log_error(s_tag, "Failed to send PONG response");
01681         }
01682         if (handle->on_ping_cb != nullptr) {
01683             handle->on_ping_cb(frame, handle->control_cb_ctx);
01684         }
01685         *result = k_ctrl_consumed;
01686         return k_rx_ok;
01687     }
01688     /* RESET -> send RESET_ACK, reset session, invoke callback, consume */
01689     if (frame->header.type == k_frame_type_reset) {
01690         rx_err_t ack_err = rx_spi_comm_send_reset_ack(handle);
01691         if (ack_err != k_rx_ok) {
01692             rx_log_error(s_tag, "Failed to send RESET_ACK, session not reset");
01693             *result = k_ctrl_consumed;
01694             return k_rx_ok; /* Don't reset session if ACK failed */
01695         }
01696         /* rx_session_reset only fails with null/uninitialized session.
01697          * handle->session is valid and initialized (checked at rx_spi_comm_init).
01698          * Invariant: reset cannot fail here. */
01699         (void)(rx_session_reset(handle->session));
01700         if (handle->on_reset_cb != nullptr) {
01701             handle->on_reset_cb(frame, handle->control_cb_ctx);
01702         }
01703         *result = k_ctrl_consumed;
01704         return k_rx_ok;
01705     }
01706     /* ACK/NACK -> retransmit subsystem or pass through */
01707     if (frame->header.type == k_frame_type_ack || frame->header.type == k_frame_type_nack) {
01708         if (handle->auto_retransmit) {
01709             internal_dispatch_ack_nack(handle, frame);
01710             *result = k_ctrl_consumed;
01711             return k_rx_ok;
01712         }
01713         /* auto_retransmit off: not consumed, return to caller as data frame */
01714         *result = k_ctrl_not_handled;
01715         return k_rx_ok;
01716     }
01717     /* Not a control frame */
01718     *result = k_ctrl_not_handled;
01719     return k_rx_ok;
01720 }
01721 /**
01722  * @brief Receive data frame from RPi5 controller via SPI (blocking with timeout)
01723  *
01724  * @details
01725  * Receives complete frame from RPi5 by polling for data availability, reading
01726  * header, reading payload + CRC, validating frame integrity, and updating RX
01727  * sequence counter. This is the primary receive function for all frame types.
01728  *
01729  * **Algorithm:**
01730  * 1. **Pre-condition 1***: Validate handle and frame pointers non-nullptr
01731  * 2. **Pre-condition 2***: Validate handle initialized
01732  * 3. **Wait for data***: Call internal_wait_for_data() with timeout
01733  * 4. **Polls RSPI peripheral for incoming data (yields CPU every 10ms)
01734  *    - Returns k_rx_err_timeout if no data within timeout_ms
01735  * 5. **Read frame header***: Call internal_read_frame_header()
01736  *    - Reads 10 bytes (SYNC + sequence + length + type + flags)
01737  *    - Validates sync word (0xAA55)
01738  *    - Extracts payload length (0-1024)
01739  * 6. **Read payload + CRC***: Call internal_spi_transfer() for remaining bytes
01740  *    - Reads payload_len + 4 bytes (CRC-32)
01741  *    - Stores in handle->rx_buffer
01742  * 7. **Decode frame***: Call internal_decode_frame()
01743  *    - Parses header fields into frame->header
01744  */

```

```

01754 * - Copies payload to frame->payload
01755 * - Validates CRC-32 (returns k_rx_err_crc_mismatch if corrupt)
01756 * 7. **Update RX sequence**: Set handle->rx_sequence = frame->sequence + 1
01757 * - Tracks expected next sequence number
01758 * - Caller can detect out-of-order/missing frames
01759 *
01760 * **Blocking Behavior**:
01761 * - **timeout_ms = 0**: Non-blocking, returns immediately if no data
01762 * - **timeout_ms > 0**: Blocks up to timeout_ms milliseconds waiting for data
01763 * - **ThreadX sleep**: Yields CPU every 10ms tick (not busy-wait)
01764 *
01765 * **Timeout Semantics**:
01766 * - k_rx_err_timeout is NOT an error in polling loops (means "no data yet")
01767 * - Typical polling: Call with timeout_ms=100 in loop, handle timeout normally
01768 *
01769 *
01770 *
01771 *
01772 *
01773 * @pre handle must be initialized via rx_spi_comm_init()
01774 * @pre frame must point to valid rx_frame_t storage (304 bytes)
01775 * @pre RSPi peripheral hardware must be initialized and enabled
01776 * @pre RPi5 controller must have transmitted complete frame
01777 * @post On success, frame contains validated header, payload, CRC
01778 * @post On success, handle->rx_sequence = frame->header.sequence + 1
01779 * @post On timeout, frame contents undefined, RX sequence unchanged
01780 * @post On CRC mismatch, frame partially populated, RX sequence unchanged
01781 * @post On protocol error, frame undefined, may need resync
01782 *
01783 * @note This function blocks (with ThreadX sleep) if data not immediately available
01784 * @note timeout_ms=0 provides non-blocking behavior for polling loops
01785 * @warning Not thread-safe - use external mutex if multiple threads receive
01786 * @warning Do not process frame if return value != k_rx_ok (data invalid)
01787 * @warning Send NACK if k_rx_err_crc_mismatch to request retransmit
01788 *
01789 * @par Performance:
01790 * - **Execution time**: 200-600 us @ 10 MHz SPI + 0-timeout_ms wait
01791 * - **14-byte frame** (ACK): ~200 us (10 us wait + 80 us read + 110 us decode)
01792 * - **100-byte frame**: ~400 us (10 us wait + 200 us read + 190 us decode)
01793 * - **279-byte frame** (max): ~600 us (10 us wait + 300 us read + 290 us decode)
01794 * - **Wait time**: Add 0-timeout_ms if data not immediately available
01795 *
01796 * @par Example - Polling Loop (Typical Usage):
01797 * @code{.c}
01798 * while (1) {
01799 *     rx_frame_t frame;
01800 *     rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 100); // 100ms timeout
01801 *
01802 *     if (err == k_rx_ok) {
01803 *         // Frame received successfully
01804 *         if (frame.header.type == k_frame_type_data) {
01805 *             process_data_frame(&frame);
01806 *             rx_spi_comm_send_ack(&g_spi_handle, frame.header.sequence);
01807 *         }
01808 *     } else if (err == k_rx_err_timeout) {
01809 *         // No data, normal in polling - continue loop
01810 *         tx_thread_sleep(1); // Sleep 10ms before retry
01811 *     } else if (err == k_rx_err_crc_mismatch) {
01812 *         // Corrupted frame, send NACK
01813 *         rx_spi_comm_send_nack(&g_spi_handle, frame.header.sequence, 0);
01814 *     } else {
01815 *         // Fatal error, log and recover
01816 *         rx_log_error("spi", "Receive error: %d", err);
01817 *     }
01818 * }
01819 * @endcode
01820 *
01821 * @par Example - Non-Blocking Check:
01822 * @code{.c}
01823 * // Check if data available without blocking
01824 * rx_frame_t frame;
01825 * rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 0); // timeout=0
01826 *
01827 * if (err == k_rx_ok) {
01828 *     rx_log_debug("spi", "Frame received: seq=%d, len=%d",
01829 *         frame.header.sequence, frame.header.length);
01830 * } else if (err == k_rx_err_timeout) {
01831 *     // No data available right now, continue
01832 * }
01833 * @endcode
01834 *
01835 * @par Example - Sequence Number Checking:
01836 * @code{.c}
01837 * static uint16_t expected_seq = 0;
01838 * rx_frame_t frame;
01839 *
01840 * rx_err_t err = rx_spi_comm_receive(&g_spi_handle, &frame, 1000);

```

```

01841 * if (err == k_rx_ok) {
01842 *   if (frame.header.sequence != expected_seq) {
01843 *     rx_log_warn("spi", "Out-of-order: expected %d, got %d",
01844 *       expected_seq, frame.header.sequence);
01845 *   }
01846 *   expected_seq = frame.header.sequence + 1; // Update for next frame
01847 * }
01848 * @endcode
01849 *
01850 * @see rx_spi_comm_send() Send frames to RPi5
01851 * @see rx_spi_comm_data_available() Check data availability without timeout
01852 * @see rx_spi_comm_send_ack() Send acknowledgment after successful receive
01853 * @see rx_spi_comm_send_nack() Send NACK after CRC mismatch
01854 * @see internal_wait_for_data() Wait for data with timeout
01855 * @see internal_decode_frame() Frame decoding and CRC validation
01856 *
01857 * @since Version 1.0.0
01858 */
01859 /**
01860 * @brief Read one complete frame from SPI into handle->rx_buffer and decode it
01861 *
01862 *
01863 *
01864 * @pre handle and frame must be non-nullptr
01865 * @post On success, frame contains validated header, payload, and CRC
01866 *
01867 * @since Version 1.0.0
01868 */
01869 static rx_err_t internal_read_and_decode_frame(rx_spi_comm_handle_t* handle,
01870                                               rx_frame_t* frame,
01871                                               const uint32_t timeout_ms)
01872 {
01873   rx_err_t err = internal_wait_for_data(handle, timeout_ms);
01874   if (err != k_rx_ok) {
01875     return err;
01876   }
01877   uint8_t header_buf[k_frame_sync_size + k_frame_header_size];
01878   uint16_t payload_len = 0;
01879   err = internal_read_frame_header(handle, header_buf, &payload_len);
01880   if (err != k_rx_ok) {
01881     return err;
01882   }
01883   const uint32_t header_len = sizeof(header_buf);
01884   const uint32_t total_size =
01885     k_frame_sync_size + k_frame_header_size + payload_len + k_frame_crc_size;
01886   const uint32_t remaining = payload_len + k_frame_crc_size;
01887   err = internal_spi_transfer(handle, nullptr, 0, handle->rx_buffer + header_len, remaining);
01888   if (err != k_rx_ok) {
01889     return err;
01890   }
01891   for (uint32_t i = 0; i < header_len; i++) {
01892     handle->rx_buffer[i] = header_buf[i];
01893   }
01894   err = internal_decode_frame(handle, frame, total_size);
01895   if (err != k_rx_ok) {
01896     rx_log_error(s_tag, "Frame decode failed");
01897     return err;
01898   }
01899   return k_rx_ok;
01900 }
01901 /**
01902 * @brief Validate received frame sequence and emit NACK on rejection
01903 *
01904 * @details
01905 *   Calls rx_session_validate_rx() to update and check the session's RX
01906 *   sequence counter. On rejection (Audit F-05), the function does NOT
01907 *   silently drop the offending frame: it emits a NACK carrying the
01908 *   rejected sequence so the remote can retransmit immediately instead
01909 *   of burning its ACK-timer budget. The NACK send is allowed to fail
01910 *   (transport not ready, etc.) and is logged as a warning; the
01911 *   original validate error remains the authoritative return value.
01912 *
01913 * @param[in,out] handle SPI comm handle (session and TX path used)
01914 * @param[in] frame Decoded frame whose header.sequence is checked
01915 *
01916 * @return rx_err_t Result of the underlying session validation
01917 * @retval k_rx_ok Sequence accepted
01918 * @retval k_rx_err_protocol_error Sequence rejected (gap/duplicate)
01919 * @retval other Propagated from rx_session_validate_rx()

```

```

01928 *
01929 * @pre handle and frame must be non-nullptr and valid
01930 * @pre handle->session must be initialized
01931 * @post On success, session RX sequence advanced
01932 * @post On failure, NACK send attempted and validate error returned verbatim
01933 *
01934 * @note Called only from rx_spi_comm_receive(); not part of the public API.
01935 * @since Version 1.0.0
01936 */
01937 static rx_err_t internal_validate_rx_sequence(rx_spi_comm_handle_t* handle, const rx_frame_t* frame)
01938 {
01939     rx_session_validate_result_t validate_result = k_session_validate_fail;
01940     rx_err_t validate_err =
01941         rx_session_validate_rx(handle->session, frame->header.sequence, &validate_result);
01942     if (validate_err != k_rx_ok) {
01943         /* Audit F-05: NACK the offending sequence so the remote can retransmit
01944          * immediately. Send-NACK can fail (transport not ready); log and keep
01945          * the original validate_err as the real cause. */
01946         rx_log_error(s_tag, "Session validate_rx returned error");
01947         const rx_err_t nack_err = rx_spi_comm_send_nack(handle, frame->header.sequence, 0);
01948         if (nack_err != k_rx_ok) {
01949             rx_log_warn(s_tag, "Failed to send NACK after sequence validation failure");
01950         }
01951         return validate_err;
01952     }
01953     (void)validate_result;
01954     return k_rx_ok;
01955 }
01956
01957 rx_err_t
01958 rx_spi_comm_receive(rx_spi_comm_handle_t* handle, rx_frame_t* frame, const uint32_t timeout_ms)
01959 {
01960     if (handle == nullptr || frame == nullptr) {
01961         return k_rx_err_invalid_arg;
01962     }
01963
01964     if (!handle->initialized) {
01965         return k_rx_err_invalid_state;
01966     }
01967
01968     /* Loop to handle control frames internally before returning data frames.
01969      * Bounded to k_max_control_frames_per_receive iterations (NASA Rule 2). */
01970     for (uint8_t ctrl_count = 0; ctrl_count < k_max_control_frames_per_receive; ctrl_count++) {
01971         rx_err_t err = internal_read_and_decode_frame(handle, frame, timeout_ms);
01972         if (err != k_rx_ok) {
01973             return err;
01974         }
01975
01976         /* Dispatch control frames (PING, RESET, ACK/NACK).
01977          * internal_dispatch_control_frame always returns k_rx_ok; errors from
01978          * sub-calls (PONG/RESET_ACK send failures) are logged and swallowed
01979          * internally. This is an invariant. */
01980         ctrl_dispatch_result_t ctrl_result = k_ctrl_not_handled;
01981         (void)internal_dispatch_control_frame(handle, frame, &ctrl_result);
01982         if (ctrl_result == k_ctrl_consumed) {
01983             continue; /* Control frame handled; loop for next frame */
01984         }
01985
01986         /* Data frame: validate sequence (NACK on rejection per Audit F-05) */
01987         return internal_validate_rx_sequence(handle, frame);
01988     }
01989
01990     /* Exceeded max control frames without receiving a data frame */
01991     rx_log_error(s_tag, "Too many control frames, no data frame received");
01992     return k_rx_err_timeout;
01993 }
01994
01995 /**
01996 * @brief Check if data is available to receive on SPI
01997 */
01998 rx_err_t rx_spi_comm_data_available(const rx_spi_comm_handle_t* handle, bool* available)
01999 {
02000     if (handle == nullptr || available == nullptr) {
02001         return k_rx_err_invalid_arg;
02002     }
02003
02004     if (!handle->initialized) {
02005         return k_rx_err_invalid_state;
02006     }
02007
02008     return rspi_peripheral_read_available(handle->channel, available);
02009 }
02010
02011 /*
=====
02012 * Utility Functions
02013 *

```

```

=====
02014 */
02015
02016 /* Sequence management is now handled by the shared rx_session_state_t.
02017 * Use rx_session_next_tx(), rx_session_validate_rx(), rx_session_reset()
02018 * via handle->session.
02019 * @see rx_session.h
02020 */
02021
02022 /*
=====
02023 * Control Frame API
02024 *
=====
02025 */
02026
02027 /**
02028 * @brief Register callbacks for PING and RESET control frames
02029 *
02030 * @details
02031 * Stores function pointers that the receive path invokes when it
02032 * decodes a PING or RESET frame. Passing NULL for a callback
02033 * disables notification for that frame type while still performing
02034 * the automatic PONG / RESET_ACK response.
02035 *
02036 *
02037 *
02038 * @pre handle != nullptr
02039 * @pre handle->initialized == true
02040 * @post Callbacks stored; invoked on next matching control frame
02041 * @post Previous callbacks overwritten
02042 *
02043 * @note Not thread-safe; register before starting the receive loop.
02044 * @see rx_spi_comm_send_pong() Auto-response sent on PING
02045 * @see rx_spi_comm_send_reset_ack() Auto-response sent on RESET
02046 * @since Version 1.0.0
02047 */
02048 rx_err_t rx_spi_comm_set_control_callbacks(rx_spi_comm_handle_t* handle,
02049                                           void (*on_ping_cb)(const rx_frame_t* frame, void* ctx),
02050                                           void (*on_reset_cb)(const rx_frame_t* frame, void* ctx),
02051                                           void* cb_ctx)
02052 {
02053     if (handle == nullptr) {
02054         return k_rx_err_invalid_arg;
02055     }
02056
02057     if (!handle->initialized) {
02058         return k_rx_err_invalid_state;
02059     }
02060
02061     handle->on_ping_cb = on_ping_cb;
02062     handle->on_reset_cb = on_reset_cb;
02063     handle->control_cb_ctx = cb_ctx;
02064
02065     return k_rx_ok;
02066 }
02067
02068 /**
02069 * @brief Send PONG frame with echoed payload
02070 *
02071 * @details
02072 * Constructs a PONG frame echoing the given payload, acquires the
02073 * next TX sequence number from the shared session, encodes the
02074 * frame, and transfers it over SPI. Typically called automatically
02075 * by the receive path in response to a PING.
02076 *
02077 *
02078 *
02079 * @pre handle != nullptr && handle->initialized
02080 * @pre payload != NULL when payload_len > 0
02081 * @post TX sequence incremented on success
02082 * @post PONG frame sent over SPI
02083 *
02084 * @note Not thread-safe; called from the SPI receive context.
02085 * @see rx_frame_create_pong() Frame construction helper
02086 * @since Version 1.0.0
02087 */
02088 rx_err_t
02089 rx_spi_comm_send_pong(rx_spi_comm_handle_t* handle, const uint8_t* payload, uint32_t payload_len)
02090 {
02091     if (handle == nullptr) {
02092         return k_rx_err_invalid_arg;
02093     }
02094
02095     if (!handle->initialized) {
02096         return k_rx_err_invalid_state;
02097     }
02098 }

```

```

02098
02099 /* Get next TX sequence from shared session.
02100  * rx_session_next_tx only fails for null/uninitialized session; handle->session
02101  * is valid (set at init time) and the handle is initialized above. Invariant. */
02102 uint16_t sequence = 0;
02103 (void)(rx_session_next_tx(handle->session, &sequence));
02104
02105 /* Create PONG frame: cannot fail since &pong_frame is non-null and payload
02106  * is validated above (nullptr allowed when payload_len == 0). */
02107 rx_frame_t pong_frame;
02108 (void)(rx_frame_create_pong(&pong_frame, sequence, payload, payload_len));
02109
02110 /* Encode into handle-owned scratch buffer (not stack -- see comment on
02111  * tx_encode_buffer in rx_spi_comm.h). encoder initialized at
02112  * rx_spi_comm_init; all pointers valid. */
02113 uint32_t wire_len = 0;
02114 (void)(rx_frame_encode(&handle->encoder, &pong_frame, handle->tx_encode_buffer, &wire_len));
02115
02116 /* Transfer via SPI */
02117 return internal_spi_transfer(handle, handle->tx_encode_buffer, wire_len, nullptr, 0);
02118 }
02119
02120 /**
02121  * @brief Send RESET_ACK frame confirming a reset request
02122  *
02123  * @details
02124  * Constructs a RESET_ACK frame (empty payload), acquires the next
02125  * TX sequence number, encodes, and transfers over SPI. Called
02126  * automatically by the receive path after processing a RESET frame.
02127  * The RESET_ACK itself does NOT reset sequence numbers; that is
02128  * handled by the session layer in response to the RESET.
02129  *
02130  *
02131  *
02132  * @pre handle != nullptr && handle->initialized
02133  * @pre A RESET frame was received and session state cleared
02134  * @post TX sequence incremented on success
02135  * @post RESET_ACK frame sent over SPI
02136  *
02137  * @note Not thread-safe; called from the SPI receive context.
02138  * @see rx_frame_create_reset_ack() Frame construction helper
02139  * @see rx_spi_comm_set_control_callbacks() Register RESET notification
02140  * @since Version 1.0.0
02141  */
02142 rx_err_t rx_spi_comm_send_reset_ack(rx_spi_comm_handle_t* handle)
02143 {
02144     if (handle == nullptr) {
02145         return k_rx_err_invalid_arg;
02146     }
02147
02148     if (!handle->initialized) {
02149         return k_rx_err_invalid_state;
02150     }
02151
02152     /* Get next TX sequence from shared session.
02153      * rx_session_next_tx only fails for null/uninitialized session; handle->session
02154      * is valid and the handle is initialized above. Invariant. */
02155     uint16_t sequence = 0;
02156     (void)(rx_session_next_tx(handle->session, &sequence));
02157
02158     /* Create RESET_ACK frame: cannot fail since &reset_ack_frame is non-null. */
02159     rx_frame_t reset_ack_frame;
02160     (void)(rx_frame_create_reset_ack(&reset_ack_frame, sequence));
02161
02162     /* Encode: encoder initialized at rx_spi_comm_init; all pointers valid. */
02163     uint8_t wire_buffer[k_frame_min_size];
02164     uint32_t wire_len = 0;
02165
02166     (void)(rx_frame_encode(&handle->encoder, &reset_ack_frame, wire_buffer, &wire_len));
02167
02168     /* Post-condition: RESET_ACK frame encoded to minimum size */
02169
02170     /* Transfer via SPI */
02171     return internal_spi_transfer(handle, wire_buffer, wire_len, nullptr, 0);
02172 }
02173
02174 /**
=====
02175  * Retransmission - Internal Helpers
02176  *
=====
02177  */
02178
02179 /**
02180  * @brief Retransmit the buffered frame via SPI
02181  *
02182  *

```

```

02183 *
02184 * @pre handle->retry_pending == true
02185 * @pre handle->retry_buffer contains valid encoded frame
02186 * @post retry_count incremented on success, retry_pending cleared on limit
02187 *
02188 * @since Version 1.0.0
02189 */
02190 static rx_err_t internal_retransmit_frame(rx_spi_comm_handle_t* handle)
02191 {
02192     /* Check retry limit */
02193     if (handle->retry_count >= handle->retransmit_cfg.max_retries) {
02194         rx_log_error(s_tag, "Retransmit limit exceeded");
02195         handle->retry_pending = false;
02196         handle->retry_count = 0;
02197         return k_rx_err_retry_limit;
02198     }
02199
02200     /* Set retransmit flag in buffered frame (flags byte is at offset 7).
02201     * retry_wire_len >= k_frame_min_size > k_hdr_flags always for valid encoded frames. */
02202     handle->retry_buffer[k_hdr_flags] |= k_frame_flag_retransmit;
02203
02204     /* Retransmit via SPI */
02205     rx_err_t err =
02206         internal_spi_transfer(handle, handle->retry_buffer, handle->retry_wire_len, nullptr, 0);
02207     if (err != k_rx_ok) {
02208         rx_log_error(s_tag, "Retransmit SPI transfer failed");
02209         return err;
02210     }
02211
02212     handle->retry_count++;
02213 #ifdef __RX__
02214     handle->retry_send_time_ms = (uint32_t)(tx_time_get() * k_threadx_ms_per_tick);
02215 #else
02216     handle->retry_send_time_ms = 0;
02217 #endif
02218
02219     return k_rx_ok;
02220 }
02221
02222 /*
=====
02223 * Retransmission API - Public Functions
02224 *
=====
02225 */
02226 /**
02227 * @brief Process pending retransmissions based on timeout
02228 *
02229 *
02230 *
02231 *
02232 * @pre handle must be non-NULL and initialized
02233 * @post retry_count incremented on retransmit, pending cleared on limit
02234 *
02235 * @note Safe to call when auto_retransmit is disabled (no-op)
02236 *
02237 * @since Version 1.0.0
02238 */
02239 rx_err_t rx_spi_comm_process_retransmits(rx_spi_comm_handle_t* handle,
02240                                           const uint32_t current_time_ms)
02241 {
02242     if (handle == nullptr) {
02243         return k_rx_err_invalid_arg;
02244     }
02245
02246     if (!handle->initialized) {
02247         return k_rx_err_invalid_state;
02248     }
02249
02250     /* No-op when retransmit disabled or nothing pending */
02251     if (!handle->auto_retransmit || !handle->retry_pending) {
02252         return k_rx_ok;
02253     }
02254
02255     /* Compute exponential backoff: ack_timeout_ms * 2^retry_count, capped.
02256     * Clamp shift to 31 to avoid undefined behavior when retry_count >= 32. */
02257     uint8_t shift = handle->retry_count;
02258     if (handle->retry_count >= k_max_backoff_shift) {
02259         shift = k_max_backoff_shift;
02260     }
02261     uint32_t backoff_ms = (uint32_t)handle->retransmit_cfg.ack_timeout_ms « shift;
02262
02263     if (backoff_ms > handle->retransmit_cfg.max_backoff_ms) {
02264         backoff_ms = handle->retransmit_cfg.max_backoff_ms;
02265     }
02266
02267     /* Check if timeout has elapsed */

```

```

02268  const uint32_t elapsed_ms = current_time_ms - handle->retry_send_time_ms;
02269  if (elapsed_ms < backoff_ms) {
02270      return k_rx_ok; /* Not yet time to retry */
02271  }
02272
02273  const rx_err_t err = internal_retransmit_frame(handle);
02274
02275  /* Override send time with caller-provided time (more accurate than internal timestamp) */
02276  handle->retry_send_time_ms = current_time_ms;
02277
02278  return err;
02279 }
02280
02281 /**
02282  * @brief Enable or disable automatic retransmission at runtime
02283  *
02284  *
02285  *
02286  * @pre handle must be non-NULL and initialized
02287  * @post auto_retransmit flag and config updated
02288  *
02289  * @since Version 1.0.0
02290  */
02291 rx_err_t rx_spi_comm_set_auto_retransmit(rx_spi_comm_handle_t* handle,
02292                                           const bool enabled,
02293                                           const rx_spi_comm_retransmit_config_t* config)
02294 {
02295     if (handle == nullptr) {
02296         return k_rx_err_invalid_arg;
02297     }
02298
02299     if (!handle->initialized) {
02300         return k_rx_err_invalid_state;
02301     }
02302
02303     handle->auto_retransmit = enabled;
02304
02305     /* Apply new config if provided */
02306     if (config != nullptr) {
02307         handle->retransmit_cfg = *config;
02308     }
02309
02310     /* Apply defaults for zero fields */
02311     if (handle->retransmit_cfg.max_retries == 0) {
02312         handle->retransmit_cfg.max_retries = k_retransmit_default_max_retries;
02313     }
02314     if (handle->retransmit_cfg.ack_timeout_ms == 0) {
02315         handle->retransmit_cfg.ack_timeout_ms = k_retransmit_default_ack_timeout_ms;
02316     }
02317     if (handle->retransmit_cfg.max_backoff_ms == 0) {
02318         handle->retransmit_cfg.max_backoff_ms = k_retransmit_default_max_backoff_ms;
02319     }
02320
02321     /* Clear pending retry state when disabling */
02322     if (!enabled) {
02323         handle->retry_pending = false;
02324         handle->retry_count = 0;
02325     }
02326
02327     return k_rx_ok;
02328 }
02329
02330 /**
02331  * @brief Register ACK/NACK notification callbacks
02332  *
02333  *
02334  *
02335  * @since Version 1.0.0
02336  */
02337 rx_err_t rx_spi_comm_set_retransmit_callbacks(rx_spi_comm_handle_t* handle,
02338                                               void (*on_ack_cb)(uint16_t sequence, void* ctx),
02339                                               void (*on_nack_cb)(uint16_t sequence, void* ctx),
02340                                               void* ctx)
02341 {
02342     if (handle == nullptr) {
02343         return k_rx_err_invalid_arg;
02344     }
02345
02346     if (!handle->initialized) {
02347         return k_rx_err_invalid_state;
02348     }
02349
02350     handle->on_ack_cb = on_ack_cb;
02351     handle->on_nack_cb = on_nack_cb;
02352     handle->retransmit_cb_ctx = ctx;
02353
02354     return k_rx_ok;

```

```
02355 }
```

